

EMLYON BUSINESS SCHOOL | MSc Data Science & AI Strategy

Evidence-Gated Autonomy for Retail Replenishment

A Comparative Study Using M5 and FreshRetailNet-50K

Master's dissertation • Completed bounded two-benchmark study • October 2026

This manuscript integrates the completed M5 studies with a bounded FreshRetailNet experiment. All 50,000 source series were profiled for integrity; forecasting and replenishment results concern the disclosed 30-series computational panel. Observed, inferred and simulated quantities remain distinct.

Research areas Retail analytics · operations and supply-chain management · AI governance · business decision automation

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Abstract

What should a retailer check before allowing an artificial intelligence system to place an order? Better demand forecasts are useful, but they are only one part of that decision. Inventory records can be late or incomplete, supplier rules can be difficult to interpret, and sales histories often describe what happened to sell rather than what customers wanted to buy. These weaknesses matter more as automation moves from recommending quantities to committing expenditure without direct human review.

This dissertation investigates whether a tool-grounded large language model (LLM) multi-agent system can support retail replenishment without making the organisation dependent on unverified AI judgements. It proposes *evidence-gated autonomy*: automated execution is granted only when the underlying business data, demand estimates, constraints, and proposed action pass explicit checks. Forecasting and optimization remain computational tools; language models interpret ambiguous documents, coordinate exceptions, and create auditable explanations. A rule-based system with the same evidence checks is retained as a demanding baseline, so that any additional value attributed to an LLM must be demonstrated rather than presumed.

The research design is extended from one public benchmark to two. **M5** supplies a long, hierarchical record of daily retail sales and enables established comparisons of forecasting, ordering, robustness, and bullwhip amplification. **FreshRetailNet-50K**, released by Wang et al. (2025), provides 50,000 store-product series from fresh retail, with hourly sales and stockout indicators, promotions, and weather. It makes a second form of evidential uncertainty observable: sales can be systematically censored when goods are unavailable. The same literature- and standards-informed canonical data model is used for both datasets, with a clearly identified availability-event and perishability extension. Neither public release provides the complete supplier and inventory state needed for a purchasing experiment; those missing components are simulated and treated as assumptions, not real-world measurements.

The evaluation connects M5 engineering evidence to a completed FreshRetailNet study on 30 training-selected store–product series. The new study contains 360 conventional replenishment runs, 32 matched parser and LLM runs, and 1,140 independent policy experiments. Recovery was measured against artificially hidden sales; four forecasts were scored on a protected seven-day holdout. The language model used the same source checks, numerical tools and safeguards as a capable parser. Cost and fill were identical in 16 of 16 matched FreshRetailNet pairs; all matched M5 pairs were also identical. Inventory-record repair changed FreshRetailNet fault-scenario cost by 0.00% and fill by 0.00 percentage points. Shared benefits belong to the recovery architecture, rather than an isolated language-model advantage. The findings support auditable automation and availability-aware analysis. They do not measure real retailer profit, naturally lost demand or human trust. Additional language value requires fair testing on harder, independently held-out business documents and field outcomes. A separate 480-run follow-up was motivated by the inherited spending cap's failure to pass clean orders. A registered training-only calibration selected a cap of 1,850; the evaluation reuses the holdout and is explicitly exploratory. Its 30 simulation seeds per cell do not turn that post-primary question into an untouched confirmation.

Executive evidence summary

The FreshRetailNet study is complete on a frozen panel of 30 store–product series. It adds 360 conventional replenishment runs, 32 matched parser and LLM runs, and 1,140 independent policy

experiments to the existing M5 evidence. Each main simulation run uses all 30 series. The source contains 50,000 series, so this panel covers 0.06% of the release. It supports a bounded engineering conclusion rather than a population estimate.

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

The central attribution check found identical cost and fill in 16 of 16 matched parser–LLM scenario and seed pairs. The language model was able to participate in a verified replenishment workflow. The tested supplier grammar did not, however, demonstrate an additional economic benefit from choosing an LLM over a capable parser.

Under an injected inventory-field failure, verified record repair changed mean LLM-arm cost from 23,331.29 to 23,331.29 (+0.00%) and fill from 45.2% to 45.2% (+0.00 percentage points). The parser received the same repair tools. Any shared improvement belongs to the recovery architecture, rather than to the language model alone.

Fresh retail adds a second problem: a trustworthy stockout record can still leave demand partly unobserved. This study evaluates recovery on artificially hidden sales, forecasts a protected seven-day holdout, and examines perishability inside disclosed simulators. It does not measure true lost demand during natural stockouts, real retailer profit, actual waste savings, or changes in human trust.

The calibrated controller still fills 79.19% of normal simulated demand, compared with 85.03% for conventional B3 over the same 30 seeds. Its lower indexed cost comes with lower service and remaining holds. Calibration improves an unusable gate; it does not demonstrate a dominant LLM policy.

In that exploratory follow-up, the original normal-operation seed pair changed from 45.2% to 80.2% fill, with held days 14 to 3. This is a verified cap-transfer effect, not an incremental LLM advantage: parser–LLM cost and fill are equal in 240/240 follow-up pairs.

A separately registered follow-up used a training-only spending-cap formula and 30 simulation seeds per cell, producing 480 additional runs. It is exploratory because the fixed-gate failure motivated it after primary observations, and it reuses the holdout. Its complete outputs appear separately; none replaces the primary failure.

CHAPTER 1 Introduction

1.1 Replenishment as a Sequential Decision under Uncertainty

The difference between sales and customer demand makes this decision harder than a conventional forecast-to-order diagram suggests. In a store with available stock, observed sales may be a reasonable signal of demand. When a product sells out at midday, however, the remainder of the day's demand disappears from the transaction record. The resulting error can reinforce itself: an algorithm learns that sales are low and responds by ordering too little again. FreshRetailNet-50K allows this mechanism to be tested using recorded stockout indicators rather than assuming that every observed zero means that customers were not interested (Wang et al., 2025).

Inventory replenishment is a repeated business decision made under uncertainty. A retailer has to estimate future demand, understand what stock is actually available, account for goods already on order, respect supplier and budget constraints, and decide how much to buy before the consequences of that decision are fully known. Ordering too much ties up working capital and can lead to markdowns, obsolescence, or waste. Ordering too little creates stockouts, lost sales, and poor customer experience. The decision is therefore not simply about forecast accuracy; it is about balancing service, cost, cash, and operational risk over time.

Classical inventory theory provides a strong foundation for this problem. Base-stock policies, (s, S) policies, multi-echelon models, and stochastic dynamic programming provide transparent rules under clearly defined assumptions (Clark & Scarf, 1960; Zipkin, 2000). These models remain valuable because they make the trade-offs explicit and, unlike a generative model, they can guarantee feasibility when the assumptions and inputs are valid. Retail environments, however, regularly challenge those assumptions through changing assortments, intermittent demand, promotions, uncertain lead times, pack-size restrictions, supplier minimums, and incomplete information about stock and open orders.

1.2 Two Gaps: Semantic and Evidential

A third concern links the semantic and evidential gaps: **sales censoring**. A sales feed can be current and internally consistent while still giving an incomplete picture of demand. This is not necessarily a broken data pipeline. It is a business observation made under limited availability. The revised study explicitly separates an incorrect inventory record from a correctly recorded stockout, because they require different responses. The former may justify holding an order until state is reconciled; the latter may justify revising the demand estimate while preserving the stockout flag as valid evidence (Wang et al., 2025; Little & Rubin, 2019).

Modern retailers already have access to strong forecasting and optimization methods. Probabilistic forecasting models can estimate distributions rather than point forecasts, while mathematical optimization can convert those forecasts into order quantities subject to capacity, budget, pack, and service constraints. Yet two practical gaps remain between a model output and an executable purchase order.

The first is a **semantic gap**. Business constraints are not always stored as clean parameters. A minimum order may be defined at supplier level rather than SKU level; a case-pack rule may use a different unit from the selling unit; an assortment decision may determine whether an item is eligible for a store at all;

and a temporary capacity change may arrive in a document rather than in a database field. The challenge is not only to read these rules, but to translate them into precise, testable constraints.

The second is an **evidential gap**. The optimization model normally treats its inventory position, open orders, prices, and sales history as facts. Research on inventory record inaccuracy shows that this assumption is often unsafe. DeHoratius and Raman (2008), for example, found substantial discrepancies between recorded and actual inventory in a large retail sample, while other work shows that even modest inaccuracies can materially affect automated replenishment decisions (Kang & Gershwin, 2005; DeHoratius, Mersereau, & Schrage, 2008). Data-integration research adds another layer: when information is assembled from heterogeneous systems, quality problems can arise through missing values, duplicates, failed ETL stages, inconsistent keys, stale data, or inconsistent representation (Rahm & Do, 2000; Woodall et al., 2016; Souibgui et al., 2019).

This dissertation argues that agentic automation in replenishment has to address both gaps. An AI system should not merely understand the rule and optimize the order. It should first establish whether the evidence on which the order is based is sufficiently trustworthy.

1.3 Operational Grounding from Retail Standards and Prior Research

The operational design of this study is grounded in a synthesis of retail standards, supply-chain research, and data-quality literature. The ARTS Operational Data Model provides a retail-specific view of items, supplier relationships, sales transactions, inventory, receipts, and transfers, and is explicitly intended to support integration across retail applications (Object Management Group, n.d.). GS1 standards make a complementary distinction between relatively stable master data, business transaction data, and time-stamped visibility events, while the GS1 Global Data Model standardizes foundational product attributes used across the product life cycle (GS1, 2022, 2026). Together, these sources provide a defensible basis for the common entities and lineage concepts used in the dissertation's canonical schema.

The literature also suggests that the reliability of those entities cannot be taken for granted. Wang and Strong (1996) frame data quality as broader than accuracy alone, including dimensions such as completeness, timeliness, and consistency. Research on ETL and data integration shows that heterogeneous sources introduce duplicate records, missing values, schema conflicts, transformation errors, and delays (Rahm & Do, 2000; Yang, Ge, & Helfert, 2017; Souibgui et al., 2019). In retail operations, inaccurate inventory records are particularly important because an automated ordering policy can repeatedly act on a false stock position (DeHoratius & Raman, 2008).

These findings lead to three design choices that recur throughout the dissertation. First, the replenishment system operates on a common, versioned decision state rather than reading directly from heterogeneous source systems. Second, data quality is treated as an operational input to the decision rather than as a separate reporting concern. Third, a decision trace records not only the recommendation but also the data version, constraints, model outputs, and verification steps that produced it. In other words, the research starts from a managerial question—*when should the organization trust automation enough to let it act?*—and turns that question into a technical control architecture.

1.4 The Promise and Perils of LLM Agents in Supply Chains

LLM agents are attractive in this setting because they can work across the boundary between structured computation and unstructured business information. ReAct demonstrated how a language model can alternate between reasoning and external tool use (Yao et al., 2023), while AutoGen formalized multi-

agent collaboration with different roles and tools (Wu et al., 2023). In supply-chain settings, systems such as OptiGuide use a language model as an interface to established optimization tools rather than as a substitute for them (Li et al., 2023). More recent work explores multi-agent inventory decisions, consensus, memory, and solver-assisted repair (Quan & Liu, 2024; Jannelli et al., 2024; Yoshizato et al., 2026; Ao, Simchi-Levi, & Wang, 2026).

At the same time, the risks are significant. Agentic models can be inconsistent across repeated runs, can reproduce human-like biases in inventory decisions, and can generate convincing explanations that do not faithfully reflect the cause of the output (Turpin et al., 2023; Zhao et al., 2025; Long et al., 2026). Documents supplied by external parties can also contain instructions that function as indirect prompt injections (Greshake et al., 2023). These limitations suggest a conservative design principle: the LLM should interpret and coordinate, but deterministic tools should decide whether the numbers are valid, the constraints are satisfied, and the action is permitted.

1.5 Problem Statement and Scope

The core problem is not to decide whether LLMs can calculate inventory orders. Mature forecasting and optimization systems already do that. The managerial question is whether a language-based coordination layer improves the quality, safety, and auditability of decisions sufficiently to justify its additional cost, complexity, and governance requirements. An agent may be especially useful when a supplier changes terms in an email or two systems provide contradictory evidence, but it may contribute little when the task is already covered by a reliable parser or a deterministic invariant.

The study therefore examines two forms of uncertainty together. **State uncertainty** concerns whether the system has a faithful representation of stock, open orders, and operating constraints. **Demand uncertainty under censoring** concerns the demand that cannot be observed after stockouts. These are not interchangeable. In M5, the former can be challenged through controlled perturbations, but natural stockouts cannot be verified from the public release. FreshRetailNet-50K supplies real availability annotations and contextual signals that make the second problem more directly testable, while still lacking numerical stock balances and actual purchase-order ledgers.

The formal scope is item-location replenishment with stochastic demand, lead times, order-pack and budget constraints, graded autonomy, and traceable decisions. M5 supports a longer-horizon, multi-store study of cost, service, and order variability. FreshRetailNet supports short-horizon, stockout-aware demand recovery and *simulated* replenishment of perishables, including expiry and waste scenarios. Neither public benchmark identifies the true demand that went unserved during a stockout, and neither can establish actual commercial returns from a hypothetical alternative purchasing policy. Supplier rules, inventories, purchase orders, lead times, costs, shelf lives, and operational faults absent from the released data will be introduced transparently as experimental scenarios, not misrepresented as observed facts.

1.6 Research Questions and Hypotheses

The overall research question is: **Under what conditions does an evidence-gated LLM multi-agent system create additional operational and managerial value over well-designed deterministic automation in retail replenishment, and how does the answer change when stockouts censor observed demand?**

The study addresses eight linked questions:

1. **RQ1 — Replenishment performance:** How do classical inventory policies, forecast-plus-optimizer systems, rule-gated automation, single LLM agents, and the full multi-agent system compare on cost, service, order variability, and reliability within each benchmark?
2. **RQ2 — Evidence gating:** Under injected inventory-state and constraint faults, does checking evidence before execution reduce harmful orders without imposing excessive delay or missed-sales costs?
3. **RQ3 — Censored demand:** On FreshRetailNet-50K, does stockout-aware demand recovery improve held-out forecasting and downstream simulated replenishment decisions relative to using recorded sales as if they were unconstrained demand?
4. **RQ4 — Availability versus error:** Can the system distinguish valid stockout observations from corrupted data, and does that distinction help avoid both unsafe orders and unnecessary escalations?
5. **RQ5 — Semantic value:** When supplier and assortment constraints are expressed in increasingly ambiguous documents, what value does an LLM add beyond a capable deterministic parser using the same structured tools?
6. **RQ6 — Multi-agent organization:** Do specialized roles, typed shared state, critic checks, and constrained memory produce measurable benefits after accounting for latency, token cost, and coordination failures?
7. **RQ7 — Trace quality:** Do structured, lineage-bearing records explain which evidence and tools actually influenced decisions, including recovered demand and stockout information?
8. **RQ8 — Managerial generalizability:** Which conclusions are stable across the two datasets, and which depend on perishable product economics, censoring rates, time horizons, or assumptions in the simulator?

These questions lead to hypotheses that will be evaluated rather than assumed:

- **H1:** Under clean and well-structured inputs, the full LLM architecture will not systematically beat a strong deterministic forecast-plus-optimizer on mean cost; any additional benefits should arise primarily in exception handling.
- **H2:** Deterministic evidence gates will reduce invalid execution under detectable injected faults relative to ungated automation.
- **H3:** For FreshRetailNet, properly evaluated stockout-aware demand recovery will reduce systematic underforecasting and improve the service–waste trade-off over raw-sales training in at least some censoring regimes.
- **H4:** Treating trustworthy stockout indicators as a data fault will cause unnecessary holds; separating *censoring* from *corruption* will improve calibrated autonomy.
- **H5:** A capable rules-based parser will match the LLM on templated supplier clauses, while potential LLM gains will become visible, if at all, on independently evaluated unfamiliar documents.
- **H6:** Typed state and independent verification will reduce schema, unit, and execution errors relative to free-form multi-agent conversation, at a measurable coordination cost.
- **H7:** A lineage-rich decision trace will improve fault diagnosis and justified accept-or-override decisions relative to a fluent free-text explanation.
- **H8:** The value of evidence gating will be more robust across datasets than the value of any particular LLM architecture; no universal superiority claim will be made unless the paired estimates support it.

1.7 Contributions and Structure

This dissertation contributes a management-oriented framework for assessing autonomous replenishment rather than a claim that more AI is necessarily better. It develops a standards- and literature-grounded data representation with explicit observed and simulated fields; adapts the same research artifact to M5 and FreshRetailNet-50K; introduces a separate stockout-aware demand-recovery workflow; tests autonomy decisions against data-quality and constraint evidence; and proposes a reproducible comparison against conventional automation with economically interpretable outcomes. It also evaluates whether decision explanations can be replayed and audited by people responsible for purchasing.

Chapter 2 reviews operations management, data quality, censored demand, agentic AI, and human oversight. Chapter 3 explains how the shared data model and quality controls follow from published

retail and information-system standards. Chapter 4 defines the artifact, decision permissions, and dataset-specific modules. Chapter 5 specifies the two-benchmark evaluation, including how FreshRetailNet's stockout annotations are used without claiming access to unobserved demand. Chapter 6 separates results already obtained in the small M5 pilot from hypotheses awaiting full evaluation. Chapter 7 addresses governance and limitations; Chapter 8 develops the contribution for managers and researchers. The appendices provide executable-study specifications, field mappings, and an evidence-status register.

CHAPTER 2 Literature Review

2.1 Inventory Control Theory and the Bullwhip Effect

Inventory theory provides the economic core of this dissertation. At its simplest, replenishment balances the cost of holding stock against the cost of running short. In multi-period and multi-location settings, that trade-off becomes more difficult because today's order changes tomorrow's inventory position and can affect other parts of the supply chain. Clark and Scarf (1960) established important results for multi-echelon systems, while later work consolidated practical policies such as base-stock and (s, S) control (Zipkin, 2000).

A second strand of literature shows why locally reasonable decisions can create instability at system level. Sterman (1989) demonstrated how decision delays and misperceptions generate oscillation in the Beer Game. Lee, Padmanabhan, and Whang (1997) described the bullwhip effect through demand-signal processing, order batching, rationing games, and price variation, and Chen et al. (2000) quantified the role of forecasting and lead time. This matters for AI-enabled replenishment because a policy should not be judged only by the accuracy of one order. A system that appears efficient item by item may still generate excessive upstream volatility.

Recent research suggests that agentic AI does not automatically remove this problem. Long et al. (2026) report that stochastic LLM decision-making can itself become a source of order variability, producing what they call an agent bullwhip effect. The implication for this dissertation is straightforward: reliability must be evaluated over repeated runs and over the full sequence of decisions, not inferred from a few plausible examples.

Lateral transshipment provides another useful control mechanism. When stores belong to the same network, a local shortage can sometimes be addressed by moving inventory rather than placing a new supplier order. Paterson et al. (2011) review a substantial literature on this topic. The simulator therefore allows transfers between eligible locations as a secondary action, which makes the setting closer to a networked retail decision than a single-stockpoint problem.

2.2 Inventory Record Inaccuracy, Censored Demand, and Data Quality

Fresh-retail stockouts offer an unusually clear example of data that are *accurately recorded yet decisionally incomplete*. A zero in a sales record may mean that no customer wanted a product, but it may also mean the item was unavailable. Wang et al. (2025) document this problem in a large fresh-retail dataset with hourly availability indicators and show why demand recovery and forecasting should be treated as linked tasks. A later operational study by Azar (2026) demonstrates the value of strict train-validation-test separation, leakage-free feature calculation, and carefully qualified profit proxies on the same data. These studies inform the second benchmark here, but neither directly evaluates whether a language-model agent improves the governance of replenishment.

The quality of the recorded state is central to automated replenishment. DeHoratius and Raman (2008) found that a large share of audited store-item inventory records in their retail sample differed from the physical quantity. Kang and Gershwin (2005) showed that such discrepancies can have an outsized effect on stockouts because an automated system may continue to believe that stock is available when it is not. DeHoratius, Mersereau, and Schrage (2008) further show that explicitly accounting for record

uncertainty can improve both replenishment and audit decisions. Together, these studies justify treating inventory accuracy as part of the decision problem rather than as a separate control activity.

Sales data create another evidential problem. A sale records a transaction, not necessarily the full demand that would have occurred if stock had always been available. In lost-sales settings, demand is censored during stockouts, which can bias forecasts downward (Nahmias, 1994). Intermittent demand adds further difficulty because many item-location series contain long sequences of zeros, for which traditional forecasting methods can perform poorly (Croston, 1972; Syntetos & Boylan, 2005).

Information-systems research broadens the discussion beyond inventory accuracy. Wang and Strong (1996) argue that users assess data through dimensions including accuracy, completeness, timeliness, and consistency. Data-integration research reaches a similar conclusion from a process perspective: heterogeneous sources create missing values, duplicates, inconsistent representations, schema conflicts, and transformation failures, so data quality has to be managed throughout the integration workflow rather than checked only at the end (Rahm & Do, 2000; Yang et al., 2017; Souibgui et al., 2019). Woodall et al. (2016), studying ETL deployments in large organizations, report failures that affect timeliness, accessibility, and representational consistency. These findings motivate the state certificate introduced in Chapter 4.

2.3 Probabilistic, Hierarchical, and Intermittent Retail Forecasting

The two datasets offer complementary forms of forecasting evidence. M5 contains longer, daily sales histories and a recognized multilevel evaluation design, which makes it useful for testing long-horizon inventory policies and their robustness to distribution shifts. FreshRetailNet-50K has a much shorter history but provides a unique *availability mask* that identifies which hourly observations were affected by stockouts. Demand recovery must therefore be evaluated separately from predictive forecasting: imputing a plausible value for an unobserved hour is not the same as observing the number of customers who would have purchased it (Wang et al., 2025).

For perishable goods, forecast bias is not merely an accuracy issue. Persistently ordering too little loses sales; ordering too much incurs expiry or markdown costs. The classic literature on perishable inventory treats finite shelf life as part of the decision problem (Nahmias, 1982). This motivates reporting waste and service together on FreshRetailNet, using explicit simulated shelf-life assumptions rather than attributing an unobserved expiry date to the source dataset.

A replenishment decision needs more than a point estimate of next week's sales. The uncertainty around the forecast determines safety stock, service probability, and the economic value of waiting or escalating. Probabilistic forecasting is therefore a better fit for the decision problem. DeepAR learns probabilistic forecasts across related series (Salinas et al., 2020), while the Temporal Fusion Transformer combines multi-horizon forecasting with variable selection and attention mechanisms (Lim et al., 2021). More recent time-series foundation models such as Chronos and Moirai provide useful zero-shot or lightly adapted baselines (Ansari et al., 2024; Woo et al., 2024).

The M5 competition is particularly relevant because it combines a large hierarchy, intermittent item-level sales, prices, and calendar effects. It includes 30,490 bottom-level item-store series that aggregate into 42,840 series, and its uncertainty track evaluates multiple forecast quantiles (Makridakis, Spiliotis, & Assimakopoulos, 2022a; Makridakis et al., 2022b). Forecast coherence across levels also matters in retail: item-store forecasts should not imply totals that contradict store, department, or category forecasts.

Trace-minimization methods provide one way of reconciling such hierarchies (Wickramasuriya, Athanasopoulos, & Hyndman, 2019).

In this dissertation, the language model does not calculate a forecast in prose. The forecasting component returns a versioned artifact containing quantiles or scenarios, calibration diagnostics, and data-quality flags. This separation is intentional: generative models may help select or explain a forecasting method, but the numerical forecast remains a tool output that can be reproduced and evaluated independently.

2.4 Tool-Augmented LLMs and Multi-Agent Coordination

LLM agents combine a language model with tools, memory, observations, and an action loop. ReAct demonstrated a general pattern in which reasoning is interleaved with external actions (Yao et al., 2023). AgentBench and AgentBoard show that performance in longer tasks depends on grounding, planning, and recovery from intermediate failures rather than on answer quality alone (Liu, X. et al., 2024; Ma et al., 2024). AutoGen provides a framework in which multiple agents can be given different roles and tool permissions (Wu et al., 2023).

These systems create opportunities but also new coordination risks. More agents do not necessarily mean better decisions. Each additional hand-off can propagate an error, create inconsistent state, increase latency, and make it harder to determine which component caused the final action. For a business process such as replenishment, the question is therefore not whether a multi-agent design is more sophisticated, but whether specialization produces measurable gains that justify its coordination cost.

Planning research supports a hybrid approach. LLM+P translates natural language into a formal planning representation and delegates search to a classical planner (Liu, B. et al., 2023). OptiGuide similarly uses an LLM to interact with an established optimization model rather than replacing it (Li et al., 2023). Simchi-Levi et al. (2025) argue for the use of LLMs as interfaces and coordinators around quantitative supply-chain tools. This dissertation adopts the same principle: language models handle ambiguity and orchestration; formal tools handle arithmetic, feasibility, and certification.

2.5 LLM Agents for Supply-Chain and Inventory Decisions

Early studies of LLM agents in supply chains show that the technology is capable of useful coordination, but the evidence is still emerging. InvAgent applies a multi-agent approach to inventory management (Quan & Liu, 2024), while Jannelli et al. (2024) study consensus-seeking among supply-chain agents. Yoshizato et al. (2026) report benefits from structured decision prompts and retrieval of previous cases. These studies support the use of specialization and memory, but they generally work in settings where the state and the business constraints are already clean enough to be presented to the agent.

Other studies highlight the need for stronger safeguards. AIM-Bench identifies decision biases in agentic inventory managers (Zhao et al., 2025). OptiRepair uses solver feedback and explicit inventory-theory checks to diagnose infeasible optimization models (Ao et al., 2026). The lesson is important for this dissertation: an LLM judge should not be the final authority on whether a replenishment plan is operationally valid. Validation should be expressed through tests that can be executed independently of the model.

2.6 Explainability, Decision Traces, and Faithfulness

Business users often ask an automated system to explain why a recommendation changed. A natural-language rationale can be helpful, but it can also create false confidence. Research on chain-of-thought faithfulness shows that a model can produce a fluent explanation that does not faithfully reveal the factors that caused the answer (Turpin et al., 2023; Arcuschin et al., 2025). This makes explanation quality a governance issue rather than only a usability issue.

The dissertation therefore distinguishes between a *rationale* and a *decision trace*. A rationale is text. A trace is a structured record of the evidence and tools used to make the decision: the snapshot, forecast artifact, active constraints, solver result, critic tests, approval status, and execution receipt. The text shown to a reviewer is generated from that record rather than treated as evidence in its own right. Faithfulness can then be tested by replaying the decision, removing evidence, or changing a cited factor and checking whether the action responds as expected.

2.7 Human–Automation Interaction and Graded Autonomy

Human-factors research has long shown that automation can be misused, disused, or over-trusted (Parasuraman & Riley, 1997). Parasuraman, Sheridan, and Wickens (2000) describe different levels of automation across information acquisition, analysis, decision selection, and action implementation. This framework is useful for AI governance because it avoids treating automation as a binary choice between manual and fully autonomous work.

The architecture in this dissertation follows that logic. A decision can remain advisory, require explicit approval, execute automatically within a bounded policy, or—in simulation only—operate fully autonomously. The level is recomputed for each decision. This is a meaningful departure from assigning a fixed autonomy level to an entire product category: the same item may be safe to reorder automatically on one day and require human review on another because the evidence has deteriorated.

2.8 Demand Censoring in Fresh Retail

Missing data and censored demand are related but distinct concepts. Under stockouts, the unobserved part of demand is influenced by the process that caused the product to become unavailable. Promotions, holidays, weather, and the time of day may affect both how quickly inventory is consumed and how much demand remains unobserved. The standard assumption that missing values can be filled without modelling the missingness mechanism is therefore questionable (Little & Rubin, 2019; Wang et al., 2025).

FreshRetailNet-50K was created specifically to study this problem. Its published design contains detailed hourly sales, stockout status, and covariates for 50,000 store-product series. The original benchmark reconstructs demand during stockouts and then uses the recovered series for forecasting. The authors report a 2.73% improvement in their own prediction metric and a marked reduction in downward bias; these are findings from their study, *not* evidence about the multi-agent framework proposed here (Wang et al., 2025). Crucially, the true amount of demand during a real stockout remains unobserved, so recovery accuracy must also be tested on controlled masks of previously observable sales.

Research on perishable inventory adds a second dimension: the economic consequence of error depends on how quickly a product loses value (Nahmias, 1982). For this dissertation, expiry, purchase prices, shortage penalties, and order deliveries are simulation variables, while availability annotations, past

sales, promotions, and historical weather are genuine dataset fields. Keeping those roles separate prevents a simulated profit measure from being misreported as a measured retailer profit.

2.9 Leakage-Free Evaluation and Counterfactual Limits

A forecast can appear to perform exceptionally well if information from the prediction period leaks into its features. Leakage can arise when a modeller computes store means or recent sales lags using the complete data file before splitting it, or when forecasts for a seven-day horizon incorporate actual sales from earlier days of that same future horizon. Azar (2026) documents precisely these risks for FreshRetailNet and provides a reproducible, out-of-sample evaluation protocol. The present study follows the principle of train-only feature fitting, validation-only model and policy selection, and a single sealed evaluation period.

The distinction between *observational* forecasting and *counterfactual* replenishment is equally important. Historical stockout flags were produced under a retailer's historical inventory policy. If a simulated agent chooses to buy more stock, the observed historical sales and stockout sequence cannot simply be replayed as though nothing changed. A counterfactual ordering experiment must use an explicitly modelled latent-demand process, and its conclusions are conditional on that model. For this reason, reported findings will be divided into outcomes directly measurable from the public release, outcomes from controlled missingness tests, and outcomes from simulated ordering interventions.

2.10 Research Gap and Two-Benchmark Contribution

Inventory theory explains how to calculate an order when the state and the rules are correctly specified. Data-quality research shows why such state may be unreliable. FreshRetailNet makes a separate but important defect observable: even a correctly stored sales figure can understate demand because a product was unavailable. LLM-agent research explores how language models can coordinate tools and actors, but it does not establish that an LLM adds value over a capable deterministic controller once the numerical work and data checks are held constant.

The gap addressed here is therefore **not** a lack of forecasting models or agent frameworks. It is the absence of a controlled, management-relevant assessment that examines *data validity*, *demand visibility*, *constraint interpretation*, *permission to act*, and *auditability* together, with appropriate non-LLM comparisons. Using M5 and FreshRetailNet-50K within one evidence-gated architecture tests how much of the proposed value persists across different information environments. The research aims to produce both technical evaluation and a practical adoption framework for managers who must decide when autonomous purchasing is warranted.

CHAPTER 3 Operational Grounding: Replenishment Data in Practice

3.1 Setting and Method

The addition of FreshRetailNet-50K makes the abstraction more demanding. The canonical model must support one dataset with real product prices and daily unit sales and another with hourly stockout status and globally scaled sales quantities. It is consequently designed around a small shared set of entities plus documented extensions, not around an artificial claim that the two public datasets provide identical business information. Standard concepts provide the entities; dataset cards provide the observed fields; research assumptions supply only the missing information required by a controlled simulator.

This chapter translates the literature reviewed in Chapter 2 into a concrete representation of the information that a replenishment system needs. It is intentionally based on published research and established retail data standards rather than on the internal architecture of any one company. The objective is not to claim that all retailers use the same database design. It is to identify a defensible minimum set of business objects and quality risks that recur across retail operations and that can therefore support a controlled research artifact.

The synthesis uses four types of evidence. First, the ARTS Operational Data Model provides a mature retail-specific vocabulary for items, vendors, sales, inventory, receipts, and transfers, and is explicitly designed to help retailers integrate operational applications (Object Management Group, n.d.). Second, GS1 standards distinguish master data, transaction data, and supply-chain visibility events, while the Global Data Model standardizes foundational product attributes and the EPCIS standard captures what happened, when, where, and in what business context (GS1, 2022, 2026). Third, the academic inventory literature identifies decision variables that matter for replenishment, including inventory position, lead time, demand uncertainty, pack sizes, supplier constraints, assortment decisions, and transshipment. Fourth, the data-quality and ETL literature identifies common risks introduced when information is integrated across heterogeneous systems (Wang & Strong, 1996; Rahm & Do, 2000; Woodall et al., 2016; Yang et al., 2017; Souibgui et al., 2019).

This approach is consistent with design-science research: the artifact is built from a clearly defined problem environment, but the design decisions are made traceable to prior knowledge rather than justified through personal experience alone (Hevner et al., 2004). The resulting schema and failure taxonomy should therefore be read as a literature- and standards-informed research abstraction. Where the literature provides strong empirical evidence—such as inventory record inaccuracy—that evidence is cited directly. Where a failure mode is a logical implementation stress case rather than a well-quantified population phenomenon, it is labelled and evaluated through sensitivity analysis rather than presented as an observed frequency.

3.2 The Canonical Replenishment Data Model

Two further entities are necessary for the new benchmark. **Availability events** represent intervals during which an item is or is not offered for sale at a location; they are event evidence and cannot be replaced by a numerical stock-balance table. **Perishability rules** describe shelf-life assumptions, expiry or write-off treatment, and the economic cost of unsold stock. Availability events can be populated directly from FreshRetailNet's released hourly stock status, while specific shelf lives and spoilage rates must be generated as explicit experimental parameters. A source-type and unit-type field distinguishes original

observations from synthetic values and prevents scaled sales amounts from silently being interpreted as unscaled physical units (GS1, 2022; Wang et al., 2025; Nahmias, 1982).

The canonical schema is the interface through which every agent sees the retail environment. The word *canonical* is used in the integration sense: heterogeneous source representations are mapped into a common set of business entities before the decision process begins. The schema is not proposed as a replacement for ARTS, GS1, ERP, or warehouse-management models. It is a narrower research model assembled from concepts that recur across those standards and across replenishment theory.

ARTS supports the inclusion of retail items, item-supplier relationships, inventory, receiving documents, transfers, prices, and sales transactions. GS1 supports the separation of stable master data from transaction and visibility-event data, and provides standardized identifiers and product attributes. Retail assortment research adds the idea that an item-location pair may be eligible or ineligible depending on the planned assortment (Rajaram, 2001; Mantrala et al., 2009). Replenishment and forecasting research motivates time-indexed demand, inventory position, open orders, lead time, and product hierarchy. The resulting model is shown below.

Entity	Role in replenishment	Main literature / standards basis
Locations	Stores, warehouses, and fulfilment nodes; region and cluster membership	ARTS retail business units; GS1 location/master-data concepts
Products and variants	SKU hierarchy, attributes, selling unit, lifecycle	ARTS item model; GS1 Global Data Model
Suppliers and supplier-items	Ordering relationship, lead time, pack or receiving unit, supplier-specific identifiers	ARTS Item Supplier View; GS1 party/master-data concepts
Prices	Valuation, margin context, time-valid selling prices	ARTS price and transaction structures
Sales transactions	Observed demand signal by item, location, and time	ARTS retail transaction model
Purchase orders and receipts	Pipeline inventory, ordered quantities, expected or realized arrivals	ARTS inventory control and receiving structures
Transfers	Movement of stock between locations	ARTS inventory movement structures; transshipment literature
Inventory state and snapshots	On-hand quantity used for decisions plus a versioned history for replay	Inventory-control literature; GS1 visibility-event logic; auditability requirement of this study
Assortment / eligibility	Defines whether an item is legitimately ranged at a location during a period	Retail assortment-planning literature
Reference and policy data	Units, currencies, calendars, service targets, budgets, and policy versions	GS1 master-data principles; data-quality literature

Entity	Role in replenishment	Main literature / standards basis
Availability and stockout events	Time-stamped available/unavailable flags at item–location level; observed in FreshRetailNet	GS1 visibility-event concepts; censored-demand research (Wang et al., 2025)
Perishability and shelf-life policies	Expiry age, carrying life, disposal/write-off assumptions; simulated in this study	Perishable inventory theory (Nahmias, 1982); not published FreshRetailNet item data
Context and demand-recovery artifacts	Weather, activity, holidays, promotion factors, availability masks and inferred latent demand with uncertainty	Wang et al. (2025); forecast calibration and evidence lineage

Table 3.1. Literature- and standards-informed canonical replenishment model.

Two features of the schema are particularly important for the research design. First, it separates the *business state* from the source systems that produced it. This allows the experiment to ask whether an agent can make a good decision given a defined state, without tying the results to one ERP or one retailer's data warehouse. Second, every state is versioned. The decision trace can therefore refer to the exact inventory, order, price, and eligibility snapshot that was available when a recommendation was generated.

The structure also supports a practical distinction found in GS1's architecture: relatively stable master data describe products, parties, and locations; transaction data describe commercial actions; and visibility or event data describe what happened to objects over time. This distinction is useful because different quality checks apply to each class. A duplicate product identifier is a master-data problem; a missing receipt is a transaction problem; a stale inventory snapshot is a temporal state problem.

3.3 How Replenishment State Is Produced

FreshRetailNet introduces a second provenance requirement. Its published hourly availability mask must be stored with the same date and store-product identifier as the observed sales, but a recovered estimate of latent demand must never overwrite the observed transaction. The system needs both numbers: *what customers actually purchased* and *what the statistical model estimates they might have wanted to purchase under full availability*. Keeping these separate makes forecast comparisons possible and allows a manager to see whether an order changed because the underlying data changed or because a different recovery model was applied.

A replenishment system rarely observes the physical world directly. Its decision state is assembled from systems such as point of sale, e-commerce, enterprise resource planning, warehouse management, supplier feeds, and planning applications. Data-integration research commonly represents this process as extraction from heterogeneous sources, transformation into a common structure, quality checking, and loading into a target analytical or operational model (Rahm & Do, 2000; Yang et al., 2017; Souibgui et al., 2019). The exact technology may be batch ETL, ELT, streaming, APIs, or a combination, but the

governance problem is the same: the decision system sees the result of an integration process, not the underlying reality.

For the purpose of this dissertation, the state-production process is represented through five general steps. **Master-data alignment** maps products, suppliers, and locations into common identifiers. **Transaction ingestion** loads sales, purchase orders, receipts, and transfers. **State construction** derives the current inventory position and the time series used for forecasting. **Quality certification** tests freshness, completeness, consistency, cardinality, units, and referential integrity. **Versioning** freezes the state used for a decision so that the decision can later be replayed.

This model is deliberately technology-neutral. A retailer with streaming inventory and a retailer with nightly batches may produce different freshness distributions, but both still face the need to know which sources were current, which transformations succeeded, and which version of the state informed the order. The simulator therefore uses daily snapshots because the M5 data are daily, not because daily batch processing is assumed to be universal.

3.4 A Taxonomy of State-Quality Failures

In the revised study, **genuine stockouts are not classified as data-quality failures**. They are an observable business condition that makes demand partially unobserved. A missing or contradictory stockout flag, in contrast, is a data-quality problem. This distinction affects both the quality score and the autonomy policy: the system may proceed using a calibrated, uncertainty-aware demand estimate when the availability annotation is credible, but it should hold or escalate when the underlying availability evidence itself is inconsistent. The difference is central to FreshRetailNet-specific hypothesis H4.

The state-quality layer used later in the experiments is derived from established data-quality dimensions and from failures described in the inventory and data-integration literature. Rather than presenting a long list of company-specific incidents, the taxonomy groups failures by the business property they violate. This makes it more generalizable and easier to connect to measurable checks.

Failure class	Business meaning	Why it matters for replenishment	Research grounding
Stale or late state	Inventory, orders, or sales are older than the decision assumes	The system acts on yesterday's or an incomplete view of stock and demand	Timeliness in Wang & Strong (1996); ETL delay/failure in Woodall et al. (2016)
Partial refresh	Different tables refer to different effective times	Current sales may be combined with stale stock or open orders	ETL process-quality literature; consistency and timeliness dimensions
Missing records	Expected item, location, order, or transaction rows are absent	Demand, available inventory, or pipeline stock is understated	Completeness literature; Souibgui et al. (2019)
Duplicate records	The same fact is loaded more than once	Sales, receipts, or open orders are overstated	Rahm & Do (2000); Souibgui et al. (2019)
Referential mismatch	A transaction cannot be linked to a valid item, supplier, or location	Facts can be dropped, misallocated, or excluded from forecasts	Schema/integration literature; ARTS and GS1 master-data logic

Failure class	Business meaning	Why it matters for replenishment	Research grounding
Identity inconsistency	The same entity appears under different codes or representations	Demand histories split and supplier or product totals become unreliable	Data-cleaning literature; master-data research
Unit or pack mismatch	Quantity is represented in a different unit from the one assumed	Stock, demand, or order quantities can be wrong by a conversion factor	ARTS unit-of-measure conversion; Das, Carrel, & Caplice (2021)
Currency or valuation mismatch	Cost or price uses an inconsistent currency or missing conversion	Budget and economic objective become misleading	Consistency dimension; retail price/master-data structures
Inventory record inaccuracy	Recorded on-hand differs from physical stock	Automated replenishment can repeatedly under- or over-order	Kang & Gershwin (2005); DeHoratius & Raman (2008)
Open-order uncertainty	Quantity, supplier, status, or expected arrival is incomplete or unreliable	Pipeline inventory is misrepresented and orders may be duplicated	Inventory-control logic; ARTS receiving/order structures
Historical restatement	Returns, cancellations, corrections, or late postings revise the past	Forecast training data and replayed decisions can change retrospectively	Transaction-processing logic; treated as a scenario rather than a population rate
Transformation error	A calculation or mapping creates an implausible field or distribution	Derived inventory or demand can collapse or inflate at scale	ETL quality literature; implemented as a stress case
Censored demand	Observed sales fall to zero because stock is unavailable	Forecasts can interpret a stockout as weak demand	Nahmias (1994); lost-sales and intermittent-demand literature

Table 3.2. State-quality failure taxonomy used in the research design.

The taxonomy is designed to distinguish failures that are easy to catch with deterministic checks from those that require contextual interpretation. Missing dates, duplicate keys, impossible quantities, and inconsistent units are good candidates for rules. A sudden drop in one store's activity is harder: it might be a broken feed, a store closure, or a genuine demand change. The proposed architecture uses deterministic tests first and reserves LLM-based interpretation for the residual ambiguity. This ordering is important both for safety and for managerial efficiency.

3.5 A Taxonomy of Constraints as They Arrive

None of the published FreshRetailNet fields is a supplier contract, a minimum-order term, a case-pack specification, or a lead-time record. The constraint-extraction experiments therefore use a common, separately generated document corpus for both benchmarks. The corpus will distinguish simple templates, ambiguous supplier emails, mixed-unit clauses, and contradictory workbook entries. Because the documents are synthetic, a successful extraction result must be described as performance on a controlled test set unless an independently sourced and annotated document set is subsequently added.

Inventory models often represent constraints as clean mathematical parameters. Business processes are less tidy. Some constraints sit in master data, some in supplier terms, and some in planning documents. Retail standards and operations literature also show that the unit in which a retailer sells an item may differ from the unit in which it orders or receives it. ARTS, for example, explicitly models supplier receiving units and conversion rules, while research on pack sizes shows that ordering in cases rather than individual selling units can materially alter inventory and service outcomes (Das et al., 2021).

Constraint type	Typical representation	Formal issue created	Research basis
Pack or case multiple	Supplier-item master or ordering rule	Order quantities must be integer multiples	ARTS receiving units; Das et al. (2021)
Minimum order quantity / value	Supplier terms or contract	Creates a lower bound and often couples multiple items	Supplier-constraint and joint-replenishment literature
Unit conversion	Master data or specification	Ordering, receiving, and selling units must be reconciled	ARTS unit-of-measure conversion; GS1 product data
Assortment eligibility	Range or assortment plan	Some item-location pairs should not be replenished	Rajaram (2001); Mantrala et al. (2009)
Lead time and arrival uncertainty	Supplier terms, order status, historical receipts	Pipeline stock arrives probabilistically rather than at a fixed date	Inventory-control literature
Supplier capacity	Allocation or temporary supply limit	Orders compete for a shared upper bound	Supply-chain optimization literature
Budget / open-to-buy	Financial planning rule	Items are coupled through a common spend limit	Retail planning practice; optimization literature
Lifecycle / season rule	Assortment calendar or product status	Replenishment may stop even when a local model would otherwise order	Fashion and assortment-planning literature
Transfer restriction	Network or store-cluster policy	Only selected location pairs can exchange inventory	Paterson et al. (2011)

Table 3.3. Constraint forms represented in the experimental environment.

The dissertation does not assume that an LLM should infer every business rule from free text. Where a constraint already exists as a structured field, the structured value is preferred. The language model is used when a rule must be extracted or reconciled across documents. Once extracted, the rule is converted into a typed object containing scope, value, unit, validity period, source, and confidence. Unit conversion and feasibility checks are then performed deterministically.

3.6 Decision Context Specifics

The datasets also represent different retail economics. M5 supports a longer history of replenishable goods in a general-purpose store hierarchy. FreshRetailNet describes short-lived, frequently restocked products whose available stock may be exhausted within the selling day. The fresh-retail experiment

accordingly adds availability-weighted demand recovery, shorter planning horizons, expiry decisions, and asymmetric shortage versus waste penalties. These are modeled scenario characteristics, not claims that the public release contains actual expiry outcomes.

The literature suggests several features that make retail replenishment different from a generic forecasting exercise. **Assortment changes** mean that a zero sale does not always mean zero demand; an item may simply not be ranged at that location. **Pack sizes** can force stores to hold more inventory than a continuous optimization model would recommend (Das et al., 2021). **Inventory inaccuracy** means the system's stock position may be uncertain. **Lost sales** mean demand becomes partly unobserved during stockouts. **Transfers** create an alternative to new purchasing. Finally, **managerial overrides** are not necessarily irrational: research on store ordering shows that local managers often modify automated recommendations because the formal model does not capture all operational constraints (van Donselaar et al., 2010).

These observations are important for the business-school framing of the study. The aim is not to demonstrate that a more complicated algorithm is always superior. It is to examine how an organization can combine formal decision support with human judgment without losing control over why a decision was made. The architecture is therefore evaluated not only on cost and service, but also on escalation quality, audit time, and calibrated trust.

3.7 Design Implications

A single abstract interface now supports two different types of evidence. M5 contributes unit sales, store/item hierarchies, published prices, and longer historical patterns. FreshRetailNet contributes scaled daily and hourly sales, availability observations, hierarchies, promotions, and weather. Both require synthetic on-hand balances, suppliers, open orders, and decisions to construct counterfactual replenishment experiments. The adapter for each dataset records every inferred, simulated, or missing field and exposes an explicit `source_kind` and `unit_kind` to downstream verification tools.

The literature synthesis leads directly to the design choices in Chapter 4. Table 3.4 makes that link explicit so that the architecture is traceable to prior research rather than to undocumented intuition.

Evidence from literature / standards	Design implication
Inventory records can be inaccurate and data quality includes timeliness, completeness, and consistency	Compute a state certificate before forecasting or ordering; use it in the autonomy decision
Retail information is split across master, transaction, and event data	Use a canonical schema with versioned lineage rather than allowing agents to read heterogeneous systems directly
ETL and integration can introduce missing, duplicate, late, or inconsistent data	Run deterministic checks on freshness, cardinality, referential integrity, and plausibility
Some anomalies are contextual rather than purely structural	Use the LLM for evidence-backed triage after deterministic checks, not instead of them
Pack sizes, supplier units, MOQs, and assortment rules alter feasible orders	Represent constraints as typed objects with units, scope, validity, and provenance

Evidence from literature / standards	Design implication
LLM rationales may be unfaithful	Generate explanations from a structured decision trace that can be replayed
Automation bias is a known human-factors risk	Use graded autonomy, explicit holds, and auditable approval status

Table 3.4. From literature and standards to system design.

3.8 Evidence Classification: Observed, Derived, Inferred and Simulated

Claims about AI systems become hard to evaluate when observed data and experimental assumptions are mixed. For each item-location-day, the proposed schema therefore labels evidence as **observed** (released dataset fields), **derived** (reproducible transformations such as a daily stockout fraction), **inferred** (statistical estimates of latent demand, each with uncertainty), or **simulated** (inventory, supplier deliveries, costs, shelf life, or injected data faults). These labels persist in the lineage of forecasts, constraints, and final recommendations.

Business fact	M5	FreshRetailNet-50K	Research treatment
Sales and product/location identifiers	Observed daily unit sales	Observed normalized daily/hourly sales	Map with original unit and scale metadata
Item prices	Observed selling prices, at weekly grain	No unit prices in published fields	Use observed M5 prices; proxy economics for FreshRetailNet
Stockout/availability indicator	Not published	Observed hourly flags and daily stockout-hour counts	Real mask in FreshRetailNet; simulator-generated in M5
Physical on-hand quantities	Not published	Not published	Simulated and marked as such in both
Suppliers, lead times, MOQs, purchase orders	Not published	Not published	Versioned synthetic supply/constraint layer in both
Perishability / shelf life / waste	Not published	Not published in public file	Optional M5 cost; explicit FreshRetailNet shelf-life scenarios
True lost demand during stockouts	Unknown	Unknown despite stockout flags	Statistically inferred; truth known only for artificial masking/simulation

Table 3.5. Evidence-status register for the two public benchmarks.

3.9 Why a Single Canonical Model Still Needs Dataset-Specific Policies

Common tables are valuable because they let the same data-quality checker, optimizer interface, critic, and decision-trace protocol run on both benchmarks. They do not imply identical inventory policies.

Where M5 encourages comparisons over weeks and multiple replenishment cycles, FreshRetailNet rewards intraday availability reasoning and a more explicit trade-off between waste and shortages. The artifact therefore separates *portable governance mechanisms* from *setting-specific quantitative choices*. A result that replicates across both settings is stronger evidence for an organisational design principle; a result confined to fresh retail is still valuable but should not be sold as a universal rule.

CHAPTER 4 Methodology and Architecture

4.1 Research Design

The expanded evaluation uses a comparative multi-case design within the same research artifact. In the M5 case, the system is judged on long-horizon replenishment using observed unit sales and transparent supply-state simulation. In the FreshRetailNet case, it is judged first on the statistical problem of demand invisibility and then on the consequences of that problem for short-horizon simulated ordering. The aim is *analytical replication*: hold the governance architecture constant while varying the information and business context. Performance is never pooled across the datasets without a clearly defined normalized measure, because the quantity units, observation periods, and economic assumptions differ.

The study adopts a design-science approach because the research question is concerned with both understanding a managerial problem and evaluating an artifact intended to address it (Hevner et al., 2004). The artifact is a hybrid replenishment system that combines deterministic data-quality controls, probabilistic forecasting, formal optimization, LLM-based interpretation, and an explicit autonomy policy. The objective is not to prove that one software architecture is universally optimal. It is to test whether this combination creates measurable value under the kinds of uncertainty and information-quality problems identified in the literature.

The evaluation has three complementary parts. First, a closed-loop simulator measures operating outcomes such as cost, service, stockouts, and bullwhip amplification. Second, controlled disturbance experiments test how the system behaves when demand, supply, constraints, or data quality deteriorate. Third, an audit study evaluates whether different explanation formats help human reviewers identify bad decisions and calibrate their trust. This mixed evaluation is appropriate for a business-school dissertation because technical performance alone does not establish organizational usefulness.

Development, model selection, and final evaluation are separated. Forecasting models are selected through rolling-origin validation. Autonomy thresholds are tuned on development scenarios rather than on the final test windows. All random seeds, prompt versions, model versions, solver settings, and disturbance parameters are logged so that results can be reproduced.

4.2 Formal Replenishment Problem

For FreshRetailNet, the mathematical program adds an age-structured inventory state and the option to dispose of stock that has expired. The simulator distinguishes *observed sales*, *latent demand*, *available inventory*, and *fulfilled demand*. Let D^* denote latent demand, Y observed sales, and A the historical availability indicator. Historical sales satisfy $Y \leq D^*$, but D^* is not revealed when $A = 0$. A recovery model generates a distribution for the missing component conditional on past sales, availability, promotions, holidays, and permissible weather covariates. In simulation, the policy changes available stock and therefore the future simulated stockout pattern; the historical availability mask is never treated as the outcome of a new counterfactual policy.

```
Fresh-retail extension (simulator definitions, not fields in the public dataset)
H[i,l,t,a]      simulated on-hand at age a
R[i,l,t]        simulated replenishment receipts
D*[i,l,t]       latent demand draw (unobserved in released data)
Y[i,l,t]        released observed, globally scaled sales
A[i,l,t,h]      released hourly availability/stockout status
E[i,l,t]        simulated expired quantity

served[i,l,t] = min(D*[i,l,t], available_stock[i,l,t])
```

```

lost[i,l,t] = max(0, D*[i,l,t] - served[i,l,t])
expiry[i,l,t] = quantity whose simulated age exceeds life[i]

cost_index = purchase_index + holding_index + lost_sales_penalty
            + expiry_penalty + supplier_fixed_order_cost
autonomy_level = policy(evidence_certificate, forecast_uncertainty,
                        business_risk, constraint_confidence)

```

Because FreshRetailNet does not publish absolute item-level prices or inventory units, the cost expression is a **dimensionless indexed simulation objective**, not a measured yuan or euro profit. The assumptions concerning item life, order quantities, and penalties are preregistered and varied in sensitivity analysis. M5 retains its independently specified unit-based economic simulator. Within each dataset, identical parameter settings and exogenous random streams are shared across policies.

The quantitative core is a stochastic replenishment model. The formulation is deliberately modular: a simple policy can be used for uncoupled, low-risk items, while a mixed-integer model is available when items are linked by supplier minimums, budgets, capacity, or transfer decisions. The important distinction for this dissertation is between the *true state* used by the simulator to evaluate an outcome and the *observed state* made available to the decision system.

```

Indices
i  items                l, l' locations                s  suppliers
t  periods              w  demand / lead-time scenarios
c  location clusters; cl(l) denotes the cluster of location l

State
H[i,l,t]                true on-hand inventory
Hobs[i,l,t]             observed on-hand inventory presented to the system
B[i,l,t]                backorders or lost-demand state
O[i,l,s,t]              outstanding supplier orders
e[i,l,t]                assortment eligibility in {0,1}
q[t]                   evidence / state-quality score in [0,1]

Decisions
x[i,l,s,t]              supplier order quantity
k[i,l,s,t]              integer pack multiplier
z[i,l,s,t]              item-line activation
y[i,l,l',t]             lateral transfer quantity

Representative constraints
x = pack * k
x <= M * e                # order only eligible items
supplier / order MOQ constraints
unit-of-measure conversion constraints
supplier capacity constraints
budget constraints
transfer restrictions within allowed clusters
inventory-balance and receipt equations

Objective
minimize expected purchase + holding + shortage + transfer + expedite cost
        + lambda * CVaR_alpha(total cost)

```

The CVaR term follows Rockafellar and Uryasev (2000) and gives additional weight to poor tail outcomes rather than optimizing only the average case. This is particularly relevant when evaluating agentic systems because a system with a good mean result but occasional extreme failures may be unacceptable in practice.

The design introduces two additional constructs. The first is the **autonomy policy**, which maps the evidence available for a decision into a permission level:

```

a[t] = pi(q[t], r[t], kappa[t], sigma[t])
      in {advisory, approval, bounded, full}

q      evidence / state-quality score
r      business risk of the action
kappa  confidence in the active constraint set
sigma  forecast uncertainty

```

The second is **safety-adjusted utility**. A policy is not rewarded simply for avoiding all risk through constant escalation. Escalations and holds have business costs as well. The comparison therefore combines expected operating cost, tail risk, the cost of escalation, and the cost of harmful executions.

$$U = -(\text{expected cost} + \lambda * \text{CVaR}) \\ - c_{\text{Esc}} * \text{number of escalations} \\ - c_{\text{Harm}} * \text{number of harmful executions}$$

A harmful execution is defined only inside the simulator, where the true state is known. It is an executed order or transfer that violates a true hard constraint or deviates materially from the oracle decision because the system acted on corrupted evidence. This measure is central to RQ2.

4.3 Blackboard Architecture over a Canonical Schema

The shared objects include `dataset_id`, `store_product_id`, `time_grain`, `source_kind` (observed, derived, inferred, or simulated), `unit_kind`, `snapshot_id`, and `available_at_time`. FreshRetailNet also carries `stockout_hours`, an hourly availability vector, demand-recovery model/version, and perishable-policy assumptions. This prevents a recovered demand estimate from being mistaken for an observed transaction and blocks the use of a synthetic balance as though it were a retailer-supplied stock count.

The agents share a typed, versioned decision state rather than exchanging only free-form messages. This pattern is sometimes described as a blackboard architecture: specialist components read from and write to a shared representation of the problem. In this dissertation, that shared representation is the canonical schema developed in Chapter 3.

The architectural choice has a governance purpose as well as a technical one. Every object contains units, timestamps, provenance, validity windows, and a decision identifier. Downstream tools consume those structured fields rather than relying on prose. A forecast produced from snapshot A cannot silently be combined with inventory from snapshot B. If evidence from different versions must be combined, the system has to record a reconciliation step.

The canonical interface is also what makes the experiment independent of any one retailer's technology stack. In a real deployment, ERP, WMS, POS, e-commerce, and supplier systems would be mapped into this layer. In the experiment, M5 and the simulator are mapped into the same layer. The agents therefore operate on the business concepts defined by the research design rather than on vendor-specific source tables.

4.4 Agents and Their Verifiable Outputs

In FreshRetailNet experiments, the Forecast Agent operates in two stages: a *demand recovery* stage that handles censored historical hours, followed by a *forecast* stage that predicts future demand without using future observed stockout labels. The State Reconciliation Agent verifies that hourly arrays and daily totals are well formed and that stockout flags correspond to the intended time interval. The Critic treats a reliable, recorded stockout as information about the measurement process—not automatically as a broken feed. The Orchestrator records whether a decision was based on raw sales, a particular recovery model, or a recovery model that failed calibration checks.

Specialization is used only where it creates a clear separation of responsibility. Each agent has a narrowly defined role and must produce a machine-checkable output before the workflow can proceed.

Agent	Business responsibility	Required verifiable output
State Reconciliation and Data-Quality Agent	Establishes whether the decision state is usable	State certificate with freshness, completeness, consistency, invariant results, and unresolved anomalies
Demand Forecast Agent	Produces the demand distribution used for planning	Forecast artifact with quantiles/scenarios, calibration metrics, feature window, and model version
Supplier and Constraint Agent	Converts supplier, assortment, and policy rules into structured constraints	Typed constraint set with scope, unit, source, validity, and confidence
Replenishment Optimization Agent	Generates economically efficient feasible plans	Candidate actions, objective breakdown, active constraints, solver status, and certificate
Risk and Critic Agent	Independently challenges the proposed action	Pass, veto, or escalate decision with explicit test results
Orchestrator and Autonomy Agent	Controls workflow and determines what the system is permitted to do	Committed decision and autonomy level, or an escalation packet
Execution and Monitoring Agent	Executes approved actions and records outcomes	Idempotent execution receipt and subsequent outcome events

Table 4.1. Agent roles and required outputs.

The architecture deliberately avoids giving the LLM final authority over facts that can be checked by code. The State Agent cannot simply declare a feed reliable; it must return the results of the tests. The Optimization Agent cannot claim feasibility without a solver status. The Critic cannot rewrite the proposal in place; it can only raise an objection that causes the proposal to be revised and checked again. This separation is intended to reduce the risk that a persuasive language-model response becomes a substitute for verification.

4.5 Coordination Protocol

The workflow follows a simple sequence: **OBSERVE** → **CERTIFY STATE** → **FORECAST and GROUND CONSTRAINTS** → **PROPOSE** → **VERIFY** → **COMMIT** → **EXECUTE** → **MONITOR**. Forecasting and constraint grounding can run in parallel once the state has been certified. The system does not proceed to an executable proposal when the state contains an unresolved hard failure.

Each stage has an explicit entry and exit condition. Typed schemas are validated before a hand-off. Units are normalized. Tool calls use idempotency keys where duplication would be harmful. Timeouts, retry budgets, and maximum dialogue turns prevent open-ended agent conversations. Agents have allowlisted tools and limited write permissions. These controls are ordinary software-engineering mechanisms, but in an agentic architecture they also serve as governance controls because they restrict what the model is capable of changing.

If the language model is unavailable, the system can fall back to a deterministic baseline policy, provided the state certificate passes. This is an important operational principle: the organization should not make continuity of routine replenishment dependent on the availability of a generative model.

4.6 State Reconciliation and the Data-Quality Score

For the fresh-retail case, the certificate has two independent dimensions. **Measurement integrity** checks whether the time index, hourly array length, sales values, category identifiers, and availability flags are valid. **Demand visibility** describes the fraction of the recent history affected by stockouts and the uncertainty of the recovery model. A high stockout fraction can increase forecast uncertainty without implying that the source record is corrupt. Only failed measurement-integrity checks trigger a hard state hold; high, but credible, censoring can instead lead to wider forecast intervals, tighter spending caps, or planner approval.

The state certificate is computed over the exact evidence needed for a decision. It therefore does not ask whether the entire enterprise data warehouse is healthy; it asks whether the inventory, open orders, sales history, price, and eligibility used for this order are sufficiently reliable. The certificate covers the following families of checks:

- **Freshness:** whether each relevant source is recent enough for the decision horizon.
- **Completeness:** whether expected item-location, order, and reference records are present.
- **Continuity:** whether the historical series contains unexpected gaps.
- **Cardinality and uniqueness:** whether duplicate facts or orders are present.
- **Referential integrity:** whether products, suppliers, and locations resolve correctly.
- **Units and ranges:** whether quantities, prices, and conversion factors are plausible and compatible.
- **Cross-source consistency:** whether overlapping sources disagree beyond an allowed tolerance.
- **Movement consistency:** whether the change in stock can be reconciled with sales, receipts, and transfers.
- **Distribution shifts:** whether a transformation failure has created an implausible system-wide change.
- **Censoring flags:** whether sales observations coincide with stockout periods.

Each test returns pass, warning, or hard failure together with evidence. Warnings are combined into a quality score; a hard failure can force a hold regardless of the numerical score. The LLM may then help interpret warnings that require context—for example, whether a sudden disappearance of transactions is more consistent with a closed store or a broken feed—but it cannot directly change the underlying data or override a hard failure. Its interpretation is recorded as a hypothesis, not as a new fact.

4.7 Constraint Extraction and Verification

The Supplier and Constraint Agent converts relevant rules into a standard representation containing the affected entity, scope, parameter, value, unit, conversion, aggregation level, effective period, source reference, precedence, and confidence. Structured source fields are preferred over generated interpretation. LLM extraction is used mainly for prose or semi-structured documents.

The process is **retrieve** → **extract** → **normalize** → **validate** → **cross-check** → **commit or escalate**. Normalization and dimensional analysis are deterministic. A case-pack rule must produce an integer-multiple constraint. A supplier unit must convert into the retailer's order unit through an explicit factor. A minimum that applies across a supplier order must not be silently converted into a per-item minimum. If the unit, scope, or validity interval cannot be resolved, the system escalates rather than guessing.

This design is deliberately conservative. In business terms, a wrong interpretation of a supplier rule can create a direct financial or contractual consequence. The architecture therefore treats low-confidence constraint extraction as a reason for human review, not as a request for the LLM to "try harder."

4.8 Forecasting Module

FreshRetailNet adds specific recovery candidates: a raw-sales baseline; an availability-weighted historical-rate estimator; an interpretable recovery model that adjusts hourly demand using comparable in-stock periods; and one learned temporal method such as TimesNet or a suitably resourced alternative drawn from the original benchmark (Wang et al., 2025). A Tobit-style or inverse-Mills-based recovery serves as an econometric or heuristic comparator following Azar (2026). Each method is evaluated first under artificial censoring of selected formerly in-stock periods, where the withheld sales provide a measurable target, and then by its ability to improve held-out forecasts. The study will not report ground-truth recovery accuracy for real stockout periods because their true unmet demand is unobserved.

Future weather is particularly vulnerable to leakage. Historical weather can be used to fit a model; at a forecast origin, only observed weather up to that date, an explicit weather forecast available at that time, or a calendar-conditioned climatology may enter the future horizon. Released evaluation-period weather is not silently used as if it had been forecast in advance. Known promotional plans and calendar holidays may be used prospectively only when their availability before the forecast origin is justified.

Forecasting models are trained and compared through rolling-origin validation. Candidate models include seasonal naive methods, Croston-style approaches for intermittent demand, LightGBM quantile regression, DeepAR, the Temporal Fusion Transformer, and pretrained Chronos or Moirai models. Model selection can be global, category-specific, or ensemble-based depending on validation performance.

The Forecast Agent returns probability information rather than a single best guess: at minimum a median, service-relevant upper quantiles, scenario samples, calibration coverage, and the version of the data used. Where stockout flags indicate censored observations, the pipeline applies a lost-sales correction or marks the forecast as lower-confidence. Higher-level forecasts are reconciled with item-store forecasts so that the hierarchy remains coherent.

Importantly, this dissertation does not depend on a claim that a foundation model will outperform established forecasting methods. Time-series foundation models are included as contemporary baselines. The research question concerns the decision system around the forecast, not a competition to identify one universally best forecasting architecture.

4.9 Optimization and Policy Tools

Fresh-retail replenishment includes two practical policy classes that are especially relevant when full mixed-integer optimization would be expensive or unnecessary: an order-up-to rule based on lead-time demand and an age-aware newsvendor policy balancing expected shortage against expiry. Both are implemented in deterministic code and compared against the larger optimizer using the same recovered or raw demand distribution. This allows the study to test whether improvements come from de-censoring, from a more sophisticated ordering policy, or from the LLM coordination layer. If a simpler policy matches the agent, that finding is recorded as a business-relevant outcome rather than treated as a failed experiment.

The optimization layer selects the simplest appropriate method. For an uncoupled item with a clear state and straightforward pack constraint, a policy calculator may be preferable to a full mathematical program. When supplier minimums, shared capacity, budgets, or transfers link several items or locations, a mixed-integer model is used.

The LLM does not calculate the objective value or certify feasibility. It supplies a validated problem specification to the optimization tool and interprets the result for the workflow. If the model is infeasible, solver diagnostics are used to identify conflicting constraints. The agent may suggest a resolution, but a constraint that comes from an authenticated contract or policy cannot be relaxed automatically. This follows the solver-in-the-loop principle used in work such as OptiRepair (Ao et al., 2026).

4.10 Evidence-Gated Autonomy and Safety Controls

Two forms of caution must not be confused. A corrupted feed can make a specific order unjustifiable, whereas a genuine stockout means the business has learned that observed sales are censored. The autonomy policy uses both integrity and visibility indicators: hard integrity failures stop automated execution; high uncertainty in recovered demand can downgrade execution to bounded or approval-gated mode. A reliable stockout flag by itself does not block an order. Otherwise the system would repeatedly refuse to replenish precisely the products most likely to need replenishment.

The architecture uses four levels of autonomy. **Advisory** means the system proposes an action but cannot submit it. **Approval** means a user must accept or modify the proposal. **Bounded autonomy** allows automatic execution within predefined limits. **Full autonomy** is reserved for simulation or tightly controlled testing.

The autonomy level is recomputed for each decision using the evidence-quality score, action risk, constraint confidence, and forecast uncertainty. A routine low-value order with fresh data, stable demand, and high-confidence constraints may qualify for bounded execution. The same item can be downgraded to approval if the inventory snapshot is stale or a supplier rule has changed. A hard state failure results in a hold and an escalation packet.

A hold is not assumed to be costless. Delaying an order can itself create a stockout. The policy therefore includes a hold budget and an urgency mechanism: if uncertainty persists, the system escalates rather than simply doing nothing indefinitely. The experiments vary this budget to quantify the trade-off between avoiding a wrong order and delaying a necessary one.

Hard controls include spend limits, maximum deviation from a deterministic reference policy, maximum days of supply, eligibility checks, pack-integrity checks, duplicate-order prevention, policy-version pinning, approval requirements for high-risk actions, least-privilege tool access, and immutable execution receipts. External documents are treated as data, not as instructions to the agent.

4.11 Native Structured Decision Trace with Lineage

The trace includes, when applicable, a distinct *demand recovery* block: the published stockout-hour count and availability mask, the pre-recovery observed sales, the recovery algorithm and version, the inferred demand distribution, uncertainty intervals, the forecast origin, and the evidence-access cut-off. The consumer sees both the raw and adjusted forecasts and can identify how much an order changed because of the recovery assumption. The simulator's counterfactual demand generation is labelled separately and never described as an observed retailer outcome.

Every recommendation creates a structured decision record. This is not a transcript of hidden reasoning. It is an audit object composed of observable inputs, tool outputs, tests, and approvals. A human-readable explanation is generated from that record after the decision has been assembled.

9. **Identity and timing:** decision identifier, item-location-supplier, timestamp, policy version, and granted autonomy level.
10. **Lineage:** state version, source freshness, invariant results, quality score, and unresolved anomalies.
11. **Evidence:** source identifiers or hashes, reconciliation results, forecast version, quantiles, and uncertainty status.
12. **Constraints:** normalized values, units, scope, source, validity interval, confidence, and whether the constraint was active.
13. **Alternatives:** candidate order and transfer quantities, cost and service trade-offs, and rejected options.
14. **Verification:** critic tests, stress checks, vetoes, revisions, and residual uncertainty.
15. **Commit and execution:** approval or override, idempotency key, execution response, and later outcome events.

A trace is considered complete only if the final action can be reconstructed from the referenced state, constraints, and tools with pinned versions. This requirement turns explainability into a testable property rather than a stylistic feature.

4.12 Common Decision Workflow, Two Dataset Adapters

The artifact follows one sequence in both cases: ingest → validate provenance → certify measurement integrity → assess demand visibility → forecast and ground constraints → optimize → verify → assign autonomy → log and (in simulation) execute. What changes is the adapter. The M5 adapter supplies daily unit sales, store-item price and calendar information. The FreshRetailNet adapter preserves hourly arrays, stockout status, promotional and environmental covariates, and scaled-quantity metadata. For reporting, every decision is accompanied by a machine-readable statement of the dataset fields actually used and the synthetic assumptions introduced.

This design deliberately avoids presenting multi-agent orchestration as a forecasting method in itself. The same approved forecast and constraint set can be passed to a deterministic controller and to an LLM-based controller. Only then is it possible to attribute differences in decision quality, exceptions, cost, latency, and explanation to the orchestration architecture rather than to different information or a stronger underlying forecast.

4.13 Study Controls and Governance Before Evaluation

Both benchmarks follow a frozen configuration manifest containing source version, file hashes, allowed feature timestamps, recovery model, forecast model, supply parameters, policy thresholds, seeds, prompts, parser rules, simulator version, and evaluation windows. The held-out portion of FreshRetailNet is not consulted to adjust prices, model weights, threshold rules, shelf lives, or policy selection. Any changes after inspecting final evaluation results will be documented as exploratory and checked on a new, unexamined setting where feasible. Approval-gated execution is simulated throughout; the study does not submit purchase orders to a real retailer.

CHAPTER 5 Experimental Design and Evaluation Protocol

5.1 Research Logic: A Comparative Two-Benchmark Evaluation

The purpose of using two datasets is not to make one enormous pooled forecasting experiment. M5 and FreshRetailNet-50K represent different information environments. M5 provides a long daily series, a standard retail hierarchy, and price information but no direct stockout observation. FreshRetailNet provides a short but information-rich history of fresh-retail sales, annotated hourly stockouts, and contextual covariates. Both permit the evaluation of an evidence-gated decision architecture, but each provides a different test of why that architecture might be useful. The first setting emphasizes state and supplier-constraint failures; the second allows an additional investigation of demand that goes unobserved during genuine stockouts.

The study applies the design-science logic of *building and evaluating an artifact* (Hevner et al., 2004). The core artifact is held as stable as possible: a canonical schema, integrity certificate, typed constraints, forecast tool, deterministic optimizer, critic, autonomy policy, and decision trace. Dataset-specific models are explicitly interchangeable components. M5 tests whether the governance layer remains reliable over longer replenishment sequences. FreshRetailNet tests whether it reacts appropriately when the observations are valid but the sales signal is censored. The resulting design is better suited to a business-school dissertation because it addresses a decision that managers actually face: where to invest in AI, data quality, and oversight when classical automation already works reasonably well.

Design dimension	M5 track (Track M)	FreshRetailNet track (Track F)
Real public observations	Daily unit sales, product/store hierarchy, weekly prices, events	Daily and hourly scaled sales, hourly stockouts, categories, promotions, weather, holidays
Time span	1,941 public daily training/evaluation history days	90-day training file and a separate 7-day evaluation file
Retail decision	General replenishment, optional inter-store transfers, cost/service over repeated cycles	Fresh replenishment, stockout recovery, service–waste trade-off under simulated shelf lives
Best supported additional question	Does verified state and constrained ordering reduce harmful actions?	Does credible stockout evidence change forecasts and approval-worthy orders?
Shared experimental inputs	Synthetic inventory, suppliers, ordering terms, budgets, delivery uncertainty, injected state faults and constrained document corpus	
Claim permitted	Comparative simulated operational performance; descriptive M5 pilot already available	Observed stockout/censoring patterns plus simulated operational consequences; experiments planned

Table 5.1. Complementary contributions of the two evaluation tracks.

In both tracks, the central experimental contrast is the full multi-agent proposal against a **capable, verified non-LLM controller with identical forecasts, optimization tools, and input constraints**. The design also compares both against less capable baselines and deliberately defective variants. This avoids the common mistake of crediting an LLM for the benefits of a better forecast, a different optimizer, or data-quality checks that conventional software could have performed.

5.2 Evidence Boundaries and Outcome Categories

Before any model is trained, every field used in the study is classified as observed, derived, inferred, or simulated. The released data are immutable inputs. Observed sales are never overwritten with recovery estimates, and simulated purchase prices are never represented as published business transactions. This is recorded in the experiment manifest and repeated in the lineage of every decision. An additional distinction separates *actual historical stockout observations* from *synthetic stockout outcomes created when a simulator changes orders*.

Outcome claims are separated into three tiers. **Tier A** covers data properties and predictive tests measurable from public observations, such as whether a future unstocked interval was flagged or how a forecast performs on held-out in-stock hours. **Tier B** covers controlled masking experiments where otherwise observed sales are hidden according to stockout-like patterns, so that the artificially withheld observations supply a valid target for evaluating a recovery model. **Tier C** covers simulated counterfactual purchasing, where cost, lost sales, expiration, and the effect of different autonomy rules are identified only within the assumptions of the simulator. A Tier C result may be practically informative, but it is not a measured uplift in a real retailer's revenue.

5.3 Track M: M5 Data, Preparation and Temporal Protocol

The M5 competition dataset contains daily sales of 3,049 products in ten stores in three U.S. states, yielding 30,490 item-store series and 42,840 series after hierarchical aggregation. The released sales history covers 1,941 days, with a calendar of events and weekly selling prices (Makridakis et al., 2022a, 2022b). This remains the long-horizon benchmark. The data are converted from wide daily columns into an item-store-date panel while preserving the original day index, category/department hierarchy, store/state structure, price availability, and calendar features.

For the confirmatory extension beyond the existing pilot, use chronological rolling origins with at least one untouched final evaluation period. Features, data-quality thresholds, model selection, supply parameters, and autonomy thresholds are fit or fixed on earlier windows only. The forecast horizon is 28 days for standard M5-style accuracy comparisons; replenishment simulation is evaluated daily across repeated decision windows using forecasts that would have been available at each origin. The choice of simulation window and the number of origins must be pinned before the final run. The initial M5 pilot involved only three selected FOODS items across ten stores and two seeds; it is preserved as preliminary engineering evidence and is not treated as representative of the full M5 hierarchy.

The release contains no physical stock balances, supplier arrivals, or direct stockout annotations. Historical M5 sales therefore serve as an observed-sales demand proxy for the simulation, with the limitation that stockout censoring may be present but cannot be directly identified. All initial balances, receipts, transfers, supplier capacities, and data-quality perturbations are synthetic and versioned. The

same exogenous scenario draws and demand stream are used for every policy within a paired comparison.

5.4 Track F: FreshRetailNet-50K, Provenance and Published Structure

FreshRetailNet-50K was released by Dingdong in 2025 as a benchmark for recovering demand hidden by stockouts (Wang et al., 2025). The downloaded revision contains 50,000 store–product series, 898 stores, 18 cities and 865 distinct product IDs. The paper identifies 863 SKUs and the card 865; this study records both source claims and uses the 865 IDs present in revision 08c1fab7f9257bc73679d415d65d644165d351d4.

The official public card lists 4,500,000 rows in `train` and 350,000 in `eval`, corresponding to 90 training days and a held-out seven-day evaluation period for each of 50,000 series. The files are available as Parquet from the publisher's Hugging Face repository and are licensed under CC BY 4.0. They contain no published on-hand inventory balance, supplier contract, lead time, wholesale cost, physical shelf life, purchase order, or quantity supplied. The authors describe the underlying collection systems, but those source systems are not part of the released research dataset (Dingdong-Inc, 2025; Wang et al., 2025).

The daily and hourly sales fields are reported as amounts multiplied by a global normalization coefficient. Their magnitude should not automatically be interpreted as the physical number of discrete cartons, bottles, or kilograms sold. The simulator therefore declares a normalized unit system for Track F and provides explicit sensitivity checks for scaling and rounding. It reports service rates, forecast error, waste ratios, and indexed cost rather than claiming to recover actual monetary profit. Without an authenticated conversion factor and observed purchase economics, absolute financial comparability with M5 is not scientifically justified.

The publisher’s implementation and released arrays confirm that 1 means out of stock. The 16 operating-hour slots run from 06:00 through 21:00. In training, 19.8771% of operating hours are flagged and 44.2668% of daily rows have at least one stockout hour. Daily stockout counts match the sums of hourly flags. These percentages describe different denominators, and neither reveals the unobserved quantity of lost demand.

5.5 Field-Level Mapping of FreshRetailNet to the Canonical Schema

The source fields are mapped without losing their meaning or provenance. The date and store-product identifiers form a daily observation key, while an optional hourly child table stores each hour's recorded sales and stock status. Category hierarchies and city-to-store relationships are native to the dataset. Promotions and weather are retained as covariates, not converted into procurement constraints. The resulting schema supports both recovery-aware forecasting and state-verification experiments.

Published fields (FreshRetailNet)	Canonical object	Transformation / quality assertion
city_id, store_id	Locations and geography	Pseudonymous city/store identifiers; verify each store's city mapping

Published fields (FreshRetailNet)	Canonical object	Transformation / quality assertion
management_group_id, first_category_id, second_category_id, third_category_id, product_id	Product and category hierarchy	Preserve the original five-level product/category hierarchy; verify keys and parent consistency
dt	Observation timestamp	Parse source date; assert chronological order and expected day count per series
sale_amount	Observed sales (daily)	Keep globally normalized scale metadata; compare against sum of hourly values within tolerance
hours_sale	Observed sales (hourly child records)	Explode only when needed; inspect vector length, valid range and day/hour alignment
hours_stock_status	Availability-event child records	Verify source flag polarity and temporal alignment; never equate it to numeric inventory
stock_hour6_22_cnt	Daily availability summary	Check count in 0–16 range and consistency with the relevant hourly flags
discount, activity_flag	Promotion and activity covariates	Preserve rate semantics (1.0 = no discount); mark availability of future plans explicitly
holiday_flag	Calendar covariate	Known calendar event; do not recode as source inventory
precpt, avg_temperature, avg_humidity, avg_wind_level	Environment covariates	History observed; future values allowed only as genuine or simulated ex-ante forecasts
Recovered-demand output (not a source column)	Inferred latent-demand artifact	Model/version, interval, uncertainty and source mask; preserve raw sales separately
Stock, suppliers, orders, lead times, unit prices, expiry (not source columns)	Simulated inventory/supply/expiry objects	Label <code>source_kind=simulated</code> and record scenario generator, assumptions, seed

Table 5.2. Field-level canonical mapping for the published FreshRetailNet-50K files.

Exact duplicate keys, contradictory hierarchy assignments, impossible discount values, inconsistent daily versus hourly totals, unavailable status coding, and cross-split time overlap are treated as integrity concerns. Genuine stockouts are not. When a test introduces a false stockout flag or removes an availability interval, that is a separate synthetic data-quality disturbance, labelled differently from an actual stockout recorded by the source.

5.6 Leakage-Free Temporal Splits and Feature Availability

The executed protocol retains the official 90-day training split and sealed seven-day evaluation. Four expanding validation folds begin at day indices 62, 69, 76 and 83. Panel membership uses only the first 62 days. Forecast selection uses pooled fully stocked validation WAPE; recovery selection uses pooled artificial-mask hourly MAE. Both selections are saved before final evaluation is opened.

All per-store, per-product, category, regional, day-of-week, and hourly-profile statistics are fitted from rows strictly preceding the relevant validation origin. Later rows must not affect earlier fitted transforms. Seven-day multi-step predictions are recursive: predicted values, not evaluation actuals, populate later forecast lags. Recovered latent-demand values for the training period are produced by recovery models fitted on the training portion of each fold, not on the full 97-day release.

The same timing rule applies to covariates. Calendar holidays can be genuinely known in advance. Promotional or activity plans are permitted in the future only if a realistic pre-decision schedule is specified; otherwise the experiment uses a train-only forecast or a known/unknown-information ablation. Future realized weather cannot be used as a forecast unless the experiment is explicitly labelled an oracle-covariate upper bound. Historical hourly stockout masks are allowed as training evidence; actual future masks are forbidden as planning inputs and are reserved for evaluating observational forecast behaviour.

An automated leakage audit reruns forecasts after changing or scrambling evaluation-period targets and asserts that predicted values are unchanged. Similar tests apply to withheld future stockout flags and future observed weather. Each result records the latest permissible source date, fitted feature artifact hashes, model version, and frozen policy settings. Failure of an audit invalidates the run rather than being reported as superior predictive performance.

5.7 Study F1: Recovering Demand Hidden by Real Stockouts

FreshRetailNet makes visible a measurement problem that M5 cannot isolate directly. Let $y_{i,t,h}$ be observed scaled sales and $a_{i,t,h}$ an indicator that the store-product was available at hour h . Let $d^*_{i,t,h}$ denote latent demand, which is not published. For completely available hours, observed sales may be used as a demand proxy subject to ordinary measurement limits. During out-of-stock hours, the unserved portion of demand cannot be measured. A recovery model estimates a conditional distribution for the missing part and records its uncertainty. The adjustment is made in a distinct field; it never changes the published observed sale.

```
Observed historical sales:      y[i,t,h]
Verified historical availability: a[i,t,h] ∈ {available, unavailable}
Unobserved latent demand:     d*[i,t,h] (not published)
Estimated latent demand:      d_hat[i,t,h] ~ p(d* | past y, a, context)
Recovered training target:     y if available; d_hat if censored
```

Important: observed availability does NOT provide the quantity of lost sales.
True latent-demand recovery error is measurable only for artificially masked in-stock observations or for synthetic simulation ground truth.

The executed recovery comparators are no imputation, a train-only in-stock hourly profile and historical LightGBM regression. The regression is named accurately; Tobit, TimesNet and the other ambitious candidate models were not executed within this budget. Each model fits before its validation origin. Published sales remain unchanged, with recovery estimates stored separately.

The identifiable recovery test hides deterministic contiguous blocks of 2, 4 and 8 operating hours on originally fully stocked validation days. The hidden values and original full daily amount are withheld

from reconstruction inputs. Saved outputs measure hourly and daily MAE, WAPE and signed bias with explicit denominators. The masks use reproducible hash ranking; they do not claim to reproduce the full empirical distribution of natural stockouts. These are point estimates, without an asserted interval-calibration result.

The second evaluation tests the practical forecasting consequence: does training on recovered daily histories improve seven-day-ahead forecasts on held-out *uncensored observable* outcomes relative to training on raw sales? Report results separately for in-stock evaluation hours/days and all observed evaluation sales, because the latter remains censored. The third evaluation checks plausibility and sensitivity on true historical stockout windows, but does not claim ground-truth reconstruction where no demand was measured. Sensitivity bounds, model disagreement, and plausible peak-hour shapes are more defensible than calling one imputation the actual sales the retailer lost.

5.8 Study F2: Forecasting with Censored and Recovered Histories

The four executed forecasts are seasonal naive, LightGBM on raw sales, LightGBM on hourly-profile recovered history and LightGBM on learned recovered history. LightGBM variants use the same 150-estimator budget and historical lag, date and series features. Recursive forecasts use predictions for later lags. Future realized weather, promotions and stockout masks are excluded. TFT, DeepAR and prediction-interval calibration were not executed for this new holdout.

Final results separately score all recorded daily sales and fully stocked daily outcomes, using WAPE, MAE and signed bias with their denominators. Every frozen candidate remains visible. True lost demand is unavailable during natural stockouts, so a higher prediction cannot be labelled proven recovery of real lost sales. Published-paper results remain literature evidence.

M5 retains its own probabilistic forecasting experiments with appropriate daily features and 28-day horizon, including seasonal naive, gradient boosting, and stronger candidate models if compute permits. There is no requirement that the best forecasting model be identical across the two benchmarks. What must remain identical are the provenance labels, evaluation discipline, governance gates, critic rules, and attribution logic used to compare the agent with deterministic automation.

5.9 Study F3 and Track M: Closed-Loop Replenishment Simulators

Historical records describe decisions made under the original retailer's stocking policy. Our research asks what would have happened if a different controller had purchased more or less stock. These are counterfactual questions and cannot be answered by simply replaying historical sales and stockouts unchanged. The study therefore uses *closed-loop simulators* in which the controller's orders alter future simulated on-hand stock. Demand, arrivals, sales, expiry, stockouts, and costs then evolve under an identical environment for every policy being compared. A simulator oracle can evaluate unsafe actions because it can observe its own true state; the real datasets do not contain that oracle.

For M5, the simulator is initialized after a warm-up period using lead-time demand and specified service targets. Supplier assignments, lead times, open orders, capacity, MOQ and pack rules, budgets, and optional inter-store transfers are synthetic. The sequence of daily decisions produces on-hand stock, pipeline inventory, fulfilled sales, lost sales, transfers, and costs. M5 observed unit sales act as an input demand proxy, with stockout-censoring limitations stated. The state perturbation layer changes the observed evidence available to the controller without changing the simulator's true state.

The bounded experiment gives competing policies a common observed-sales-proxy demand stream. It does not use one recovered arm's estimates as evaluation truth. The core adapter scales published normalized sales by 100 for inherited order-tool compatibility. The independent continuous calculator experiment tests common demand multipliers of 1, 1.25 and 1.5. Each policy's orders create new simulated availability; historical stockout masks are not replayed as outcomes under the changed stocking policy.

The core physical environment uses FIFO expiry and a hypothetical three-day life, while its inherited MILP remains age-unaware. A separate calculator experiment varies life across 1, 3 and 7 days, raw and recovered forecast views, demand multipliers and 30 seeds. Its continuous quantities and dimensionless cost coefficients differ from the core, so absolute outcomes are not pooled. These assumed lives are not measured Dingdong product attributes.

Simulator setting	Track M	Track F	Observed or assumed?
Decision cadence	Daily	Daily; optional intraday sensitivity	Study choice
Horizon	Multi-week; consistent with M5 evaluation	Seven days; 1–3 day reorder sensitivity	Study choice
Demand driver	Public daily observed unit sales proxy	Train-only estimated latent demand with real historical availability patterns used for fitting	Observed input / inferred counterfactual
Inventory and receipts	Simulated	Simulated	Not released in either benchmark
Expiry and spoilage	Optional sensitivity only	Age-based stock disposal and penalties	Fully simulated; no actual waste in public data
Economic valuation	Synthetic procurement and operating costs, optionally informed by observed selling prices	Dimensionless indexed costs; scaled-quantity caveat	Assumed, not company P&L

Table 5.3. Disclosed differences between the two counterfactual replenishment environments.

5.10 Versioned Supply, Constraint, Assortment and State-Quality Layers

The simulator contains three versioned disturbance families, largely retained from the original M5 design and now applied consistently to both datasets. The **supply and constraint layer** generates product-to-supplier assignments, pack multiples, individual and supplier-aggregate MOQs, unit conversions, optional capacity cuts, uncertain arrivals, and budget limits. The **assortment layer** generates temporal product-location eligibility while keeping the product hierarchies and observed source identifiers unchanged. The **integrity layer** introduces feed gaps, partial refresh, duplicate ingestion, missing location mappings, inconsistent units, contradictory purchase-order fields, or false stockout flags into the controller's observed state while retaining the simulator's true state.

Failure-class frequencies are not described as estimates from private operational telemetry. Instead, the taxonomy follows the information-systems and inventory literature reviewed in Chapters 2 and 3, while

magnitudes are defined as transparent low, medium, and high stress levels. Classes without defensible empirical prevalence assumptions are tested as explicit rare events. Every scenario records the fault, onset, duration, magnitude, detection signal, and expected business impact. A successful integrity certificate can therefore be audited against the known perturbation that created the test.

The new FreshRetailNet track requires an additional distinction. A real, published `stock_hour6_22_cnt > 0` is an *observed censoring event*, not an integrity disturbance. Synthetic corruption of that count, a contradictory hourly mask, or removal of the mask is a *state-quality disturbance*. The experiment tests whether the gate responds differently to the two cases. Genuine high censoring may rationally lead to broader uncertainty intervals or human approval; it should not automatically be treated as proof that data ingestion failed.

5.11 Policies, Agent Ablations, and Attribution Controls

To isolate where value is created, the full system is not compared only with a deliberately weak LLM. The main control is an engineered deterministic system that sees the same evidence, invokes the same forecasting models and solver, and applies the same state checks. A more elaborate model should justify its complexity through improvements on independently evaluated semantic exceptions, lower unnecessary escalation, better audits, or stronger performance under difficult scenarios.

ID	Controller / policy	Contribution isolated
B1	Seasonal naive + base-stock / order-up-to	Transparent minimum standard
B2	Gradient-boosted forecast + (s,S) or newsvendor	Stronger model with simple deterministic policy
B3	Probabilistic forecast + stochastic optimizer	Forecast-and-optimization without evidence gate or LLM
B4	B3 + deterministic integrity checks, parser, and recovery gate	Capable non-LLM control (primary comparison)
B5	Time-series foundation / advanced candidate + transparent policy	Value of forecast model complexity
B6	Single LLM agent with the same tools	Effect of role specialization
B7	Multi-agent system using free-form messages	Value of typed messages and provenance
B8	Typed multi-agent system without an independent critic	Critic's incremental contribution
B9	Typed multi-agent + critic with fixed autonomy	Evidence-gating effect
B10	Full typed, verified, evidence-gated LLM architecture	Proposed artifact including decision traces

ID	Controller / policy	Contribution isolated
B11 (Track F)	B4 with correct stockout-aware recovery but no LLM	Demand-recovery value separately from AI coordination

Table 5.4. Baseline policies and ablations; B11 is a FreshRetailNet-specific non-LLM comparator.

Track F evaluates policies under both *raw-sales forecasting* and *recovered-demand forecasting*. This creates a paired two-factor design: recovery on/off and LLM coordination on/off, while holding forecast family and optimization tools fixed. It can establish whether any improvement is a statistical benefit from de-censoring, an independent benefit from language-based exception handling, or an interaction between the two. For document interpretation, the deterministic parser must cover all known templates and be given comparable access to validation rules. The LLM is challenged on held-out semantic forms, not merely rewarded for handling examples that the parser was deliberately prevented from reading.

5.12 FreshRetailNet-Specific Experiments and Test Matrix

The six FreshRetailNet questions guide different forms of evidence. F1 and F2 receive masking and holdout forecasting tests. F3 and F6 receive independent age-aware simulations and sensitivity. F4 receives a targeted stockout-gate mechanism ablation. F5 receives a parser-versus-LLM factorial using shared forecasts and tools. The full architecture suite, independent natural supplier documents and human study remain outside the completed scope.

Experiment	Controlled comparison	Primary question and metric
F1. Recovery validity	Raw sales vs exposure correction vs regression/Tobit vs learned recovery	Artificial-mask WAPE/WPE and interval coverage; no claim of real latent-demand ground truth
F2. Forecast value	Identical forecast model on raw vs recovered training histories	Seven-day held-out forecast bias, WAPE, pinball loss, censoring-stratified results
F3. Ordering consequences	Same deterministic controller with raw vs recovered demand	Simulated cost index, fill, shortage, expiry waste, sensitivity to assumed shelf life
F4. Correct gating	Integrity gate that incorrectly blocks any stockout vs censoring-aware gate	Unnecessary holds, unsafe executions, service effects, escalation precision
F5. Agent value	Capable parser/controller vs single LLM vs full multi-agent on the same cases	Semantic extraction accuracy, harmful orders, run variance, audit quality, cost/latency
F6. Domain stress	Promotion peaks, weather uncertainty, perishability, lead-time and supplier disruptions	Robustness of rankings, CVaR, worst-case loss, oversight workload

Table 5.5. FreshRetailNet research questions; executed coverage and remaining gaps are documented in Chapter 6 and Appendix C.

F1 provides identifiable recovery targets only when sales are artificially hidden from originally available periods. **F2** evaluates actual released sales, stratified by whether their own observation window was stocked; real lost demand remains unidentified. **F3** changes stocking policy in a simulator and is consequently a statement about the simulated demand model. **F4** is a particularly direct test of managerial governance: does a system confuse valid evidence of unavailability with invalid evidence? **F5** determines whether there is any gain from the language model after a strong deterministic system already receives the recovered demand. **F6** tests whether such conclusions survive reasonable variations in the business economics.

The experiments are preregistered with an explicit primary outcome per contrast. A favourable result on one metric does not justify an unqualified conclusion if it increases waste, causes more approval bottlenecks, or relies on unrealistic generator assumptions.

Executed coverage of the FreshRetailNet questions

Question	Completed evidence	Remaining gap
F1 recovery	Artificial masking; zero/profile/LightGBM point estimates	Natural lost-demand truth, TimesNet/Tobit and calibrated intervals
F2 forecasting	Four candidates, four validation folds, seven-day holdout	Full-population modeling and probabilistic TFT/DeepAR comparisons
F3 perishable decisions	Independent continuous calculator grid; explicit conservation	An age-aware version of the inherited core MILP
F4 valid availability	60 targeted gate-ablation runs	Human interpretation and production exception costs
F5 LLM attribution	Shared-tool parser/LLM record-repair factorial	All semantic difficulty levels and raw/recovered-forecast factorial variants
F6 sensitivity	Shelf lives 1/3/7, common demand multipliers 1/1.25/1.5	A complete economics, cadence and disruption sensitivity study
Human audit	No participants; machine audit of records	Measured review speed, trust and staff effort

5.13 Operational, Forecasting and AI Governance Metrics

The study reports multiple outcomes because the management decision is inherently multi-objective. For forecasting, the shared metrics are WAPE, signed WPE, probabilistic pinball loss, interval coverage, and category/store-stratified calibration. M5 adds the competition-relevant hierarchical scales and WRMSSE where suitable. FreshRetailNet adds artificial-mask recovery accuracy, bias under varying censoring intensity, stockout-hour exposure and the proportion of the horizon with reliable observations. A reported forecast error always identifies whether the target is recorded sales, an artificially hidden originally observed value, or simulator-generated latent demand.

Operational metrics comprise fill rate, service probability, stockout incidence, mean and tail total cost, inventory turns, days of supply, and order amplification. FreshRetailNet additionally reports perishable expiry share, waste-to-fulfilled-demand ratio, the shortage–waste frontier, and simulated normalized cost per store-product-day. Its costs are indexes determined by specified simulation coefficients; M5 economic results are also model-dependent but have observed physical sales units and published prices that allow a more interpretable unit basis. No raw comparison of M5 dollars with FreshRetailNet normalized costs is presented.

Governance metrics comprise invalid or harmful executions relative to the simulator oracle, hard-constraint violations, holds caused by real faults, holds caused by genuine stockouts, false escalations, missed dangerous actions, decision latency, cost of review, valid tool-call rate, number of agent turns, run-to-run variance, token usage and estimated operating cost. The study distinguishes *successful*

verification from *correct AI semantic choice*. For instance, a critic can block an incorrectly requested repair, demonstrating that the safety architecture succeeded even though an LLM made a poor choice.

Construct	Primary reported metric	Interpretation safeguard
Forecast quality	WAPE, WPE, pinball loss	Label evaluated target and observable/censored population
Demand recovery	WAPE on artificially masked in-stock hours	Do not present real lost-demand accuracy as observed
Operational value	Cost index, fill rate, expiry ratio, CVaR	All procurement economics are simulated; indexed cost for FreshRetailNet
Evidence gating	Harmful execution and false-hold rates	Separate genuine stockouts from corrupted availability records
LLM incremental value	Paired B10–B4 benefit net of tokens and latency	Matched tools, forecasts, constraints and scenarios
Audit usefulness	Correct accept/override rate and audit time	Randomized explanation conditions where approved

Table 5.6. Primary outcomes and restrictions on interpretation.

5.14 Decision-Trace Faithfulness and Stockout Evidence

Every trace identifies the data snapshot, time cutoff, item-store key, source type, observations used, forecast version, estimated uncertainty, constraint sources, feasible alternatives, critic checks, autonomy decision and simulated execution receipt. Track F requires a separate stockout-evidence block containing the historical hourly availability mask, out-of-stock-hour count, raw observed sales, recovery model, fitted training interval, recovered demand distribution and the amount by which recovery changed the proposed purchase. A reader must be able to distinguish what the retailer recorded from what the model inferred.

Replay tests determine whether pinned evidence, tool versions, and seeds reproduce the action. Counterfactual tests modify one factor at a time: demand quantile, budget, supplier capacity, shelf life, availability integrity, or recovery uncertainty. A good trace does not merely state that stockouts matter; it must indicate whether changing the censoring information actually changes the recommendation or its autonomy level. Deletion tests remove a piece of claimed decisive evidence and check that the explanation is not contradicted by an unchanged decision. This evaluation concerns verifiable external decision artefacts, not access to the hidden internal reasoning of a neural network.

5.15 Robustness Scenarios, Distribution Shift and Business Risk

Track M retains the original scenarios for normal operations, feed gap, partial refresh, duplicate ingestion, capacity cut, supplier failure, unit mismatch, aggregate MOQ change, assortment change, extreme demand and transfers. Track F adds genuine late-day stockout pressure, promotion-induced demand spikes, weather uncertainty, sudden demand changes, high censoring, short shelf life, excessive waste penalty, delayed morning replenishment, and corrupted or absent availability annotations. Each

scenario has a documented frequency or is explicitly designated as an artificial stress case. The same perturbation seed and true simulator state are applied to all policies in a paired comparison.

To separate generalization from overfitting, the final evaluation reports results for both common scenarios and dataset-specific scenarios. A system that succeeds only when no meaningful disturbances occur should not be described as robust. Equally, a system that avoids every harmful order by refusing all decisions may sacrifice too much service to be commercially useful. The principal business trade-off is therefore between the expected cost of unnecessary action, the cost of missed action, and the cost of human intervention.

5.16 Sampling, Statistical Analysis and Confidence

Thirty matched seeds describe conventional-policy variability conditional on one panel, origin and assumed world. The live factorial has seeds 13 and 29. Days, SKU rows and cached responses are not independent replications. All seed rows and descriptive contrasts are available; no confirmatory significance, noninferiority or population-precision claim is made.

The completed grid contains 360 conventional core runs, 32 live factorial runs, 1,080 independent calculator sensitivity runs and 60 gate ablations. Each run uses 30 series over seven days. The numerical grid's 30 seeds do not create 30-seed evidence for the two-seed LLM comparison.

Cross-dataset synthesis uses the direction, effect sizes and uncertainty of *standardized within-dataset contrasts*, not a false common monetary scale. A finding is marked stable if it is supported across plausible scenario settings and both information environments; otherwise the report explains which conditions appear to drive it. The existing M5 v2 pilot had just two independent seeds and thirty hand-selected series, so it supplies engineering evidence only and receives no post hoc confirmatory p-value.

5.17 Human Audit and Management Decision Study

Subject to the institution's approval and feasible recruitment, planners or graduate students with relevant operations coursework will evaluate a balanced set of decision packets. Packets span both datasets and include correct recommendations, decisions based on a corrupted state, errors in an extracted constraint, and high-censoring FreshRetailNet cases where a valid stockout should not be mistaken for a data failure. Participants receive three explanation forms in randomized order: free-form LLM rationale, structured trace without provenance and stockout detail, and the full lineage-bearing evidence trace.

The main outcomes are time to decision, correct accept/override classification, detection of invalid state and inappropriate autonomy, calibrated confidence, and perceived effort. Cases are counterbalanced across participants to reduce learning effects. The study will make clear that the stockout and inventory values presented in simulated purchase packets may combine genuine historical availability records with hypothetical inventories and prices. Sample size will be determined using a pilot and feasible effect-size assumptions; no human-study outcome is asserted before recruitment and approval.

5.18 Implementation Protocol, Reproducibility and Stop Rules

The implementation is divided into an ingestion layer, a measurement-and-provenance layer, recovery and forecasting tools, a versioned simulator, policy controllers, decision traces and an analysis layer. Track M uses the existing M5 ingestion and pilot harness. Track F adds a source adapter for the publisher's Parquet files, checks the released schema and status coding, builds daily/hourly views, creates

train-only transforms, and produces recovered-demand and uncertainty artifacts. The same verified controller interface is reused wherever the business decision is comparable.

The technical run manifest stores dataset repository/version or file hashes, timestamp cutoffs, code revision, model and prompt versions, source-unit treatment, simulator parameters, LLM temperature and retry policy, seeds, available compute, and all generated artifact IDs. An experiment stops with an explicit error when there are inconsistent hourly arrays, a broken chronology, forbidden feature leakage, unsupported unit conversions, missing policy inputs, or invalid solver certificates. The final audit report lists every failed or excluded run and the reason; it never silently drops failures from a denominator.

The full 4.85 million daily-row release is checked for integrity. Modeling and closed-loop conclusions use a frozen 30-series panel selected from first-62-day sales and censoring evidence. The sample is balanced across three historical censoring strata by deterministic ranking and covers 0.06% of series. It is computationally purposive rather than representative.

CHAPTER 6 Experimental Findings and Managerial Implications

6.1 What Has Actually Been Measured: M5 v2 Pilot

A preliminary M5 engineering pilot was completed before the present dual-dataset redesign. The pilot report, dated 8 October 2026, records 32 completed simulated runs, 448 decision traces, 68 captured model attempts and approximately 69,073 model tokens. The source panel comprises 30 item-store series—three selected FOODS products across ten stores—and two independent simulation seeds. It is not a random or representative sample of the full M5 hierarchy. The evaluation uses artificial supplier constraints, simulated inventory and state perturbations, and 14-day decision windows. These characteristics limit the strength and scope of the findings (M5 v2 Pilot Report, 2026).

The first important descriptive result is negative: the LLM controller and a capable deterministic parser achieved **identical cost and fill-rate outcomes** in every matched scenario at the two tested recovery settings. This does not establish that language models are always redundant. It means that the specific controlled supplier grammar and decision setting used in this pilot did not demonstrate additional economic value from the language model. That finding strengthens the rationale for testing genuinely different semantic difficulty levels, including unfamiliar and conflicting documents, in the expanded experiment.

The second result concerns the *recovery architecture*, not the LLM. Under an injected derived-inventory-field collapse, adding an evidence-based recovery path reduced mean simulated operating cost from 894.30 to 796.51 (an improvement of about 10.9%) and increased fill rate from 89% to 96% (7 percentage points). The same benefit was obtained by the deterministic-parser controller. The appropriate conclusion is therefore that observable-evidence recovery was useful under that tested fault—not that an LLM created the benefit. The report records zero hard-constraint violations among committed plans in the model-enabled arms, reflecting the protections of the wider controlled system.

A third result reveals a weakness that a high extraction score could hide. The LLM recovery selector made the desired tool-selection choice on only 10 of 20 fresh choices and requested inappropriate recovery on ten stale-feed decisions that required a hold. A separate verification tool rejected those attempted repairs and prevented unsafe continuation. This illustrates why semantic competence, tool-selection reliability and execution safety must be measured separately. It also gives the two-dataset design a clear falsifiable proposition: a more sophisticated agent should be judged not just by whether it reads a number correctly, but by whether it calls the right tool and respects a decision's evidence threshold.

Finding	Observed pilot result	Permissible conclusion
LLM versus capable parser	Same matched simulated cost and fill results	No incremental LLM value shown in this controlled corpus
Recovery under inventory-field collapse	Cost 894.30 → 796.51; fill 89% → 96%	Benefit attributed to verified recovery, not uniquely to LLM
Hard-constraint verification	Zero committed-plan hard violations in model-enabled arms	Safety checks worked under the tested scenarios, not universal proof

Finding	Observed pilot result	Permissible conclusion
LLM recovery tool choice	10/20 correct; wrong repairs blocked	Verification prevented harm despite semantic selection errors
Scope	32 runs, two seeds, three FOODS items, ten stores	Exploratory engineering evidence only

Table 6.1. Status and limited interpretation of the completed M5 v2 pilot.

An earlier M5 stage also completed 372 runs: 360 numerical controls over seven days and 30 seeds, eight two-day prose-feasibility runs, and four two-day structured controls. The stages are not pooled. The first 96 numerical runs used seeds 0–7; a later 264-run extension used seeds 8–29 after that batch completed. The pooled 30-seed numerical means are therefore exploratory historical descriptions. The initial live prose experiment held every decision and therefore did not demonstrate an operational LLM saving. Its evidence motivated the bounded semantic-head and verified-repair improvements subsequently evaluated in V2.

In the initial feed-gap numerical controls, B3 committed 63 actual hard-constraint violations while B4 committed none and held 90 of 210 decision days. This supports a safety role for the evidence gate, with an explicit delay trade-off. These are original M5 results; the FreshRetailNet portability failure described next shows why a threshold should not be transferred without checking the new quantity scale.

Historical M5 numerical means: seven days, 30 seeds per cell

Policy	Scenario	Simulated USD	Fill	Violations	Held days
B1	Inventory-field fault	621.07	92.8%	0	0
B1	Stale feed	443.18	90.8%	0	0
B1	Normal	399.85	92.0%	0	0
B2	Inventory-field fault	548.97	85.2%	0	0
B2	Stale feed	440.07	77.2%	0	0
B2	Normal	399.63	81.0%	0	0
B3	Inventory-field fault	497.05	93.7%	0	0
B3	Stale feed	425.57	90.3%	63	0
B3	Normal	375.36	94.0%	0	0
B4	Inventory-field fault	369.42	90.5%	0	90
B4	Stale feed	367.80	90.5%	0	90
B4	Normal	374.00	93.9%	0	0

6.2 FreshRetailNet: observed evidence and simulated outcomes

FreshRetailNet changes the study because it provides a type of business evidence that M5 does not: real recorded availability by hour. Under M5 alone, the simulator can create stockouts and evaluate how controllers respond, but any claim about the historical occurrence of unobserved retail demand is largely assumption-based. In FreshRetailNet, the existence and timing of many stockouts are documented in the public record, even though their unserved quantities are not. The evaluation can therefore ask whether a manager's view of demand improves when the information system uses availability flags correctly, and whether that changes the decisions the manager is asked to authorize (Wang et al., 2025).

The completed checks show why a valid stockout flag calls for a different response from a broken feed. Correct availability evidence can limit what sales reveal. Demand recovery therefore supplies a separate inference. An inventory-field failure is instead held or reconciled through permitted source and movement evidence. The following results retain that distinction.

Results table. Completed scope

Experiment	Series	Seeds	Days	Runs / decisions
Core numerical policies	30	30	7	360 / 2520
Matched parser/LLM factorial	30	2	7	32 / 224
Age-aware sensitivity grid	30	30	7	1,080 runs
Stockout-gate mechanism ablation	30	30	7	60 runs
Forecast final holdout	30	Not simulation seeds	7	210 targets × 4 methods

Results table. Selected panel by first 62-day censoring stratum

Stratum	Series	Mean hourly stockout share	Mean normalized sales
0	10	12.43%	0.6318
1	10	18.31%	0.6352
2	10	33.61%	0.4843

Historical availability is observable; lost demand is not

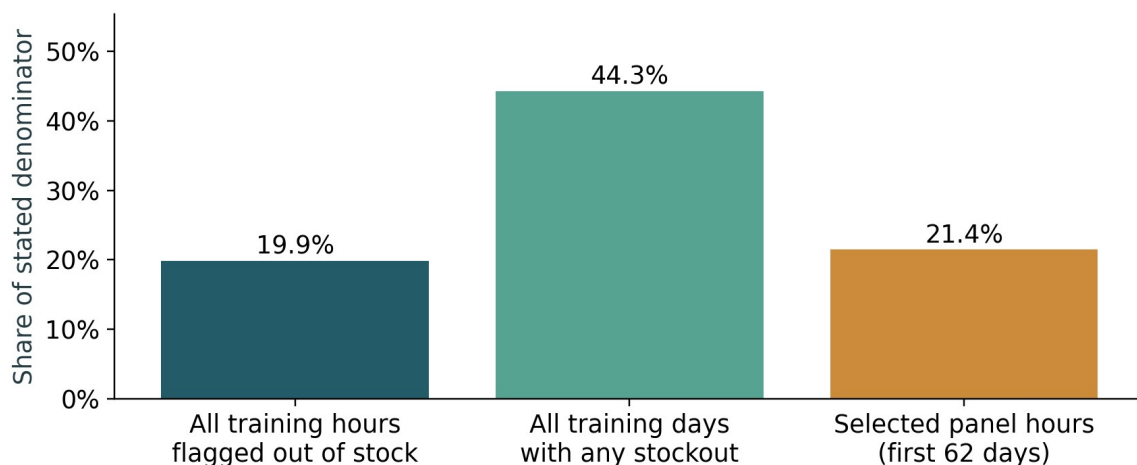


Figure 6.1. Historical availability is observable; lost demand is not. Training population rates use 72 million operating-hour flags and 4.5 million daily rows. The selected-panel rate uses its first 62 training days. Daily and hourly percentages have different denominators. None measures the quantity of naturally lost demand.

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

6.3 Demand recovery, forecast performance and perishable decisions

The saved recovery and forecast results are leakage-checked and reported below. Artificial masking supplies known targets; natural stockouts do not. Validation chose raw-history LightGBM for forecasting

and the hourly LightGBM model for recovery before the final holdout was read. Every alternative remains in the tables. The selected recovery method's modest hourly MAE advantage does not establish reliable reconstruction of naturally lost demand.

A planner needs more than an accuracy ranking. Under the default independent age-aware setting, recovered forecasts modestly reduced indexed cost and improved fill, while increasing expiry per demand. Some other cells increased cost. The service–waste frontier and sensitivity table therefore show a conditional trade-off, rather than an unconditional return from recovery.

Results table. Pooled artificial-mask operating-hour recovery

Method	N predictions	MAE	WAPE	Signed bias
Hourly LightGBM	6930	0.04230	98.33%	-83.10%
Hourly profile	6930	0.05815	135.18%	10.50%
No recovery	6930	0.04302	100.00%	-100.00%

Results table. Pooled daily reconstruction from artificial masks

Method	Masked cases	WAPE	Signed bias
Hourly LightGBM	1485	25.80%	-24.32%
Hourly profile	1485	19.62%	3.07%
No recovery	1485	29.26%	-29.26%

Recovery accuracy where a target can actually be checked

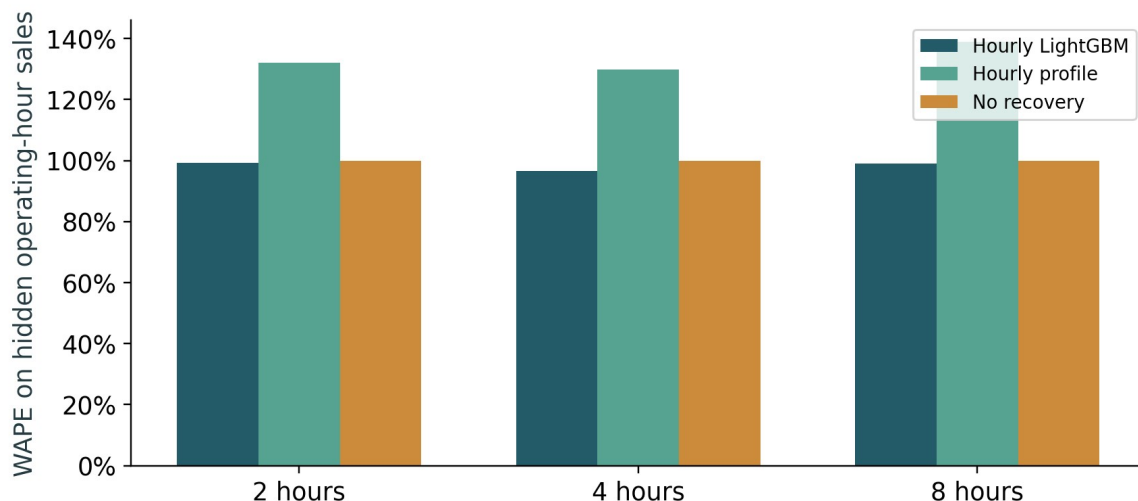


Figure 6.2. Recovery accuracy where a target can actually be checked. Originally observed sales are hidden in 2-, 4- or 8-hour validation blocks. Models fit only earlier data. The target is deliberately withheld sales, rather than unobserved natural stockout demand. Mask cases overlap and are not independent retailers.

Results table. Final seven-day forecast scores

Method	Target	N	WAPE	Bias
LightGBM, learned recovery	All sales	210	40.26%	-1.16%
LightGBM, learned recovery	Fully stocked	114	38.95%	7.11%
LightGBM, profile	All sales	210	40.61%	6.96%
LightGBM, profile	Fully stocked	114	40.07%	11.49%
LightGBM, raw	All sales	210	40.41%	-2.86%
LightGBM, raw	Fully stocked	114	38.84%	5.93%

Method	Target	N	WAPE	Bias
Seasonal naive	All sales	210	45.63%	2.61%
Seasonal naive	Fully stocked	114	42.84%	13.02%

Forecast quality depends on which sales are observable

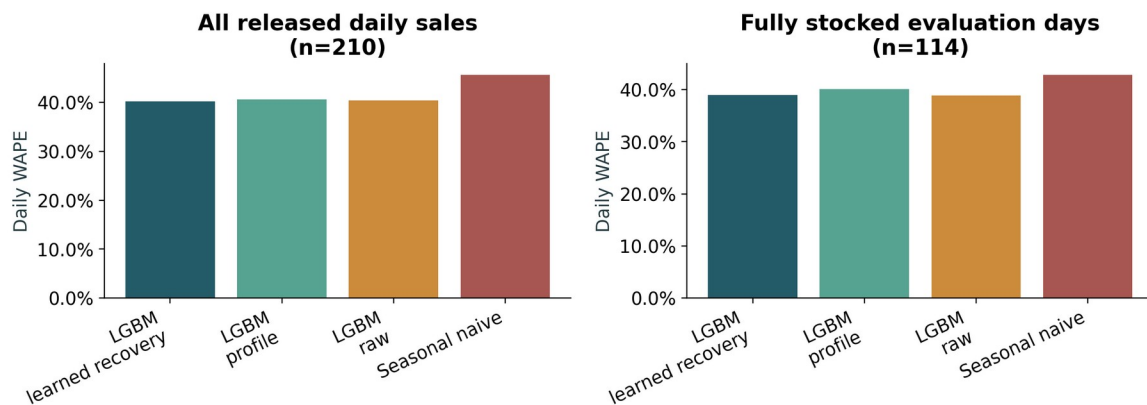


Figure 6.3. Forecast quality depends on which sales are observable. All four frozen forecasts cover the same 30 series and seven-day holdout. All-day sales can be censored; fully stocked days are conditionally observable. Future masks, weather, promotion and actual sales are never forecast features.

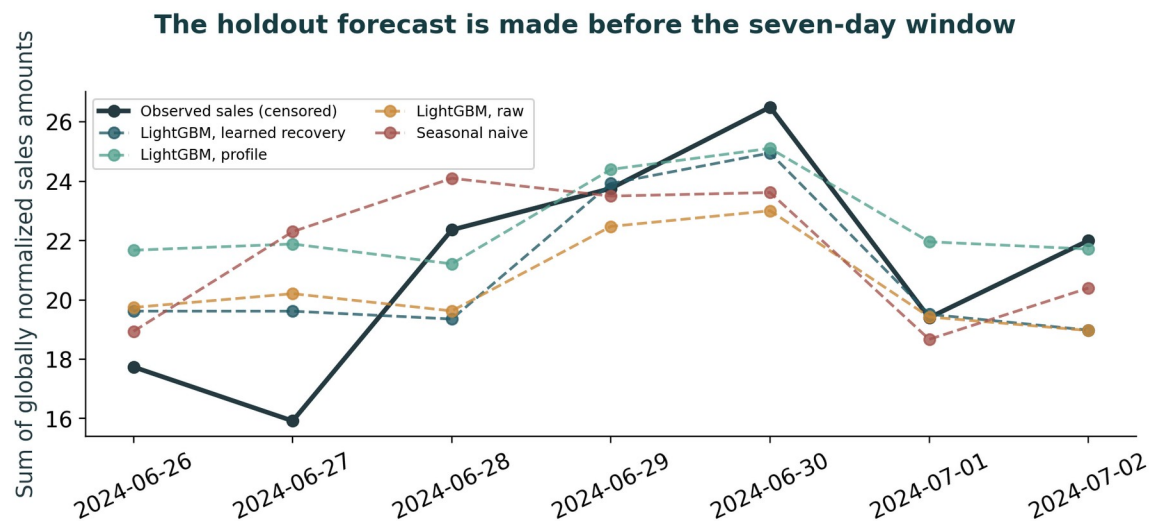


Figure 6.4. The holdout forecast is made before the seven-day window. Panel aggregate predictions are made at the final training origin. Later forecast lags use predictions. Globally normalized sales are neither physical quantities nor a currency.

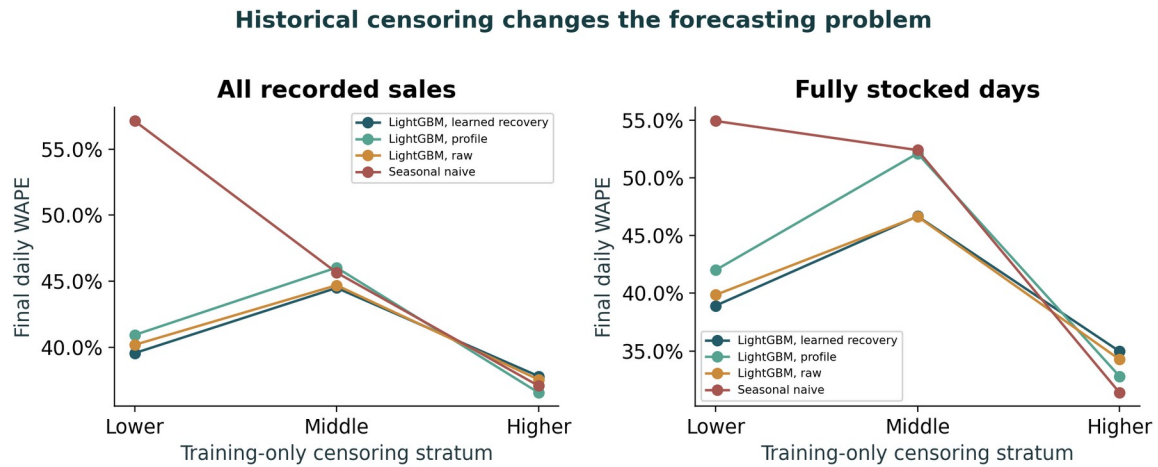


Figure 6.5. Historical censoring changes the forecasting problem. Ten series per stratum, defined before evaluation using the first 62 training days. Errors are weighted by the sum of actual scored sales within each group. Evaluation availability selects the fully stocked scoring subset only; it does not change membership or forecasts. These small strata are descriptive.

Results table. Core numerical means; counts summed across 30 seeds

Scenario	Policy	Cost index	Fill	Stockout	Violations	Held days
Normal	B1	18,290.32	77.6%	28.8%	210	0
Inventory-field fault	B1	18,610.24	78.5%	27.1%	266	0
Stale feed	B1	19,447.74	75.7%	30.0%	209	0
Normal	B2	19,685.44	69.7%	35.6%	190	0
Inventory-field fault	B2	19,172.46	72.9%	32.1%	204	0
Stale feed	B2	22,377.34	65.3%	39.3%	190	0
Normal	B3	18,794.03	85.0%	20.1%	0	0
Inventory-field fault	B3	19,026.79	85.0%	20.0%	0	0
Stale feed	B3	19,607.02	83.3%	21.7%	0	0
Normal	B4	24,013.14	43.5%	57.7%	0	210
Inventory-field fault	B4	24,013.14	43.5%	57.7%	0	210
Stale feed	B4	24,013.14	43.5%	57.7%	0	210

Results table. Actual violations and clean-reference flags are different endpoints

Policy	Rule checks violated	Reference-only flagged days	Nonzero action days	Held days
B1	685	7	630	0
B2	584	80	630	0
B3	0	380	630	0
B4	0	0	0	630

Results table. Failed active constraint checks in naive requested plans

Policy	Supplier capacity	Maximum order	Budget	Total checks
B1	623	14	48	685
B2	550	0	34	584

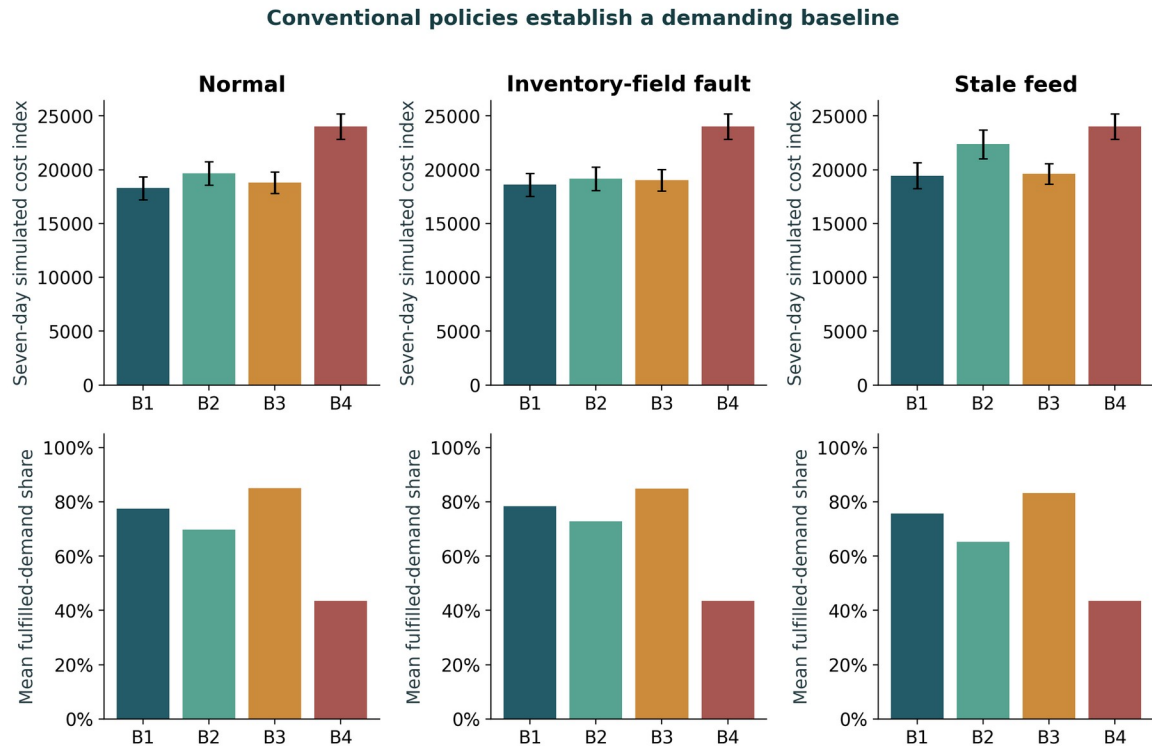


Figure 6.6. Conventional policies establish a demanding baseline. Thirty seeds per policy and scenario. Error bars show one standard deviation across conditional simulation seeds. B1: seasonal base-stock; B2: LightGBM (s,S); B3: native GRU-NB with MILP; B4: B3 with an evidence gate. Physical stock expires by FIFO, while the inherited MILP remains age-unaware.

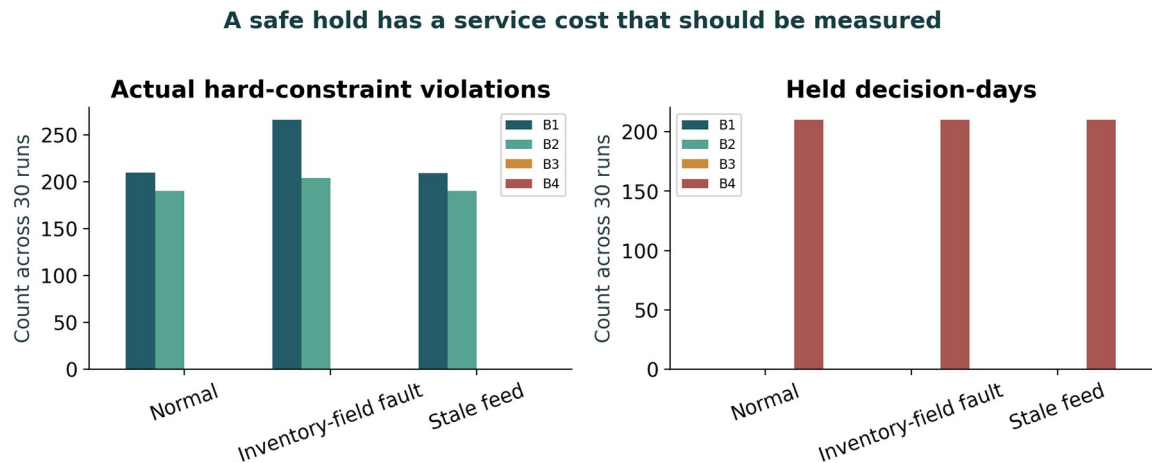


Figure 6.7. A safe hold has a service cost that should be measured. Counts are summed across 30 runs per cell. Hard violations use actual active constraints and differ from clean-reference deviations. Holds count decision days. A low violation count should be assessed alongside service.

Results table. Independent sensitivity means; full grid in CSV

Calculator	History	Life	Demand×	Cost index	Fill	Expiry share
age_aware	raw	1	1.0	444.2409	87.0%	45.1%
age_aware	raw	1	1.25	490.2626	82.1%	35.3%
age_aware	raw	1	1.5	571.9640	76.2%	28.0%
age_aware	raw	3	1.0	250.8000	96.5%	12.0%
age_aware	raw	3	1.25	303.3008	92.1%	6.6%

Calculator	History	Life	Demand×	Cost index	Fill	Expiry share
age_aware	raw	3	1.5	387.9095	86.5%	4.0%
age_aware	raw	7	1.0	208.4383	96.4%	0.7%
age_aware	raw	7	1.25	278.1695	92.1%	0.6%
age_aware	raw	7	1.5	372.4381	86.5%	0.5%
age_aware	selected_recovery	1	1.0	445.3558	87.5%	45.5%
age_aware	selected_recovery	1	1.25	489.5699	82.7%	35.6%
age_aware	selected_recovery	1	1.5	570.7272	76.7%	28.3%
age_aware	selected_recovery	3	1.0	251.1878	96.7%	12.4%
age_aware	selected_recovery	3	1.25	300.5691	92.6%	7.0%
age_aware	selected_recovery	3	1.5	383.2645	87.1%	4.3%
age_aware	selected_recovery	7	1.0	206.3261	96.7%	0.7%
age_aware	selected_recovery	7	1.25	273.6730	92.6%	0.5%
age_aware	selected_recovery	7	1.5	366.4855	87.1%	0.5%
order_up_to	raw	1	1.0	472.5299	79.4%	46.2%
order_up_to	raw	1	1.25	534.9456	74.6%	36.8%
order_up_to	raw	1	1.5	625.4762	69.2%	29.6%
order_up_to	raw	3	1.0	233.4785	91.6%	6.9%
order_up_to	raw	3	1.25	325.7840	84.4%	4.1%
order_up_to	raw	3	1.5	454.8281	76.1%	3.2%
order_up_to	raw	7	1.0	216.0284	91.6%	0.8%
order_up_to	raw	7	1.25	314.9469	84.5%	0.6%
order_up_to	raw	7	1.5	445.4677	76.3%	0.6%
order_up_to	selected_recovery	1	1.0	472.4809	80.0%	46.4%
order_up_to	selected_recovery	1	1.25	534.1117	75.1%	37.1%
order_up_to	selected_recovery	1	1.5	626.0210	69.5%	30.1%
order_up_to	selected_recovery	3	1.0	232.0698	92.1%	7.5%
order_up_to	selected_recovery	3	1.25	321.5021	85.0%	4.4%
order_up_to	selected_recovery	3	1.5	448.5044	76.8%	3.3%
order_up_to	selected_recovery	7	1.0	213.4033	92.1%	0.8%
order_up_to	selected_recovery	7	1.25	310.4848	85.0%	0.6%
order_up_to	selected_recovery	7	1.5	439.5487	76.9%	0.6%

Fresh retail exposes a service-waste trade-off



Figure 6.8. Fresh retail exposes a service–waste trade-off. Independent calculator experiment, with 30 shared seeds per cell and demand multiplier 1. Shelf lives are hypothetical. Quantity and economic conventions differ from the core MILP experiment, so their absolute results are not pooled.

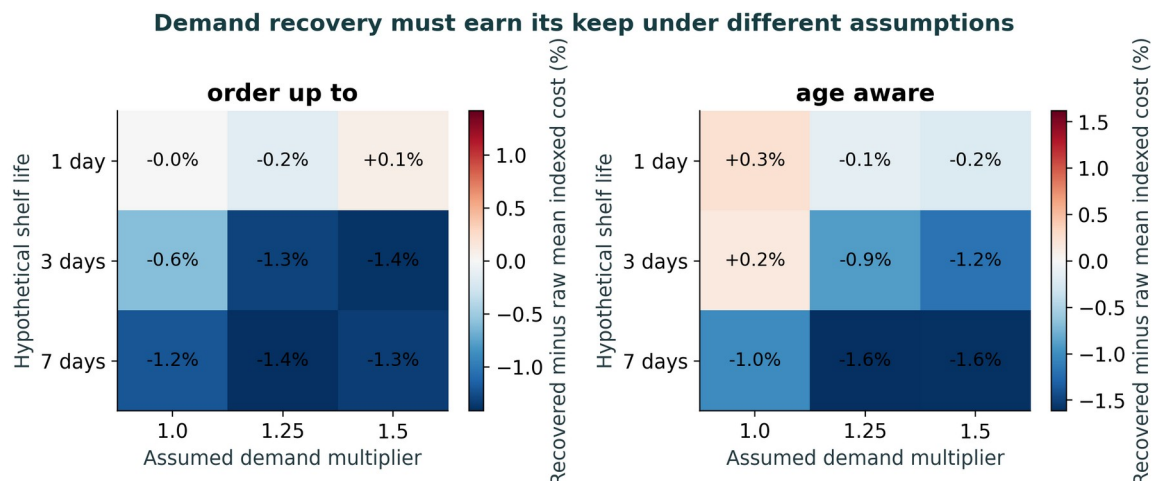


Figure 6.9. Demand recovery must earn its keep under different assumptions. Relative change from raw to recovered forecast history within each common-demand cell. Thirty seeds, shelf lives 1/3/7 days and demand multipliers 1/1.25/1.5. Negative values mean lower assumed indexed cost. They are not measured profit or purchasing savings.

6.4 The incremental contribution of the LLM

The live factorial found identical cost and fill in 16/16 matched parser–LLM pairs. Supplier texts remain synthetic and controlled. The model’s participation demonstrates bounded execution with numerical tools and independent verification, but no additional economic advantage over a capable parser on these texts.

A retailer still needs evidence that language assistance resolves exceptions that its rules cannot handle economically. Checked interpretation and exception coordination are plausible benefits. Inference delay, monitoring, failures and oversight are incremental costs. The audit retains erroneous choices even when independent tools prevent unsafe execution. Semantic accuracy and final execution safety are separate endpoints.

Results table. Agent factorial means; two seeds per row

Arm	Scenario	Cost index	Fill	Held days	Violations	Freshcalls
Parser	Normal	23,331.29	45.2%	14	0	0
Parser	Inventory-field fault	23,331.29	45.2%	14	0	0
Parser	Stale feed	23,331.29	45.2%	14	0	0
Parser	Capacity cut	21,572.97	57.4%	4	0	0
Parser + repair	Normal	23,331.29	45.2%	14	0	0
Parser + repair	Inventory-field fault	23,331.29	45.2%	14	0	0
Parser + repair	Stale feed	23,331.29	45.2%	14	0	0
Parser + repair	Capacity cut	21,572.97	57.4%	4	0	0
LLM	Normal	23,331.29	45.2%	14	0	7
LLM	Inventory-field fault	23,331.29	45.2%	14	0	0
LLM	Stale feed	23,331.29	45.2%	14	0	0
LLM	Capacity cut	21,572.97	57.4%	4	0	5
LLM + repair	Normal	23,331.29	45.2%	14	0	7
LLM + repair	Inventory-field fault	23,331.29	45.2%	14	0	6

Arm	Scenario	Cost index	Fill	Held days	Violations	Freshcalls
LLM + repair	Stale feed	23,331.29	45.2%	14	0	6
LLM + repair	Capacity cut	21,572.97	57.4%	4	0	5

Results table. Primary execution and reference-distance flags

Arm	True rule checks violated	Reference-only flagged days	Nonzero action days	Held days
LLM	0	10	10	46
LLM + repair	0	10	10	46
Parser	0	10	10	46
Parser + repair	0	10	10	46

The LLM shares the same forecasts, tools and safeguards

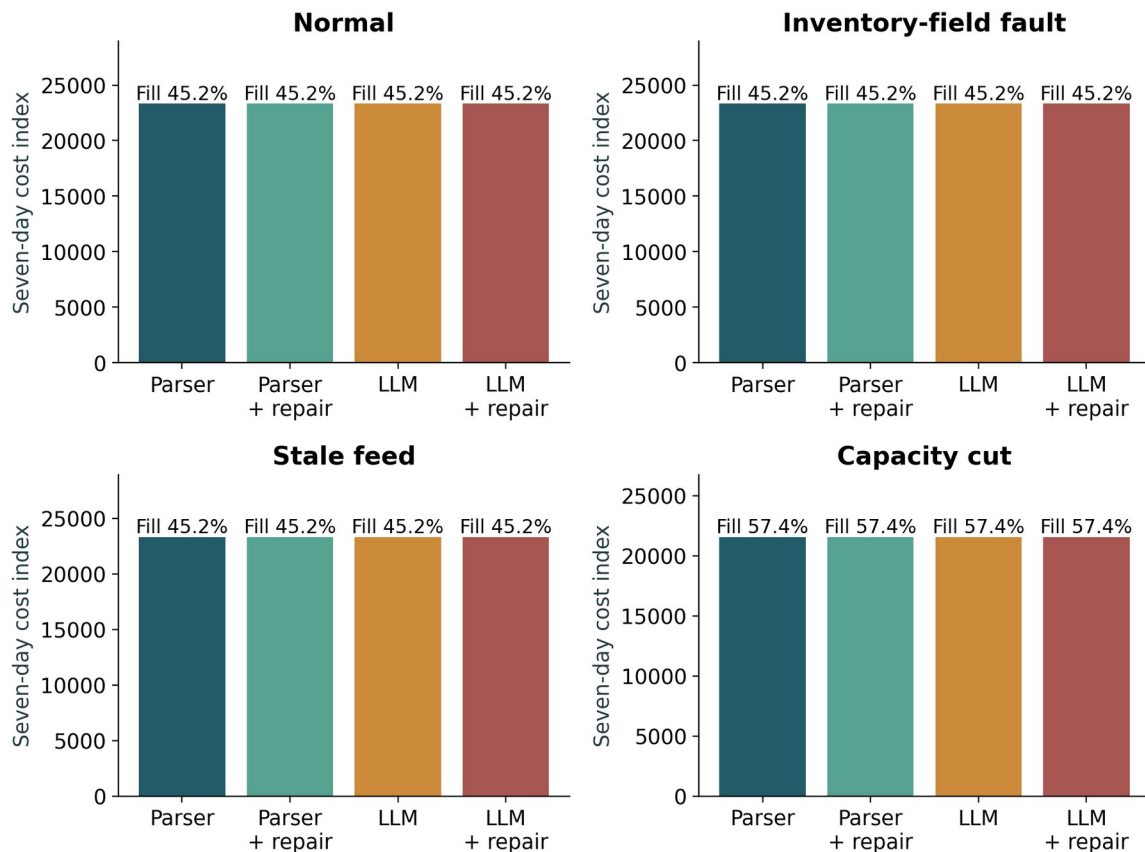


Figure 6.10. The LLM shares the same forecasts, tools and safeguards. Four matched controller/repair configurations, four scenarios and two seeds, each using seven days. Repair is verified inventory-record reconciliation. It is not statistical recovery of censored customer demand. All configurations share the numerical tools and source evidence.

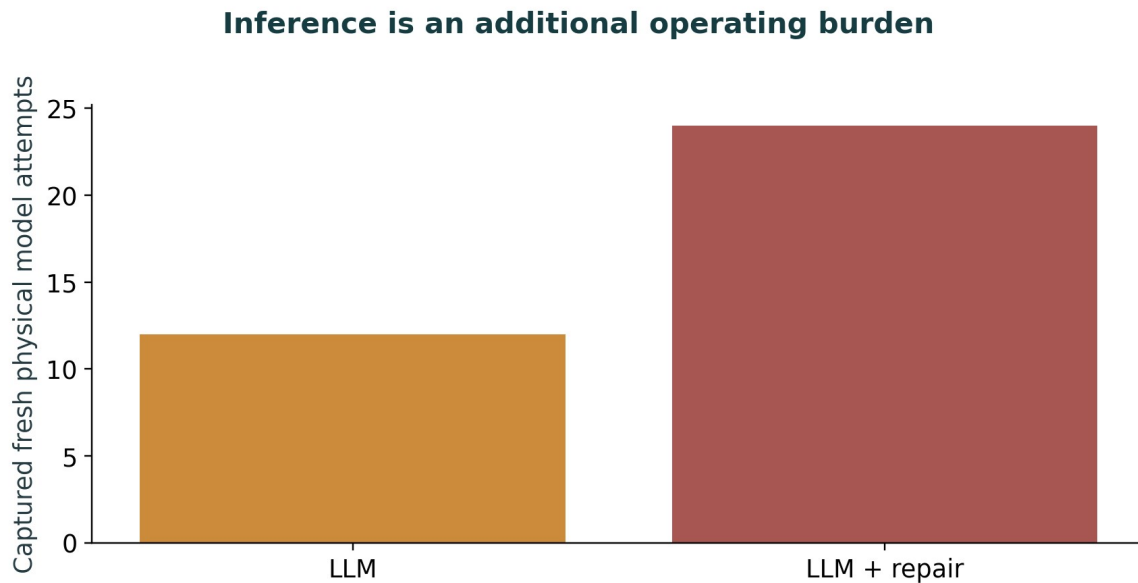


Figure 6.11. Inference is an additional operating burden. Primary registered study physical attempts from the independent audit. The exploratory follow-up has separate counts. Cache imports and typed extraction outputs are not additional calls. Local API charges are zero; CPU hosting and oversight costs are unpriced.

6.5 Evidence gating, record repair and useful holds

Under the injected inventory-field fault, enabling repair changed mean LLM-arm cost from 23,331.29 to 23,331.29 (0.00%) and fill from 45.2% to 45.2% (0.00 percentage points). The parser receives the same repair. Valid historical stockouts remain demand-visibility evidence; the deliberately naive stockout-hold rule is reported only as a targeted mechanism ablation.

The expected managerial outcome is a *graded* approach rather than universal automation or universal manual approval. Straightforward, verified orders may be executed within spending caps. Orders based on materially uncertain demand or incomplete constraints may require approval. Orders based on contradictory, untraceable state should remain on hold. The relative economic value of those thresholds must be learned from controlled comparisons, not defended solely through intuition.

A concrete simulator case shows both the value and the limit of this design. On decision day 92, the damaged derived inventory for FRN_0422@S0173 was 0, while fresh source and movement balance both showed 444. The model selected the permitted reconciliation tool. Independent checks restored 444 scaled model units, changed 27 permitted quantities and removed the integrity hard failure. The resulting order was still held: proposed spending of 1,657.60 exceeded the 1,500 autonomous cap. Correct exception handling can repair evidence without making the operating policy usable.

6.5.1 Exploratory transfer of the spending threshold

The original gate did not transfer successfully. Its inherited autonomous spending cap was 1,500 cost-index points, while a clean proposed FreshRetailNet order could require about 1,656 points. In the primary normal parser condition, 14 of 14 decision days were held. This failure is reported before any repair: an architecture can be computationally feasible and still be commercially inactive if its threshold is carried into a new quantity scale without checking.

The failure led to a separately registered follow-up. Its hypothesis was chosen after the primary study had started; the official holdout had already been used. The study therefore labels all subsequent transfer results exploratory. The original fixed-gate outputs are unchanged and remain in the report and archive.

Cap calibration uses seven earlier historical dates, training a separate GRU through day 75 and simulating June 12–18 (indices 76–82). The rule was fixed before those proposals were generated: $\min(0.8 \times \text{budget}, \text{ceil}(1.10 \times \text{the seven-proposal 95th percentile} / 50) \times 50)$. The historical 95th percentile was 1,659.635; the selected cap was 1,850, below the 2,400 maximum. The independent two-person threshold remains 2,500. All seven proposals were included, including 2 days held by the unchanged extra-spend risk rule. Proposal values, dates, model identity and selection timestamps are saved.

The transfer experiment contains 480 runs and 3,360 decision days: four configurations, four scenarios and 30 seeds. It uses the primary saved GRU tensors and the same simulator, supplier sources, LLM, prompts, constraints and checking tools. Only the spending cap and its version change semantically; output locations, cache and spend ledger are separately named for delivery and audit.

Cost and fill were identical in 240/240 parser–LLM pairs. Under the inventory-field fault, verified repair changed mean LLM cost by -11.71% and fill by +12.25 percentage points. These results distinguish fixing an unusable operating threshold from the value of the language model. They are not an untouched confirmation, and any shared repair benefit remains attributable to the architecture.

For a business, this is an implementation lesson as much as a modeling result. A limit calibrated for unit sales in one retailer can produce systematic holds after quantity normalization changes. Conservative caps remain valuable, but the business must verify that legitimate orders can pass them. That review should use historical proposals and explicit risk budgets, rather than repeatedly adjusting limits until a favourable holdout result appears.

The paired seed results show the benefit and its limits. Under the inventory-field fault, indexed cost falls in 26/30 seeds and rises in 4; fill improves in 30/30. Verified repair restores the normal-case economic and service results for every seed. The stale-feed condition is unchanged because fresh source evidence is unavailable. These are conditional simulator effects; the same recovery benefit occurs with the capable parser.

Calibration restores useful execution, but the remaining gate still carries an opportunity cost. Over the same 30 seeds, normal-operation fill is 79.19% for the calibrated controller and 85.03% for conventional B3, a -5.83-point change. Mean indexed cost is 18,100.60 against 18,794.03 (-3.69%), with 57 held days against 0. Both use the same saved forecaster, simulator and numerical tools and both have zero actual constraint violations. Lower indexed spending accompanies lower service; these values do not establish overall LLM superiority. The parser gives the same calibrated result, and this comparison remains exploratory.

Reference-only deviations are retained separately from actual active-rule violations in this follow-up as well. The legacy harmful_execution composite includes either endpoint. Its label alone cannot identify realized injury, waste or economic harm; outcomes and the underlying reason for each flag must be inspected.

The independent follow-up audit verifies 480 runs and 3,360 physical decision records. It captures 204 fresh attempts and 308,841 tokens, with 0 client/schema errors. There are 0 actual committed requested-

plan violations. Selector compliance remains a separate table entry, so a blocked semantic error is not counted as a successful choice.

The bounded numerical head returns 38/38 exact submitted terms in 24 fresh extraction responses. Separately, 90/180 fresh recovery selections follow the instruction. The 90 stale-feed mis-selections are refused by independent checks. A zero client/schema-error count therefore coexists with substantive reasoning errors; safe final execution depends on the wider control process.

Grounding was attempted on 2,820 of 3,360 follow-up decision days; 2,820 of those attempted source sets were exact. Each available set has 163 clauses, yielding 459,660 exact active-rule records. The other 540 days were state-gated before grounding and remain in the denominator of all decision days. These records combine a bounded model extraction head with deterministic parsing and verification; full-set agreement is not the model's accuracy on every clause.

On the original matched seeds 13 and 29 in normal operation, cap transfer changed mean LLM cost from 23,331.29 to 18,454.95 (-20.90%) and fill from 45.2% to 80.2% (+35.05 points). Held decision days changed from 14 to 3. These paired values isolate the cap change more clearly than comparing the primary two-seed mean with all 30 follow-up seeds, but the question remains exploratory.

Exploratory transfer means; primary results remain separate

Arm	Scenario	Runs	Cost index	Fill	Held days	Violations
LLM	Capacity cut	30	16,942.51	75.9%	15	0
LLM	Inventory-field fault	30	20,500.69	66.9%	105	0
LLM	Stale feed	30	20,500.69	66.9%	105	0
LLM	Normal	30	18,100.60	79.2%	57	0
LLM + repair	Capacity cut	30	16,942.51	75.9%	15	0
LLM + repair	Inventory-field fault	30	18,100.60	79.2%	57	0
LLM + repair	Stale feed	30	20,500.69	66.9%	105	0
LLM + repair	Normal	30	18,100.60	79.2%	57	0
Parser	Capacity cut	30	16,942.51	75.9%	15	0
Parser	Inventory-field fault	30	20,500.69	66.9%	105	0
Parser	Stale feed	30	20,500.69	66.9%	105	0
Parser	Normal	30	18,100.60	79.2%	57	0
Parser + repair	Capacity cut	30	16,942.51	75.9%	15	0
Parser + repair	Inventory-field fault	30	18,100.60	79.2%	57	0
Parser + repair	Stale feed	30	20,500.69	66.9%	105	0
Parser + repair	Normal	30	18,100.60	79.2%	57	0

Training-only proposed spending; all seven days retained

Historical decision	Spending index
76	1,658.700
77	1,657.600
78	1,659.800
79	1,657.050
80	1,649.900
81	1,506.350
82	1,659.250

Remaining clean-operation gate trade-off; 30 paired seeds, exploratory

Controller	Runs	Cost index	Fill	Held days	True violations
Conventional B3 MILP	30	18,794.03	85.03%	0	0
Calibrated LLM with the shared gate	30	18,100.60	79.19%	57	0

Exploratory actual violations versus clean-reference distance

Arm	True rule checks violated	Reference-only flagged days	Nonzero action days	Held days
LLM	0	265	558	282
LLM + repair	0	311	606	234
Parser	0	265	558	282
Parser + repair	0	311	606	234

Independent audit of the exploratory transfer

Measure	Recorded count
Completed runs	480
Decision days	3360
Fresh model attempts	204
Recorded model tokens	308841
Client or schema errors	0
Violated active-rule checks in requested plans	0
Held decision days	1032
Instruction-compliant recovery selections	90
Incorrect recovery selections	90
Matched parser-LLM pairs	240
Pairs with identical primary outcomes	240
Pairs with identical daily actions	240
Verified physical stock balances	3360
Forbidden model-context findings	0

Exploratory source grounding; conditional coverage is explicit

Configuration	Grounding attempted	Exact full source sets	State-gated days	False clauses	Omitted clauses
Parser	660	660	180	0	0
Parser + repair	750	750	90	0	0
LLM	660	660	180	0	0
LLM + repair	750	750	90	0	0

Gate transfer on the same two original seeds; exploratory holdout reuse

Arm	Scenario	Cost change	Fill change (pp)	Held days before/after
LLM	Capacity cut	-21.24%	+18.64	4/1
LLM	Inventory-field fault	-14.97%	+23.62	14/7
LLM	Stale feed	-14.97%	+23.62	14/7
LLM	Normal	-20.90%	+35.05	14/3
LLM + repair	Capacity cut	-21.24%	+18.64	4/1
LLM + repair	Inventory-field fault	-20.90%	+35.05	14/3
LLM + repair	Stale feed	-14.97%	+23.62	14/7
LLM + repair	Normal	-20.90%	+35.05	14/3
Parser	Capacity cut	-21.24%	+18.64	4/1
Parser	Inventory-field fault	-14.97%	+23.62	14/7
Parser	Stale feed	-14.97%	+23.62	14/7
Parser	Normal	-20.90%	+35.05	14/3

Arm	Scenario	Cost change	Fill change (pp)	Held days before/after
Parser + repair	Capacity cut	-21.24%	+18.64	4/1
Parser + repair	Inventory-field fault	-20.90%	+35.05	14/3
Parser + repair	Stale feed	-14.97%	+23.62	14/7
Parser + repair	Normal	-20.90%	+35.05	14/3

A spending threshold needs calibration to its new scale

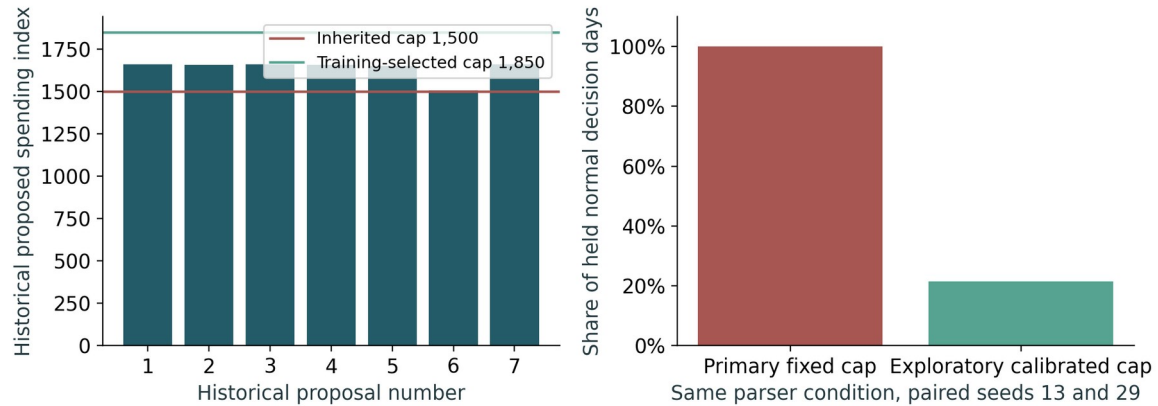


Figure 6.12. A spending threshold needs calibration to its new scale. The right panel compares the same original seeds, 13 and 29, with 14 normal parser decision days under each cap. Full 30-seed means appear in the following figure. The fixed-gate failure motivated this exploratory follow-up after primary evaluation began. Seven historical calibration proposals use training-only dates and a separately fitted model. The cap formula was registered before calibration. Reusing the official evaluation window means the follow-up is not confirmatory.

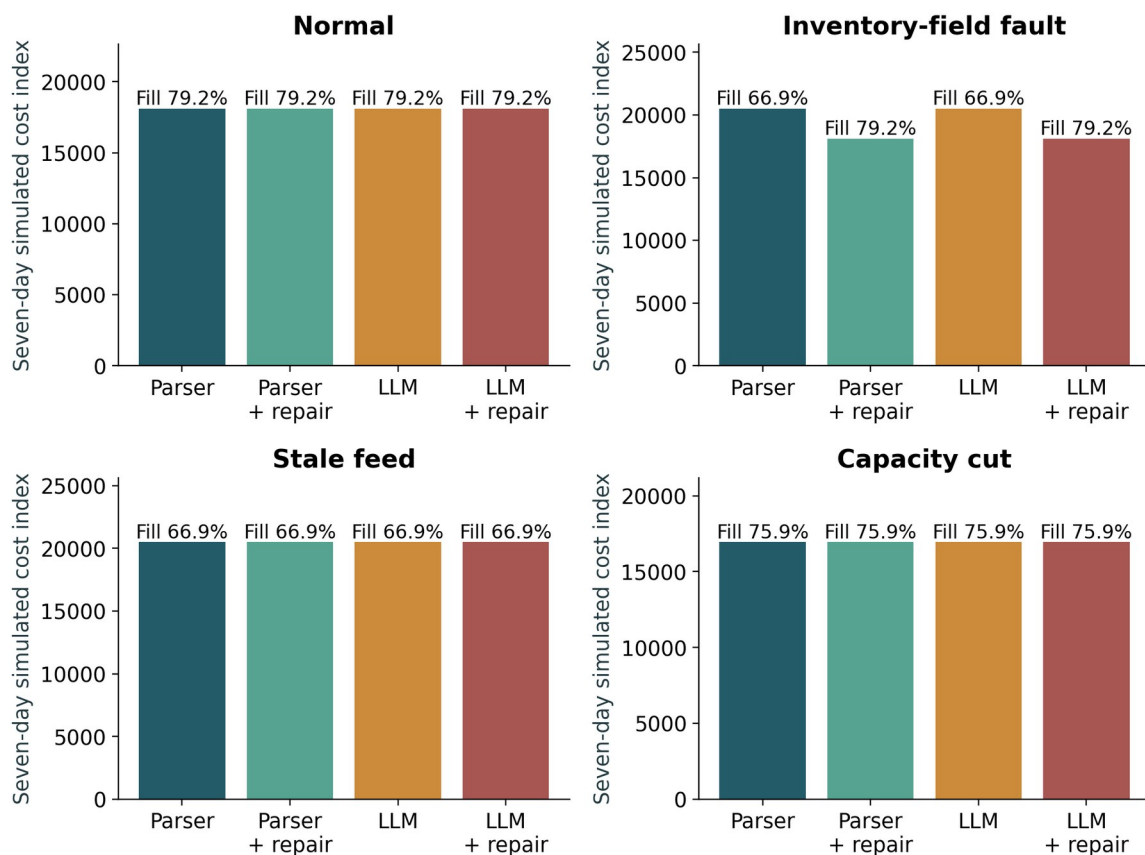
Exploratory calibrated transfer: useful execution and fair attribution

Figure 6.13. Exploratory calibrated transfer: useful execution and fair attribution. 30 paired simulator seeds per configuration and scenario, all 30 series and seven days. The original saved forecast, prompts, model and numerical tools are retained. Only the spending threshold and its version change semantically. This follow-up was chosen after inspecting primary portability and reuses the holdout.

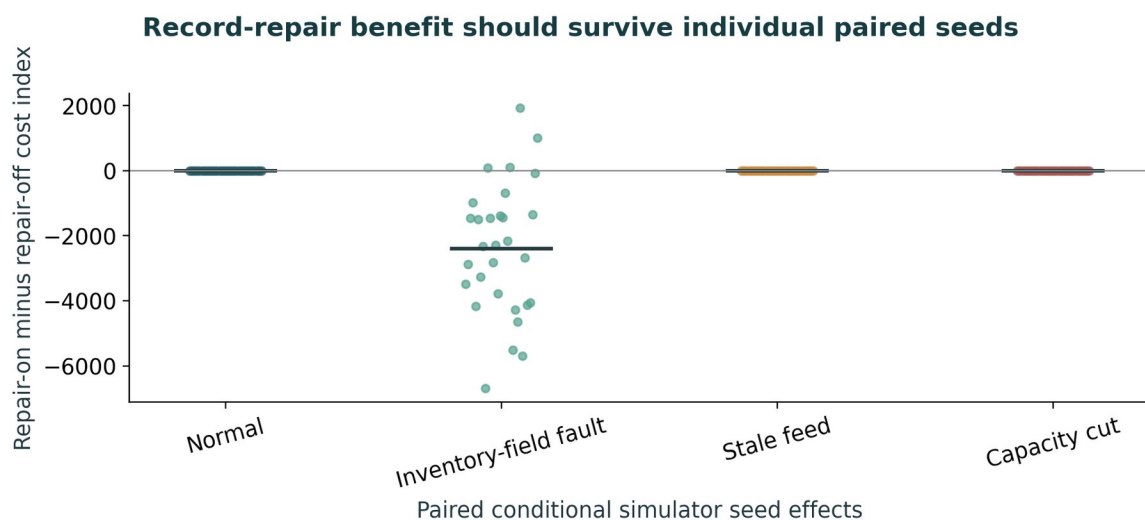


Figure 6.14. Record-repair benefit should survive individual paired seeds. Each point is a matched simulator seed in the exploratory follow-up, with a short line for its mean. Seeds vary a shared assumed environment, not independent retailers or historical forecast origins. The graph makes adverse and null effects visible without a confirmatory population claim.

6.6 Reliability and the limits of multi-agent attribution

The system has typed responsibilities and independent verification, but this study does not isolate the number of agents or execute every single-agent and free-form-chat comparison proposed earlier. Measured reliability comes from source grounding, completed traces, model calls and committed outcomes. Cache imports are not new trials. Conclusions concern the verified workflow rather than an untested benefit of additional agents.

In the primary live study, 24 physical extraction calls returned 38 exact numerical terms out of 38. That is the bounded numerical head, not the whole supplier corpus. Twelve fresh recovery selections were instruction-compliant in six cases; six wrong selections were refused by independent checks. Across 224 agent decisions, 188 attempted source grounding and all 188 source sets were exact. Each set has 163 clauses, yielding 30,644 checked active clauses; 36 state-gated decisions did not attempt grounding. Exact conditional source coverage must not be relabelled unconditional model accuracy.

The primary agent configurations made 40 nonzero action-days and held 184 decision-days in total. The nonzero actions occur under capacity cuts, whose smaller orders can pass the inherited spending cap. Normal, field-fault and stale-feed scenarios are fully held. These successful bounded actions demonstrate feasibility in a narrow condition, while the normal-operation failure motivates the explicitly separate transfer follow-up.

All 40 primary nonzero agent action-days carry a reference-distance flag, with zero actual constraint violations. The legacy composite therefore labels them harmful_execution even though the flag arises from plan distance alone. The report retains those counts and makes no claim of zero modeled risk or zero disagreement with the reference. Feasibility, clean-reference agreement and achieved service are distinct endpoints.

Results table. Primary registered study audit: distinct operational counts

Measure	Recorded count
Completed runs	392
Decision days	2744
Conventional core runs	360
Matched controller runs	32
Fresh model attempts	36
Fresh model responses	36
Recorded model tokens	37506
Client or schema errors	0
Fresh numerical-extraction responses	24
Fresh recovery selections	12
Violated active-rule checks in requested plans	1269
Held decision days	814
Instruction-compliant recovery selections	6
Incorrect recovery selections	6
Matched parser-LLM pairs	16
Pairs with identical primary outcomes	16
Pairs with identical daily actions	16
Verified physical stock balances	2744
Forbidden model-context findings	0

Results table. Source grounding coverage; full columns in the audit CSV

Configuration	Grounding attempted	Exact full source sets	State-gated days	False clauses	Omitted clauses
numerical	2340	2340	180	0	0
Parser	44	44	12	0	0
Parser + repair	50	50	6	0	0
LLM	44	44	12	0	0
LLM + repair	50	50	6	0	0

6.7 Decision traces and human review: evidence and future validation

A good explanation should help a person challenge a recommendation rather than simply make it sound persuasive. On M5, the crucial information may be the inventory snapshot, supplier rule, or held-order rationale. On FreshRetailNet, it may be the number of stockout hours, the difference between raw and recovered demand, and how much the proposed order changes under plausible recovery alternatives. A trace that labels observed, inferred, and simulated fields may allow a reviewer to identify questionable assumptions more quickly than a fluent paragraph without evidence.

No formal human study was conducted. Structured traces make decisions reviewable and their references auditable. They do not by themselves prove faster review, lower staff effort or better calibrated trust. Those remain participant-study outcomes for later validation.

6.8 Cross-Dataset Synthesis and Practical Adoption

The two datasets support a useful but bounded synthesis. Verified record repair can restore continuity under a known fault; the LLM does not add measured cost or fill value over the capable parser on the tested grammar. FreshRetailNet contributes observed stockout timing, artificial-mask tests and perishable trade-offs. These are information and architecture findings, rather than proof that the language model forecasts better. Cross-dataset contrasts remain descriptive because horizons and economics differ.

For an organisation considering investment, a sensible staged path is to establish data lineage and basic inventory checks, make the distinction between observed sales and censored demand where needed, document supplier and assortment constraints, benchmark against a capable deterministic controller, and only then add LLM assistance for exceptions where its net value is demonstrated. Autonomy should expand through audited thresholds, not simply because the technology makes automated execution possible. Chapter 8 returns to this adoption principle as the central business contribution of the study.

Results table. Within-dataset inventory-repair contrasts

Dataset	Scenario	Days	Cost change	Fill change(pp)	Identical parser/LLM pairs
M5	Normal	14	0.00%	0.00	4/4
M5	Inventory-field fault	14	-10.93%	7.00	4/4
M5	Stale feed	14	0.00%	0.00	4/4
M5	Capacity cut	14	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Normal	7	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Inventory-field fault	7	0.00%	0.00	4/4
FreshRetailNet (fixed gate)	Stale feed	7	0.00%	0.00	4/4

Dataset	Scenario	Days	Cost change	Fill change(pp)	Identical parser/LLM pairs
FreshRetailNet (fixed gate)	Capacity cut	7	0.00%	0.00	4/4
FreshRetailNet (calibrated gate)	Normal	7	0.00%	0.00	60/60
FreshRetailNet (calibrated gate)	Inventory-field fault	7	-11.71%	12.25	60/60
FreshRetailNet (calibrated gate)	Stale feed	7	0.00%	0.00	60/60
FreshRetailNet (calibrated gate)	Capacity cut	7	0.00%	0.00	60/60

Two information environments; two separately estimated effects

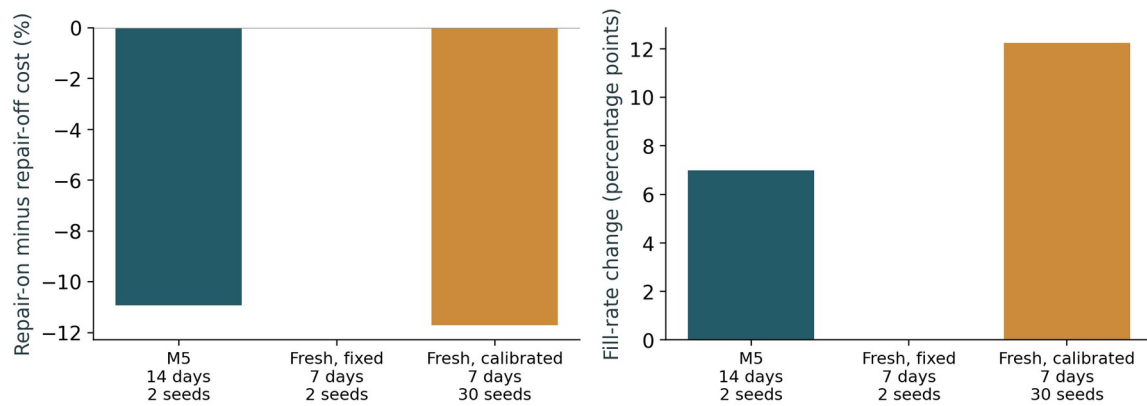


Figure 6.15. Two information environments; two separately estimated effects. Separate within-dataset repair effects under an inventory-field fault. M5 uses 14 days and FreshRetailNet seven; their quantity units, expiry assumptions and economics differ. The calibrated result is an exploratory reused-holdout follow-up with 30 simulator seeds; the other bars have two. These bars do not estimate a commercial return or the causal effect of changing datasets.

The business case supported by this evidence is for a controlled replenishment process. A retailer needs dependable stock evidence, a clearly labelled estimate when availability hides demand, typed supplier rules and checks before spending. An LLM can participate inside that process. The experiments have not established that it should replace the conventional numerical tools or a capable rules engine.

Useful language tasks remain plausible: translating supplier clauses into checkable fields, requesting a permitted exception tool, and producing a decision packet connected to its sources. The study demonstrates bounded feasibility for a controlled synthetic grammar. It does not measure planner time savings, human trust, unfamiliar natural-contract performance or production ROI. Those are distinct investment tests.

A sensible deployment starts in shadow mode with actual inventories and supplier documents, then moves to reviewed orders and spending caps. Compare exception-resolution quality and delay against the existing system. Expand autonomy only where the measured additional benefit covers error, oversight, hosting and integration costs. Where a capable parser gives the same business outcome more cheaply, it remains a reasonable default.

Results table. Answers to the retained research questions

Question	Completed answer	Scope limit
RQ1: performance	Conventional policies and matched parser/LLM outcomes are reported for each core scenario.	All single-agent and free-form architectures remain unexecuted.
RQ2: evidence gate	M5 detects known faults; FreshRetailNet exposes spending-cap portability and delay costs.	Effects depend on thresholds and simulator assumptions.
RQ3: censored demand	Recovery is measurable on masking; final forecast benefit is mixed; default service/expiry trade-off is small.	Natural unserved demand has no identified target.
RQ4: availability vs error	The explicit stockout-gate ablation shows a valid observation need not justify an integrity hold.	The naive comparator is a mechanism test, not a capable production parser.
RQ5: semantic value	Cost and fill match the capable parser on the disclosed synthetic grammar.	Unfamiliar, conflicting natural documents are not tested.
RQ6: agent organization	Typed outputs and independent checks are operationally audited.	No head-to-head test isolates additional agent roles.
RQ7: trace quality	Source-linked records, physical balances and attempts are machine-audited.	Human diagnosis and trust were not measured.
RQ8: generalizability	Within-dataset effects and scale-transfer failure are compared descriptively.	One small panel and origin cannot establish retailer-wide external validity.

Results table. Hypotheses: descriptive evidence status, without confirmatory tests

Hypothesis	Status	Reason
H1	Descriptively consistent	Matched grammar outcomes show no additional LLM cost/fill advantage; this is not a noninferiority test.
H2	Mechanism verified; incremental benefit bounded	Detected evidence faults trigger holds or verified repair. Conventional MILP already has zero actual rule violations; cap portability restricts useful execution.
H3	Mixed and conditional	Recovery selection, forecast accuracy and service–expiry trade-offs differ; natural demand recovery is not proved.
H4	Supported within the targeted ablation	Holding on valid stockout observations worsens simulated service against the censoring-aware rule.
H5	Supported on templates; remaining clause untested	The capable parser matches the LLM. Gains on unfamiliar documents were not evaluated.
H6	Untested as a comparative hypothesis	No free-form multi-agent baseline isolates typed-state or specialization effects.
H7	Untested	No participant study measures fault diagnosis, acceptance or trust.
H8	Conditional engineering evidence	Known-fault controls are useful, but the transferred cap fails until a separately calibrated exploratory follow-up.

CHAPTER 7 Ethical, Safety, and Governance Considerations

7.1 Automation Bias and Accountability

FreshRetailNet adds a particular source of automation bias: a planner may interpret a recovered-demand number as though it had been observed in transactions. The proposed interface must label every recovery estimate as an inference with uncertainty and show the original stockout evidence that motivated it. A model that reports “120 lost units” without uncertainty or a public ground-truth basis can create stronger, but unjustified, confidence in a purchase recommendation. The manager remains accountable for risk limits and approval policy even when the calculation is automated.

A polished AI recommendation can appear more authoritative than it deserves. Human-factors research shows that users may over-rely on automated outputs, especially when the system presents itself with confidence (Parasuraman & Riley, 1997). The design therefore treats explanation as evidence for review rather than as a tool for persuasion. Failed checks, uncertainty, alternatives, and escalation reasons should be shown alongside the recommended action.

Accountability also requires a clear record of who or what made the final decision. The trace distinguishes between a system-generated proposal, a human-approved order, a human-modified order, a held decision, and a bounded-autonomous execution. Procurement policy remains an organizational responsibility even when the computational steps are automated.

7.2 Data Quality as a Safety Property

The added benchmark makes clear that data quality and demand visibility are not the same property. A correct stockout indicator is useful evidence, although it reveals that the recorded sales value is only a lower bound on demand. The quality gate should therefore distinguish corrupt or missing availability data from genuine inventory unavailability. Over-aggressive holds during real stockouts can themselves worsen stockout persistence, while unwarranted demand imputation may increase spoilage. Both trade-offs are explicit in the evaluation.

Evidence gating shifts some risk from acting on bad information to delaying an action. The primary cap-transfer failure makes that trade-off visible: a controller can satisfy its checks while systematically withholding otherwise feasible orders. Holds should therefore be reported beside service, cost and the time needed to resolve an exception. A held decision means the permitted action lacks sufficient support under the current policy; it does not establish that ordering nothing is the best business decision.

The dissertation uses only public benchmark data and synthetic operational layers. No confidential retailer sales, supplier terms, or production telemetry are required for the reported experiment. This improves reproducibility and reduces privacy and commercial-confidentiality concerns.

7.3 Security and Prompt Injection

Dataset governance matters as well. FreshRetailNet uses pseudonymized identifiers and a public CC BY 4.0 release. The dissertation cites the publisher, records the dataset and code versions, and does not attempt to identify stores, individuals, or unreleased commercial information. Public supplier-text stress tests are synthetic and must be separated from any actual business documents. A future deployment

would also require controls over contracts, inventory records, model providers, and purchase-order permissions that cannot be evaluated from these public files alone.

Supplier documents, spreadsheets, and operational notes are untrusted inputs. Research on indirect prompt injection shows that content retrieved by an LLM-enabled application can contain instructions designed to influence the model's behavior (Greshake et al., 2023). The architecture therefore separates data from system instructions, restricts tools by role, validates structured outputs, and requires authenticated sources before a document can change an executable constraint.

Execution endpoints use least privilege and idempotency keys. A language model that interprets a contract does not automatically receive permission to submit a purchase order. This separation of interpretation from execution is as important as the prompt-level defenses.

7.4 Supplier Concentration

Optimization can unintentionally reinforce supplier concentration when one supplier appears consistently cheaper or more reliable. The issue is secondary in a replenishment study where many products may be effectively single-sourced, but it remains a governance concern. Where supplier choice exists, concentration should be measured and, if diversification is a policy objective, represented explicitly as a constraint rather than expected to emerge from the cost function.

7.5 Environmental and Economic Externalities

Fresh-retail waste is a central business and sustainability dimension. Since the released dataset does not provide measured product expiry or disposed quantities, waste is evaluated through simulated shelf lives and explicitly stated disposal penalties. Those scenario results may show how governance and demand recovery shift the service–waste trade-off, but they cannot be claimed as measured reductions in food waste at Dingdong. Any claim about the sustainability benefit of an LLM should also account for its computational cost and whether a deterministic alternative provides comparable outcomes.

Higher service levels can require more stock, storage, transport, and expedited shipments. Conversely, poor replenishment can create waste through obsolescence or markdowns. Where data are available, environmental costs can be added to the objective or reported separately. The computational cost of agentic AI is also recorded through model calls, token usage, and latency so that the system can be compared with simpler alternatives.

7.6 Limitations

The design has five main limitations. First, neither benchmark supplies the full inventory-and-procurement state required to observe real autonomous purchasing: stock balances, suppliers, lead times, purchase orders and procurement costs are simulated. Second, even FreshRetailNet's genuine stockout flags do not reveal the quantity of latent customer demand during unavailable hours; recovery can be assessed directly only on artificially hidden observations and indirectly through held-out forecasting. Third, FreshRetailNet's global sales normalization prevents a straightforward interpretation of its values as physical unit counts, and its seven-day held-out period limits longer-run causal claims. Its brief history also restricts seasonal extrapolation.

Fourth, outcomes from simulated alternative replenishment policies are conditional on assumptions about demand, delivery, prices, shelf lives and objectives. The independent calculator sensitivity does not share

the core MILP's economic scale or target timing. Fifth, supplier documents use a controlled synthetic grammar and a single small local LLM. No human study was conducted. The primary agent experiment has two seeds; the calibrated follow-up has 30 but reuses the holdout after the portability failure was observed. These studies do not establish population significance or production returns.

The valid claims concern disclosed engineering comparisons: recovery and forecasting on observable targets, conditional policy outcomes in simulation, and the behaviour of structured checks under known test faults. The negative findings also matter. A conventional constrained optimizer already avoided the tested hard-rule violations, and the capable parser matched the LLM's cost and fill. External business impact requires independent inventory, procurement and outcome data from a partner retailer.

CHAPTER 8 Conclusion

8.1 Contribution and Answer to the Management Question

This dissertation asks when an organisation should allow an AI system to turn a stock recommendation into a purchase. The proposed answer is deliberately conditional. Autonomy should be supported by evidence about the underlying state, the uncertainty of demand, the validity of business constraints, and the consequences of being wrong. Forecasting and optimization already provide much of the numerical decision-making; LLMs may add value in interpreting ambiguous information and coordinating exceptions, but their additional benefit must be measured against an equally well-equipped deterministic alternative.

The redesigned study uses two public benchmarks to examine this proposition under distinct conditions. M5 supplies the longer, hierarchical sales environment in which inventory policies, supplier constraints, and operational failures can be simulated and compared. FreshRetailNet-50K adds a realistic measurement complication: the public data contain hourly records of genuine stockouts that can suppress observed sales. This makes it possible to evaluate when a demand-recovery step changes forecasts and simulated ordering decisions. The study treats the datasets as complementary rather than interchangeable and keeps observed, inferred and simulated information separate.

The research contribution is both technical and managerial: a common, auditable decision architecture; a way to distinguish bad data from censored-but-valid observations; a test design that attributes benefits to forecasting, recovery, validation, or language-based coordination rather than conflating them; and a staged governance framework for adopting autonomous replenishment. The completed M5 pilot already demonstrates why attribution is necessary: its beneficial state recovery was also achieved by a deterministic parser, and its LLM made incorrect recovery-tool selections that independent checks prevented from becoming unsafe executions.

The primary transfer result also changes the managerial answer. The inherited 1,500-point autonomous spending cap held every normal-operation decision, even when source evidence was clean. On the same original seeds, historical calibration to 1,850 points changed mean fill from 45.2% to 80.2% and held days from 14 to 3. The exploratory follow-up found identical cost and fill in 240/240 parser–LLM pairs. Useful autonomy therefore requires operating limits that fit the business scale, as well as trustworthy evidence. The gain is shared by the verified architecture; it is not an isolated language-model return. The calibrated controller’s full 30-seed normal fill was 79.2%, below the conventional constrained B3 baseline’s 85.0%. Its lower indexed cost accompanies lower service, so calibration does not remove the remaining opportunity cost of the gate.

8.2 Completed evidence and its limits

Both benchmarks now contribute completed engineering evidence. The M5 archive retains the original numerical and feasibility study and the 32-run V2 pilot. FreshRetailNet adds 360 numerical runs, 32 matched live runs, 1,080 independent calculator sensitivity runs, 60 gate ablations, and protected forecast and recovery tests on 30 series. The evidence is conditional on the disclosed panel and assumptions. It does not validate all 50,000 series, natural lost demand, commercial waste savings or human trust. Incremental language-model superiority over a capable parser was not demonstrated by matched cost and fill. A separate 480-run follow-up was motivated by the inherited spending cap’s failure to pass clean

orders. A registered training-only calibration selected a cap of 1,850; the evaluation reuses the holdout and is explicitly exploratory. Its 30 simulation seeds per cell do not turn that post-primary question into an untouched confirmation.

Neither public benchmark reveals every business quantity needed to evaluate actual purchase orders. True lost demand is unobserved during natural stockouts, and the cost of an alternative inventory policy can only be evaluated inside an assumed counterfactual world. The completed checks support the implemented feature and evidence boundaries. They do not remove the need to test those assumptions against independent operational outcomes.

8.3 Future Research and Managerial Implementation

The next priority is broader independent product, store and time panels, real inventory and procurement ground truth, and difficult natural supplier documents with fair parser coverage. Recovery needs external checks where demand was not observed. Human review and partner shadow-mode studies could measure whether checked language assistance improves exception resolution enough to justify its operating burden.

For organisations, the practical adoption sequence should be staged. Start with inventory data reconciliation, timestamp and unit checks, and complete decision lineage. Where stockouts cause demand censoring, add a properly calibrated recovery model and show its uncertainty rather than silently replacing observed sales. Represent supplier and assortment rules as typed constraints. Test an LLM on the exceptions conventional software cannot reliably parse, retaining independent verification and clearly defined approval rights. Only then should the system progress to bounded autonomous purchase execution, and only after empirical evidence shows that its benefits exceed its error, oversight and operating costs.

The longer-term contribution would be validation with genuine retailer procurement and inventory outcomes, including measured spoilage, realized supplier lead times, purchase-order modifications, and staff adoption. Such validation could turn this careful simulation and benchmark research into an evidence-based guide for deploying trustworthy autonomous operational systems.

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Appendix A Reproducible Implementation Plan

A.1 Implementation Phases and Acceptance Criteria

The implementation program below records what the bounded study completed and what remains outside its scope. Source validation, the 30-series adapters, point-recovery and forecast tests, physical expiry, controlled agent comparisons and descriptive reporting are complete. Full architecture ablations and human participants remain unevaluated.

Phase	Output	Acceptance gate	Reported status
1. Data acquisition	Pinned M5 and FreshRetailNet source manifests	Licenses, source hashes, schema checks, distinct series and date coverage reported	Complete source profiles and pinned hashes
2. Canonical adapters	Mapped daily sales, hourly availability and hierarchy tables	Keys, scale metadata, stockout polarity, child-array consistency verified	Complete; simulated fields explicitly labelled
3. Recovery models	Raw, exposure-based, regression and learned recovery artefacts	Artificial-mask targets identified without using future observations	Three point-estimate methods; natural truth unavailable
4. Forecasting	Train-only fitted transformations and seven-day/28-day forecasts	Eval-target and eval-weather perturbation tests do not change prior predictions	Four forecasts on 30 series; no full probabilistic suite
5. Simulators	Age-aware Track F simulator and existing Track M simulator	Inventory balance, receipts, expiry, deterministic paired replay	FIFO and separate calculators; core MILP is age-unaware
6. Controls and agents	B1–B11 policy implementations	Matching tools and constraints; hold/approval/execution traces complete	B1/B2/B3/B4 and matched B10; other architecture families untested
7. Statistical report	Frozen contrasts, CIs, paired outcomes and exclusions	Independent seed count, unit of analysis and negative results disclosed	Descriptive paired effects; conditional seed variation
8. Human review	Counterbalanced audit packets and approval protocol	Ethical approval and recruitment before claiming participant results	Not conducted; no participant claim

Table A.1. Implementation and evidence gates for the expanded study.

A.2 Dataset Loader and Validation Pseudocode

```
# Pseudocode: exact package versions and hashes are pinned before execution
frn = load_dataset("Dingdong-Inc/FreshRetailNet-50K")
assert required_fields <= set(frn["train"].column_names)
assert len(frn["train"]) == 4_500_000 # verify against pinned release
assert len(frn["eval"]) == 350_000
assert no_overlapping_store_product_dates(frn["train"], frn["eval"])
assert daily_series_complete_or_flagged(frn)
assert check_hourly_sales_alignment(frn)
assert stockout_count_matches_hourly_mask(frn, hours=range(6, 22))
```

```

train, val_folds, test = make_temporal_split(frn, holdout="official_eval")
for fold in val_folds:
    rec = fit_recovery(fold.train_only)
    feat = fit_features(fold.train_only, rec)
    preds = predict_recursive_7d(feat, fold.future_exante_covariates)
    assert unaffected_by_scrambling_future_actuals(preds)
select_models_and_policies_using_validation_only()
run_final_test_once()

for dataset in ("M5", "FreshRetailNet-50K"):
    for origin, scenario, seed in preregistered_runs(dataset):
        shared_world = make_simulation_world(origin, scenario, seed)
        for policy in ("B4", "B10", "B11_if_fresh"):
            outcome = simulate(policy, shared_world.clone())
            validate_solver_certificates(outcome)
            write_immutable_trace(outcome)

```

This pseudocode explains the timing and validation rules; the executable scripts and saved manifests are the authoritative implementation. The downloaded release passed key, chronology, hourly sales and flag-alignment checks. Discount values outside [0,1] were retained and reported, with discount excluded from forecasting features. Every exception and exclusion remains in the source-validation receipts.

A.3 Configuration Manifest

```

research_version: freshretailnet-bounded-v1
status: completed_bounded_panel
source_revision: 08c1fab7f9257bc73679d415d65d644165d351d4
source_license: CC-BY-4.0
source_validation: 4,850,000 daily rows / 50,000 series
modeling_panel: 30 training-selected series / 0.06% of release
final_horizon: 7 days after 90 training days
primary_numerical: 360 runs / 30 simulation seeds
primary_agent_factorial: 32 runs / 2 simulation seeds
independent_policy_sensitivity: 1,080 runs
stockout_gate_ablation: 60 runs
quantity_scale_core: 100 * globally_normalized_sales
economic_unit: simulated cost index
future_observed_features: prohibited
natural_lost_demand_ground_truth: unavailable
human_participants: not conducted
authoritative_manifest: results/freshretailnet/protocol.json
exploratory_gate_transfer: 480 runs / 30 seeds
transfer_status: post-primary hypothesis / reused holdout
training_selected_spend_cap: 1850.0

```

A.4 Required Tables and Plots in the Final Results

The completed figures and tables cover source availability and panel composition, identifiable artificial-mask recovery, daily reconstruction, protected forecasts and training censoring strata, core cost/service/safety, matched parser-LLM outcomes, independent service-expiry and demand sensitivity, operating attempts and within-dataset M5 comparisons. Appendices retain all individual core runs and original M5 figures. Full sensitivity and exploratory transfer rows are accompanying CSVs. Human explanation outcomes, a full semantic-difficulty suite and production financial returns are not included because they were not measured.

Appendix B Source-Field Checklist and Research Assumptions

B.1 Verified FreshRetailNet Public Fields

The 19 published columns were verified in the pinned Parquet files: `city_id`, `store_id`, `management_group_id`, `first_category_id`, `second_category_id`, `third_category_id`, `product_id`, `dt`, `sale_amount`, `hours_sale`, `stock_hour6_22_cnt`, `hours_stock_status`, `discount`, `holiday_flag`, `activity_flag`, `precpt`, `avg_temperature`, `avg_humidity` and `avg_wind_level`. Sales/hierarchy and historical availability are used by the bounded adapter. Future realized contextual fields are not planning features (Dingdong-Inc, 2025).

B.2 Fields That Must Never Be Described as Observed

Neither public dataset provides the complete procurement ledger needed for real replenishment outcomes. For FreshRetailNet in particular, any numerical current on-hand balance, supplier assignment, MOQ, purchase-order receipt, wholesale price, actual shelf life, expiry/waste quantity, or recovered latent demand is an experimental assumption or inference. The true counterfactual profit of alternative purchase policies is unavailable. This distinction must appear in the main text, tables, figures, and decision-trace interface.

B.3 Benchmark Counting and Data Version Caveats

The published FreshRetailNet paper reports 863 perishable SKUs, while the dataset card reports 865. The dissertation reports the distinct identifier count computed from its pinned file, alongside those two publication values. The dataset card's approximately 20% stockout frequency refers to hourly exposure; a daily row with at least one unavailable hour represents a different rate. The train/evaluation counts are 4,500,000 and 350,000 rows in the documented release, and these figures are verified programmatically before they are included as results.

B.4 Interpretation of Financial and Sustainability Outcomes

FreshRetailNet sales are normalized, and actual spoilage outcomes are not released. The dissertation therefore reports operating-cost indexes and simulated expiry effects rather than real gross margin, revenue, or tonnes of avoided food waste. A scenario can be compared fairly across two controllers only when they share the same scaling convention, demand generator, and economic assumptions. There is no valid conversion from those model outputs to a real company's financial savings without independent evidence.

Appendix C Completed Studies and Remaining Evidence Gaps

C.1 Completed Evidence

The archived pilot report and result CSV supplied with the dissertation record 32 completed M5 simulator runs across four scenarios (normal, capacity cut, feed gap, and derived-field collapse), two seeds, and four controller/recovery arms. They contain 448 decision traces and 68 captured LLM attempts. The study panel has only thirty item-store series. The matched LLM and deterministic-parser outcomes were identical. Recovery improved results under the inventory-derived-field collapse in both controller families. Verification prevented ten inappropriate recovery requests from the model selector. These findings are reported as descriptive pilot evidence only.

C.2 Not Yet Evaluated

Unevaluated commitments include full-population FreshRetailNet modeling, ground truth for naturally unserved demand, all B1–B11 architecture ablations, independent heterogeneous supplier-language tests, a formal human audit and actual ERP purchasing. Completed 30-series forecasts, recovery tests and simulator runs are reported in the following appendices. Unexecuted tests have no invented outcome.

C.3 Reporting Safeguard

Final manuscript revisions should promote a planned hypothesis into a results claim only when it is supported by a saved run artifact, a verified source sample, and the relevant analysis. A result from the FreshRetailNet paper by Wang et al. (2025) or from Azar (2026) may be discussed in the literature review but must not be re-labelled as an experiment conducted for this dissertation. Likewise, the existing small M5 pilot cannot be described as a complete 30,490-series evaluation.

Appendix D FreshRetailNet Results and Artifact Register

D.1 Artifact guide and exact run-level results

The accompanying files preserve every completed individual run. Numerical, live-agent and independent-policy CSVs have 360, 32 and 1,140 rows respectively. Forecast files contain series/day predictions, observable targets and flags used only for scoring. Artificial-mask files retain each hidden target. Audit files contain model calls, grounding, selector and decision records.

The report directory contains FreshRetailNet_numerical_all_runs.csv, FreshRetailNet_agent_all_runs.csv and FreshRetailNet_policy_sensitivity_all_runs.csv, as well as their descriptive means and matched contrasts. M5_FreshRetailNet_descriptive_comparison.csv records the compatible relative comparisons. The full evidence archive adds raw objects, saved models, frozen code, source receipts and failed attempts.

Selected computational panel

Series	Stratum	Train mean sales	Train stockout share
FRN_0247@S0020	2	0.38677	22.28%
FRN_0363@S0021	2	0.39774	26.31%
FRN_0210@S0054	2	0.33565	23.69%
FRN_0218@S0137	1	0.90758	17.34%
FRN_0635@S0156	1	0.81226	18.25%
FRN_0422@S0173	1	0.93710	20.26%
FRN_0738@S0181	2	0.28823	23.69%
FRN_0190@S0191	0	0.83548	8.87%
FRN_0295@S0202	1	0.37645	18.95%
FRN_0653@S0202	0	0.35194	14.82%
FRN_0635@S0219	1	0.42403	17.84%
FRN_0473@S0231	0	0.51613	15.02%
FRN_0802@S0259	0	0.84839	15.93%
FRN_0769@S0298	0	0.70000	12.80%
FRN_0843@S0300	2	0.54355	32.66%
FRN_0682@S0301	0	0.96290	9.78%
FRN_0834@S0334	2	0.51290	60.38%
FRN_0802@S0355	1	0.72097	16.43%
FRN_0810@S0410	0	0.48903	9.98%
FRN_0715@S0413	2	0.35452	23.69%
FRN_0650@S0423	0	0.45597	14.72%
FRN_0729@S0474	0	0.49355	10.99%
FRN_0218@S0482	0	0.66452	11.39%
FRN_0620@S0484	1	0.60774	19.35%
FRN_0577@S0521	1	0.42242	17.34%
FRN_0834@S0649	2	0.92419	45.77%
FRN_0560@S0805	1	0.57726	17.74%
FRN_0074@S0851	2	0.55387	22.18%
FRN_0402@S0861	1	0.56629	19.56%
FRN_0379@S0877	2	0.54516	55.44%

D.2 Core individual runs — Normal

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	17,735.329	78.80%	27.62%	0	7
B1	1	16,579.451	81.03%	27.62%	0	6
B1	2	17,420.022	79.04%	28.57%	0	7
B1	3	18,125.015	77.93%	30.00%	0	6
B1	4	19,701.915	74.99%	30.95%	0	7
B1	5	17,780.512	77.92%	29.05%	0	7
B1	6	19,138.125	75.20%	31.90%	0	7
B1	7	18,583.141	77.28%	29.05%	0	7
B1	8	17,731.543	78.18%	29.05%	0	7
B1	9	20,937.615	70.63%	35.71%	0	7
B1	10	16,176.249	81.99%	23.33%	0	7
B1	11	19,119.515	75.78%	31.43%	0	7
B1	12	19,894.785	73.82%	30.00%	0	7
B1	13	18,242.236	77.23%	29.52%	0	7
B1	14	19,224.607	76.77%	30.00%	0	7
B1	15	18,838.073	76.21%	26.67%	0	8
B1	16	18,331.092	76.75%	26.67%	0	7
B1	17	17,990.827	78.62%	31.43%	0	7
B1	18	18,288.650	77.44%	30.95%	0	7
B1	19	16,198.008	82.75%	24.76%	0	7
B1	20	18,173.944	78.53%	26.19%	0	7
B1	21	18,464.016	76.82%	27.14%	0	7
B1	22	17,714.833	79.14%	25.24%	0	7
B1	23	18,081.225	77.84%	29.52%	0	7
B1	24	17,709.263	78.39%	29.52%	0	7
B1	25	18,315.159	76.78%	26.67%	0	8
B1	26	17,002.835	81.65%	23.81%	0	7
B1	27	18,872.682	77.91%	29.52%	0	7
B1	28	19,199.945	75.61%	33.81%	0	7
B1	29	19,139.064	75.72%	28.10%	0	7
B2	0	18,714.549	73.04%	32.86%	0	6
B2	1	19,450.479	70.16%	36.67%	0	4
B2	2	18,221.380	71.89%	33.33%	0	7
B2	3	20,319.804	68.74%	35.71%	0	6
B2	4	20,524.152	68.68%	35.24%	0	7
B2	5	20,375.280	67.35%	37.14%	0	5
B2	6	20,247.988	69.68%	36.19%	0	7
B2	7	20,150.511	68.26%	35.71%	0	7
B2	8	20,921.230	65.55%	38.10%	0	7
B2	9	22,216.301	64.51%	38.10%	0	7
B2	10	18,691.531	72.37%	33.81%	0	6
B2	11	19,969.856	68.91%	35.71%	0	4
B2	12	20,875.138	67.17%	38.57%	0	7
B2	13	19,058.313	71.17%	37.14%	0	5
B2	14	21,143.615	67.12%	36.67%	0	6
B2	15	20,538.898	66.91%	37.62%	0	7
B2	16	19,760.069	69.79%	32.38%	0	6
B2	17	17,376.287	75.41%	33.81%	0	7
B2	18	19,133.461	69.75%	35.24%	0	6

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B2	19	17,718.816	74.84%	30.95%	0	6
B2	20	20,228.331	69.08%	38.57%	0	7
B2	21	20,047.329	68.73%	37.14%	0	6
B2	22	19,655.185	70.05%	34.76%	0	6
B2	23	18,658.486	71.88%	33.81%	0	7
B2	24	20,149.638	68.19%	39.05%	0	7
B2	25	18,722.456	71.34%	32.86%	0	6
B2	26	19,227.461	71.22%	34.76%	0	7
B2	27	18,135.282	73.70%	34.29%	0	7
B2	28	20,076.943	68.79%	37.62%	0	7
B2	29	20,254.479	67.94%	35.24%	0	7
B3	0	18,834.188	85.17%	19.52%	0	0
B3	1	18,489.906	86.17%	17.14%	0	0
B3	2	18,667.147	84.51%	17.62%	0	0
B3	3	19,054.028	83.53%	24.29%	0	0
B3	4	20,285.256	81.83%	26.19%	0	0
B3	5	19,219.266	84.43%	20.95%	0	0
B3	6	17,918.294	88.27%	16.19%	0	0
B3	7	18,421.132	85.98%	18.57%	0	0
B3	8	18,481.642	85.25%	20.95%	0	0
B3	9	20,756.792	80.59%	27.62%	0	0
B3	10	18,482.335	86.34%	18.10%	0	0
B3	11	20,111.100	81.98%	21.43%	0	0
B3	12	20,453.425	82.34%	21.90%	0	0
B3	13	17,996.392	85.59%	19.52%	0	0
B3	14	19,358.248	83.05%	22.86%	0	0
B3	15	17,988.876	86.53%	15.71%	0	0
B3	16	18,656.536	84.80%	19.05%	0	0
B3	17	17,781.322	87.27%	20.00%	0	0
B3	18	20,510.611	80.13%	24.76%	0	0
B3	19	16,054.654	91.33%	16.67%	0	0
B3	20	18,379.175	85.74%	20.48%	0	0
B3	21	19,573.899	83.19%	23.81%	0	0
B3	22	18,493.814	85.83%	17.62%	0	0
B3	23	17,458.419	88.26%	17.14%	0	0
B3	24	18,711.980	85.16%	20.48%	0	0
B3	25	18,223.855	86.36%	15.71%	0	0
B3	26	18,638.899	85.59%	20.00%	0	0
B3	27	18,396.860	85.75%	18.57%	0	0
B3	28	18,730.647	85.91%	20.48%	0	0
B3	29	19,692.291	83.95%	19.05%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

D.2 Core individual runs — Inventory-field fault

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	17,611.796	80.89%	23.81%	0	9
B1	1	17,589.209	80.09%	26.67%	0	7
B1	2	18,685.214	78.41%	25.71%	0	10
B1	3	17,961.443	80.07%	29.05%	0	9
B1	4	19,659.989	76.54%	28.10%	0	10
B1	5	17,568.577	80.82%	29.05%	0	8
B1	6	18,918.969	76.56%	29.52%	0	10
B1	7	18,806.993	78.62%	26.67%	0	10
B1	8	19,059.460	75.85%	31.90%	0	9
B1	9	21,289.694	71.65%	33.33%	0	8
B1	10	17,013.616	81.32%	23.33%	0	8
B1	11	18,275.851	78.94%	29.52%	0	9
B1	12	19,382.016	77.21%	28.10%	0	9
B1	13	18,608.672	77.57%	28.57%	0	9
B1	14	19,359.835	79.00%	26.19%	0	8
B1	15	18,303.688	79.36%	24.76%	0	12
B1	16	19,139.488	75.91%	27.62%	0	9
B1	17	19,185.711	77.25%	30.95%	0	8
B1	18	18,520.027	78.35%	28.10%	0	7
B1	19	17,086.143	82.92%	20.48%	0	9
B1	20	19,148.857	79.62%	24.76%	0	8
B1	21	17,851.136	79.99%	25.24%	0	8
B1	22	17,942.610	79.55%	24.29%	0	10
B1	23	19,336.243	76.16%	28.57%	0	8
B1	24	17,585.124	80.62%	25.71%	0	9
B1	25	19,130.153	77.19%	25.71%	0	10
B1	26	16,293.325	85.01%	20.00%	0	8
B1	27	18,338.343	80.33%	25.24%	0	9

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	28	20,166.117	75.06%	30.95%	0	8
B1	29	20,488.995	73.67%	30.95%	0	10
B2	0	19,224.390	73.48%	31.43%	0	6
B2	1	18,885.377	73.69%	34.29%	0	7
B2	2	18,694.716	73.38%	30.95%	0	7
B2	3	19,100.233	73.24%	32.86%	0	8
B2	4	19,636.017	72.01%	30.95%	0	8
B2	5	19,643.144	71.80%	33.81%	0	6
B2	6	19,168.154	72.87%	32.86%	0	7
B2	7	19,560.190	72.15%	33.33%	0	8
B2	8	20,989.762	66.70%	36.67%	0	8
B2	9	21,858.340	66.05%	35.71%	0	8
B2	10	18,715.354	72.90%	33.81%	0	7
B2	11	18,352.859	73.56%	30.00%	0	5
B2	12	19,607.219	72.28%	33.33%	0	6
B2	13	18,729.918	73.90%	31.90%	0	7
B2	14	20,543.536	70.27%	34.29%	0	6
B2	15	18,394.891	74.70%	30.48%	0	7
B2	16	19,953.326	71.05%	31.43%	0	8
B2	17	16,720.617	79.08%	29.52%	0	5
B2	18	19,900.279	69.34%	32.86%	0	6
B2	19	16,832.209	79.17%	25.71%	0	7
B2	20	19,189.160	74.64%	32.38%	0	7
B2	21	19,267.769	73.14%	33.81%	0	6
B2	22	18,473.638	74.54%	30.48%	0	7
B2	23	19,915.621	70.34%	33.81%	0	8
B2	24	19,043.478	72.56%	35.71%	0	7
B2	25	18,339.779	75.53%	26.19%	0	7
B2	26	18,807.756	74.88%	30.48%	0	6
B2	27	17,459.968	77.19%	29.05%	0	5
B2	28	19,465.057	72.23%	32.86%	0	5
B2	29	20,701.076	69.50%	32.86%	0	9
B3	0	19,159.301	84.76%	20.48%	0	0
B3	1	18,207.028	86.83%	17.62%	0	0
B3	2	18,910.859	84.39%	17.14%	0	0
B3	3	18,824.738	84.91%	23.81%	0	0
B3	4	20,468.676	82.03%	24.76%	0	0
B3	5	19,807.642	83.44%	21.43%	0	0
B3	6	18,098.312	88.55%	14.76%	0	0
B3	7	19,213.589	85.01%	18.57%	0	0
B3	8	19,358.964	83.92%	21.90%	0	0
B3	9	20,934.487	80.31%	26.19%	0	0
B3	10	18,774.688	86.04%	17.62%	0	0
B3	11	20,378.474	82.13%	21.43%	0	0
B3	12	19,802.567	84.18%	21.43%	0	0
B3	13	18,029.206	85.94%	20.00%	0	0
B3	14	20,067.941	82.05%	24.29%	0	0
B3	15	18,315.320	86.83%	15.24%	0	0
B3	16	18,836.141	84.45%	19.52%	0	0
B3	17	17,830.934	87.53%	20.48%	0	0
B3	18	19,853.349	81.93%	22.86%	0	0
B3	19	16,183.520	91.47%	15.71%	0	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B3	20	18,550.147	85.44%	20.00%	0	0
B3	21	19,678.908	83.43%	23.33%	0	0
B3	22	18,593.378	86.19%	18.10%	0	0
B3	23	18,055.491	87.26%	18.10%	0	0
B3	24	18,953.864	85.31%	20.48%	0	0
B3	25	18,434.459	86.27%	15.71%	0	0
B3	26	18,888.526	86.04%	19.52%	0	0
B3	27	18,410.566	85.89%	18.10%	0	0
B3	28	19,522.936	84.32%	22.38%	0	0
B3	29	20,659.723	82.44%	20.00%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

D.2 Core individual runs — Stale feed

All 30 seeds per policy

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	0	19,512.420	76.23%	29.52%	0	7
B1	1	17,789.008	78.99%	29.52%	0	6
B1	2	18,464.022	76.88%	27.62%	0	7
B1	3	18,885.500	77.44%	29.52%	0	7
B1	4	21,575.320	71.89%	34.29%	0	7
B1	5	18,314.971	77.65%	30.95%	0	7
B1	6	20,251.288	72.72%	31.43%	0	7

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B1	7	19,261.653	76.31%	29.52%	0	7
B1	8	20,248.948	72.98%	33.33%	0	7
B1	9	21,721.574	70.06%	36.67%	0	7
B1	10	17,104.856	80.92%	24.76%	0	7
B1	11	19,604.691	74.55%	33.33%	0	7
B1	12	22,158.547	69.08%	32.38%	0	7
B1	13	19,903.064	73.67%	31.90%	0	7
B1	14	20,886.125	74.43%	29.52%	0	7
B1	15	19,400.510	75.49%	30.00%	0	8
B1	16	20,417.276	71.54%	30.00%	0	5
B1	17	19,236.356	77.49%	30.48%	0	7
B1	18	19,505.589	75.26%	32.38%	0	7
B1	19	17,243.807	81.15%	26.19%	0	7
B1	20	18,857.179	77.74%	26.19%	0	7
B1	21	19,556.668	75.18%	29.52%	0	7
B1	22	18,601.854	78.16%	26.67%	0	7
B1	23	19,337.286	74.87%	31.90%	0	7
B1	24	19,385.310	74.23%	32.38%	0	7
B1	25	18,607.195	77.44%	26.67%	0	8
B1	26	18,492.408	78.89%	24.29%	0	7
B1	27	19,230.189	77.53%	28.10%	0	7
B1	28	19,309.966	76.31%	30.95%	0	7
B1	29	20,568.612	74.55%	30.48%	0	7
B2	0	23,223.296	65.09%	37.14%	0	7
B2	1	21,682.569	66.33%	39.52%	0	6
B2	2	21,849.802	64.26%	40.00%	0	7
B2	3	22,811.511	64.25%	40.48%	0	8
B2	4	22,376.619	65.28%	39.52%	0	7
B2	5	21,317.351	64.60%	40.48%	0	3
B2	6	21,716.401	66.17%	39.52%	0	8
B2	7	23,160.512	62.08%	40.48%	0	7
B2	8	25,032.830	59.20%	43.33%	0	6
B2	9	24,327.803	63.19%	40.95%	0	7
B2	10	21,270.958	67.97%	38.57%	0	6
B2	11	22,663.677	62.45%	40.48%	0	3
B2	12	24,527.174	60.83%	43.33%	0	7
B2	13	21,185.771	67.82%	36.67%	0	6
B2	14	24,928.814	62.76%	41.43%	0	7
B2	15	22,112.482	65.63%	40.48%	0	7
B2	16	22,560.230	62.85%	38.57%	0	6
B2	17	20,873.872	71.40%	38.10%	0	7
B2	18	22,333.899	63.95%	37.62%	0	4
B2	19	20,311.691	69.85%	34.76%	0	6
B2	20	21,149.794	69.26%	38.10%	0	7
B2	21	24,461.539	60.01%	45.24%	0	6
B2	22	21,273.712	67.88%	37.14%	0	5
B2	23	21,481.098	65.38%	42.38%	0	7
B2	24	22,977.143	62.41%	42.86%	0	6
B2	25	20,330.263	71.55%	33.33%	0	7
B2	26	22,702.083	65.02%	38.57%	0	6
B2	27	20,486.263	71.82%	34.29%	0	7
B2	28	23,435.644	64.00%	40.48%	0	7

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B2	29	22,755.512	66.08%	36.67%	0	7
B3	0	20,299.390	81.93%	23.33%	0	0
B3	1	19,181.198	84.36%	19.52%	0	0
B3	2	19,816.853	81.88%	20.48%	0	0
B3	3	20,192.821	81.83%	25.71%	0	0
B3	4	20,119.272	81.46%	27.62%	0	0
B3	5	18,839.755	84.72%	21.43%	0	0
B3	6	19,174.011	85.43%	18.57%	0	0
B3	7	18,986.265	85.21%	19.05%	0	0
B3	8	20,914.570	80.35%	24.29%	0	0
B3	9	20,949.736	80.29%	28.10%	0	0
B3	10	18,760.641	84.71%	20.00%	0	0
B3	11	20,951.077	81.03%	21.90%	0	0
B3	12	21,062.438	80.61%	24.29%	0	0
B3	13	18,908.825	83.84%	21.43%	0	0
B3	14	20,392.447	81.27%	25.71%	0	0
B3	15	18,900.140	85.14%	16.19%	0	0
B3	16	19,650.705	82.26%	19.52%	0	0
B3	17	18,012.580	86.77%	21.43%	0	0
B3	18	21,277.099	78.80%	25.24%	0	0
B3	19	17,616.039	88.01%	18.10%	0	0
B3	20	18,487.261	85.48%	20.48%	0	0
B3	21	20,091.646	82.37%	24.76%	0	0
B3	22	19,692.357	83.04%	19.52%	0	0
B3	23	18,598.392	85.74%	19.05%	0	0
B3	24	19,229.893	83.89%	21.43%	0	0
B3	25	19,035.589	84.51%	17.14%	0	0
B3	26	20,289.185	81.74%	23.81%	0	0
B3	27	18,815.932	85.06%	18.57%	0	0
B3	28	19,332.990	84.30%	23.33%	0	0
B3	29	20,631.400	82.26%	20.95%	0	0
B4	0	24,488.115	42.27%	60.95%	7	0
B4	1	25,521.012	39.25%	60.95%	7	0
B4	2	23,654.760	43.89%	56.67%	7	0
B4	3	23,251.410	45.19%	59.05%	7	0
B4	4	24,019.237	44.22%	55.24%	7	0
B4	5	24,508.003	41.92%	60.95%	7	0
B4	6	22,696.454	47.58%	52.86%	7	0
B4	7	23,110.907	45.83%	54.76%	7	0
B4	8	25,396.775	39.78%	60.95%	7	0
B4	9	26,169.138	38.52%	61.90%	7	0
B4	10	23,095.977	45.08%	56.19%	7	0
B4	11	23,592.960	45.42%	56.19%	7	0
B4	12	25,421.616	40.38%	60.95%	7	0
B4	13	22,258.569	47.43%	53.33%	7	0
B4	14	25,075.476	40.86%	60.95%	7	0
B4	15	23,274.089	45.64%	55.24%	7	0
B4	16	24,461.748	42.52%	57.62%	7	0
B4	17	24,095.772	43.55%	56.67%	7	0
B4	18	23,666.492	44.01%	56.67%	7	0
B4	19	22,742.703	45.99%	55.24%	7	0
B4	20	24,289.215	42.78%	61.43%	7	0

Policy	Seed	Cost	Fill	Stockout	Held	Violations
B4	21	24,624.152	42.41%	59.05%	7	0
B4	22	22,626.153	46.72%	55.24%	7	0
B4	23	23,949.690	43.01%	59.52%	7	0
B4	24	22,468.061	47.18%	57.62%	7	0
B4	25	24,722.401	41.14%	55.71%	7	0
B4	26	23,381.589	45.43%	56.67%	7	0
B4	27	22,401.393	47.23%	53.81%	7	0
B4	28	27,026.395	36.69%	61.90%	7	0
B4	29	24,404.020	42.93%	57.62%	7	0

D.3 Live-agent individual runs

All 32 live factorial runs

Arm	Scenario	Seed	Cost	Fill	Held	Violations
LLM	Capacity cut	13	20,589.055	59.41%	2	0
LLM	Capacity cut	29	22,556.878	55.46%	2	0
LLM	Inventory-field fault	13	22,258.569	47.43%	7	0
LLM	Inventory-field fault	29	24,404.020	42.93%	7	0
LLM	Stale feed	13	22,258.569	47.43%	7	0
LLM	Stale feed	29	24,404.020	42.93%	7	0
LLM	Normal	13	22,258.569	47.43%	7	0
LLM	Normal	29	24,404.020	42.93%	7	0
LLM + repair	Capacity cut	13	20,589.055	59.41%	2	0
LLM + repair	Capacity cut	29	22,556.878	55.46%	2	0
LLM + repair	Inventory-field fault	13	22,258.569	47.43%	7	0
LLM + repair	Inventory-field fault	29	24,404.020	42.93%	7	0
LLM + repair	Stale feed	13	22,258.569	47.43%	7	0
LLM + repair	Stale feed	29	24,404.020	42.93%	7	0
LLM + repair	Normal	13	22,258.569	47.43%	7	0
LLM + repair	Normal	29	24,404.020	42.93%	7	0
Parser	Capacity cut	13	20,589.055	59.41%	2	0
Parser	Capacity cut	29	22,556.878	55.46%	2	0
Parser	Inventory-field fault	13	22,258.569	47.43%	7	0
Parser	Inventory-field fault	29	24,404.020	42.93%	7	0
Parser	Stale feed	13	22,258.569	47.43%	7	0
Parser	Stale feed	29	24,404.020	42.93%	7	0
Parser	Normal	13	22,258.569	47.43%	7	0
Parser	Normal	29	24,404.020	42.93%	7	0
Parser + repair	Capacity cut	13	20,589.055	59.41%	2	0
Parser + repair	Capacity cut	29	22,556.878	55.46%	2	0
Parser + repair	Inventory-field fault	13	22,258.569	47.43%	7	0
Parser + repair	Inventory-field fault	29	24,404.020	42.93%	7	0
Parser + repair	Stale feed	13	22,258.569	47.43%	7	0
Parser + repair	Stale feed	29	24,404.020	42.93%	7	0
Parser + repair	Normal	13	22,258.569	47.43%	7	0
Parser + repair	Normal	29	24,404.020	42.93%	7	0

Appendix E Complete M5 Figure Register

The previous M5 studies remain part of the comparative record. These original scientific figures are preserved from their final reports and describe earlier M5 windows. They are not FreshRetailNet results. The repository retains both original M5 PDFs, all 404 individual runs and the complete audit archives.

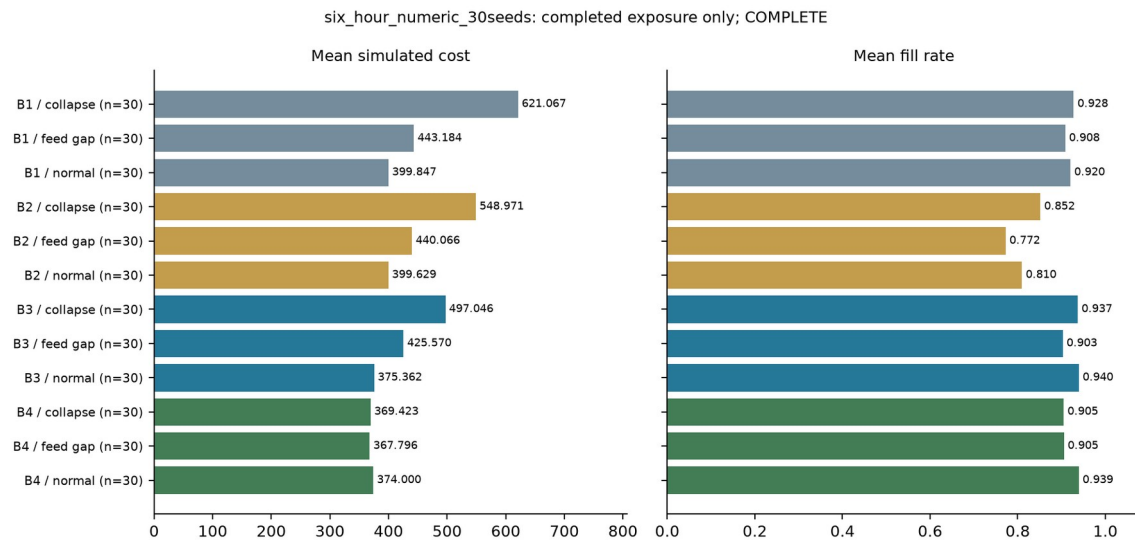


Figure E.1. Initial numerical study: cost and service across the three scenarios. Thirty seeds per policy/scenario; seven days; structured rules. These are simulated trade-offs, not LLM outcomes.

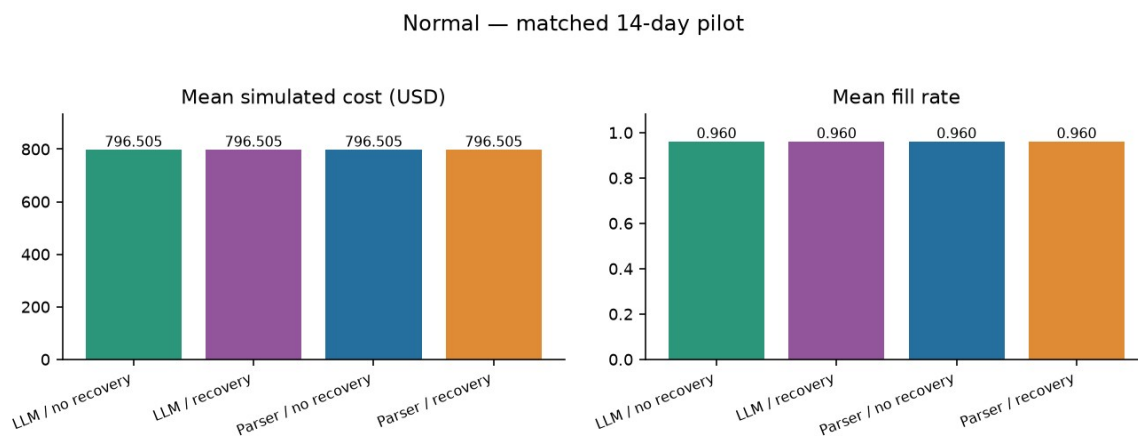


Figure E.2. V2 normal operation: cost and fill by configuration. Each configuration has two seeds and fourteen days. The LLM and parser means are identical.

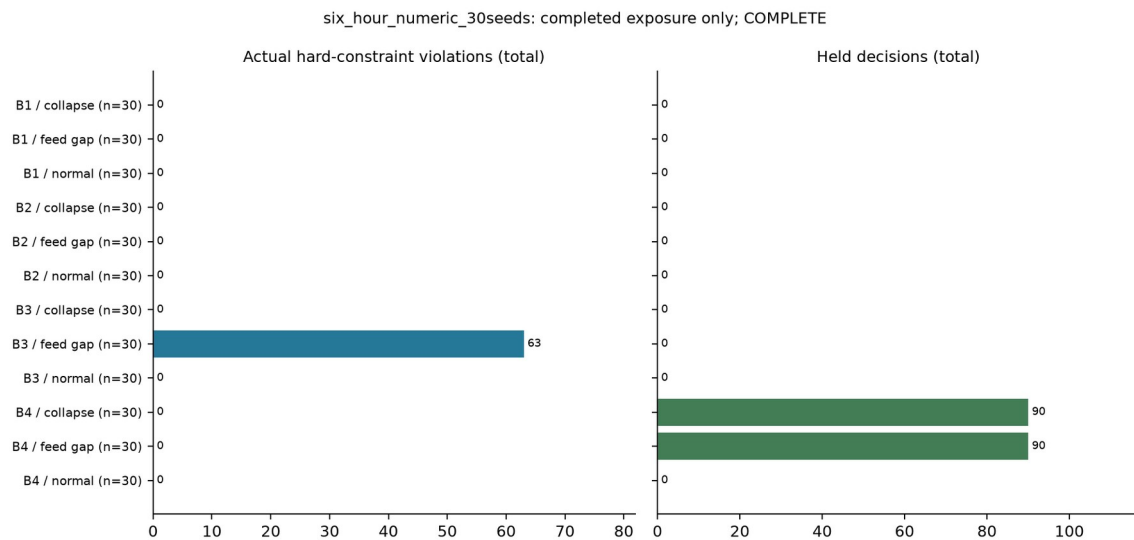


Figure E.3. Initial numerical study: hard violations and held decisions. Holds and violations are different endpoints; reference deviations are reported separately.

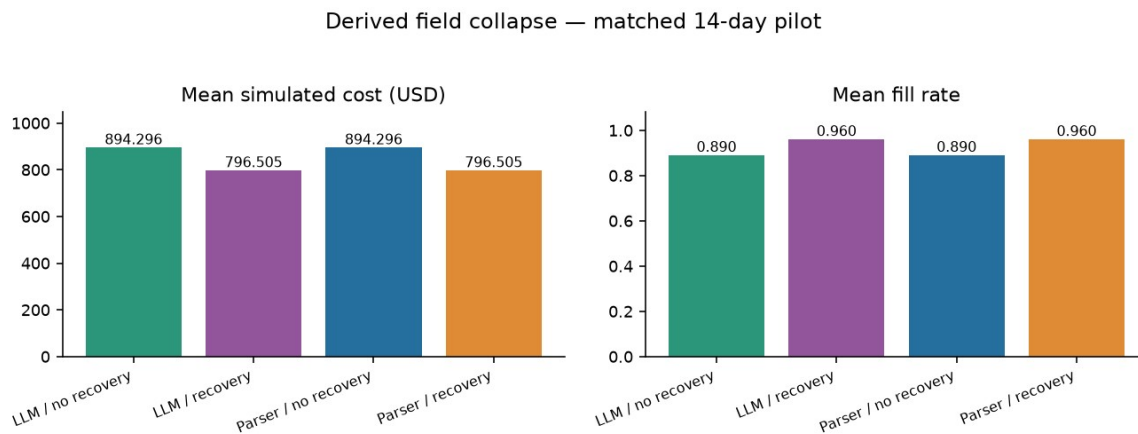


Figure E.4. V2 derived-field collapse: verified recovery restores cost and service in both parser and LLM configurations. The improvement is attributable to recovery.

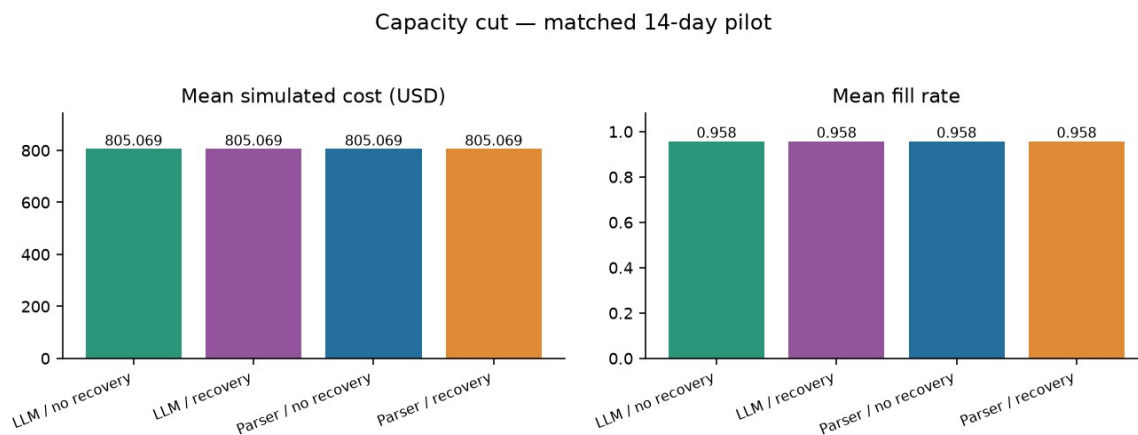


Figure E.5. V2 supplier capacity cut: all configurations respect the active constraints. The LLM supplies a checked numeric term, not the final order quantity.

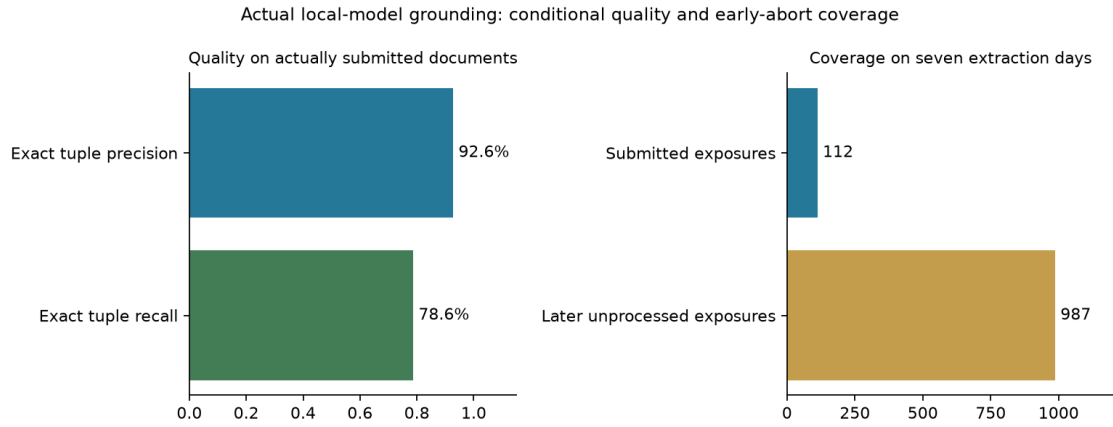


Figure E.6. Initial live-model grounding: conditional tuple accuracy and document coverage. Accuracy is measured on submitted rules; unprocessed sources do not count as successful extraction.

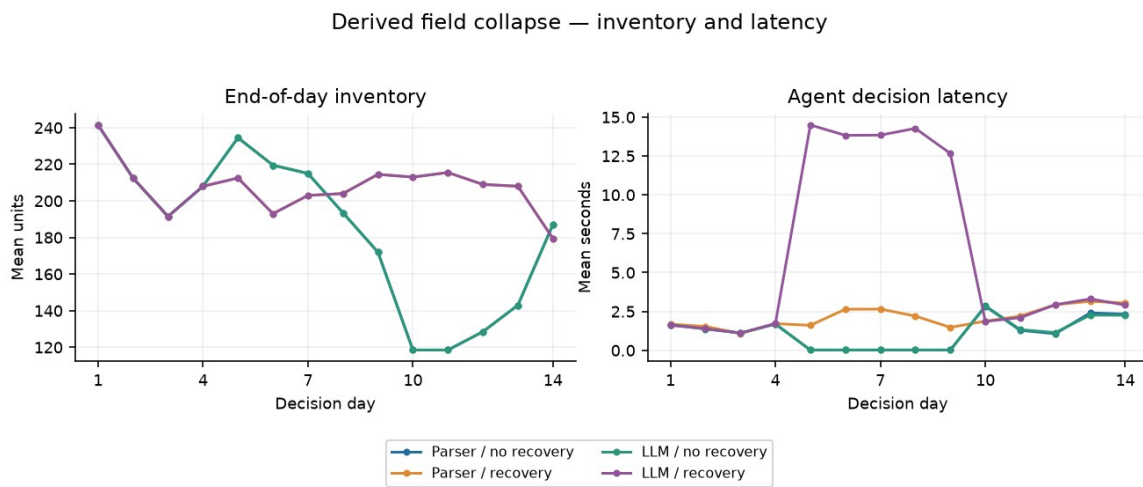


Figure E.7. V2 derived-field collapse: inventory and decision latency. Service recovery and inference effort should be assessed together; local execution has no priced hosting estimate.

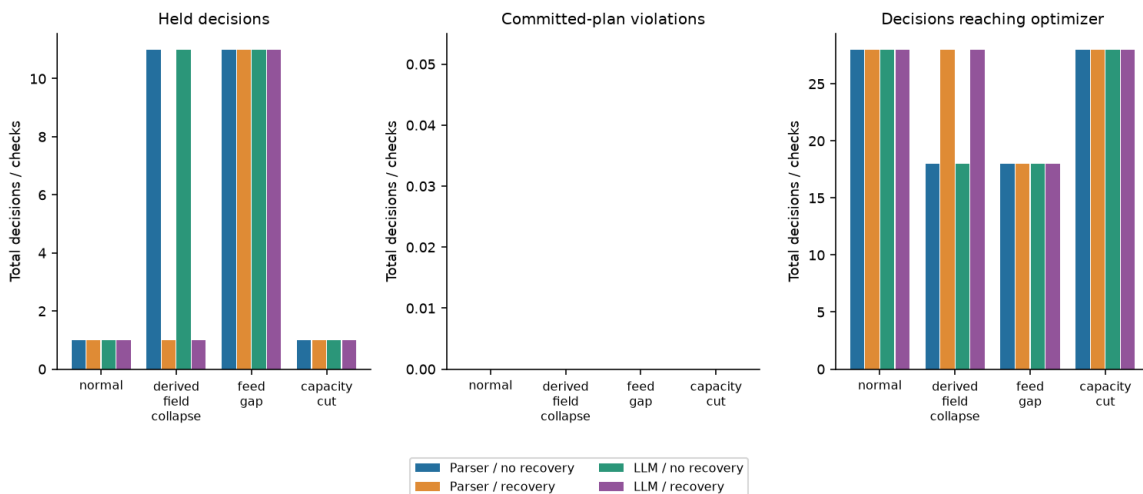


Figure E.8. V2 execution and safety: nonzero replenishment, holds and committed-plan violations. Zero committed violations does not imply zero semantic model errors.

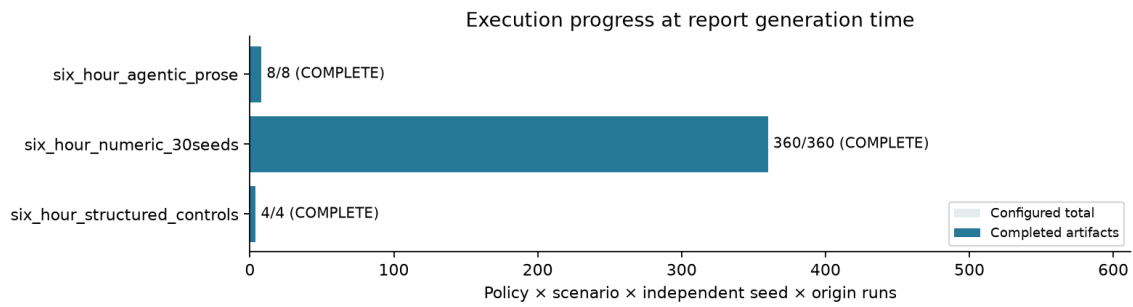


Figure E.9. Initial study completion by stage. Completed controls are not language-model runs; merged and worker copies are counted once.

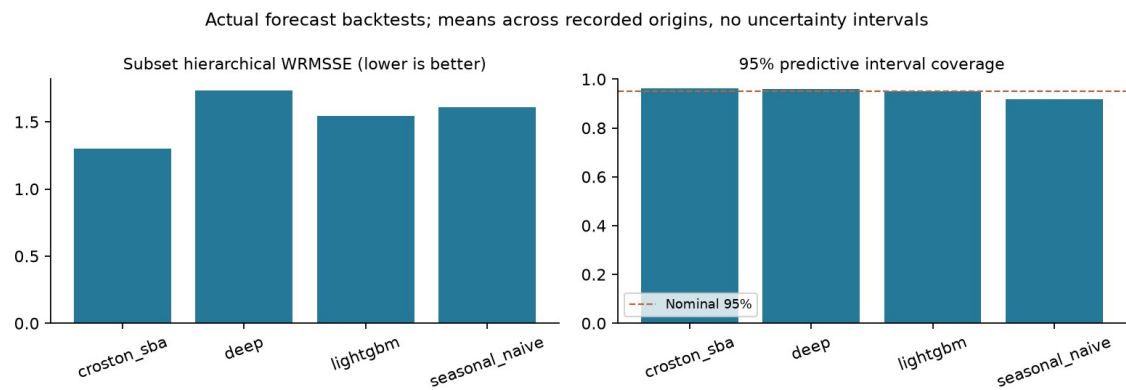


Figure E.10. Forecast scores using the preferred pre-study holdout origins where available. Separate origins and models are descriptive; no confidence interval is implied.

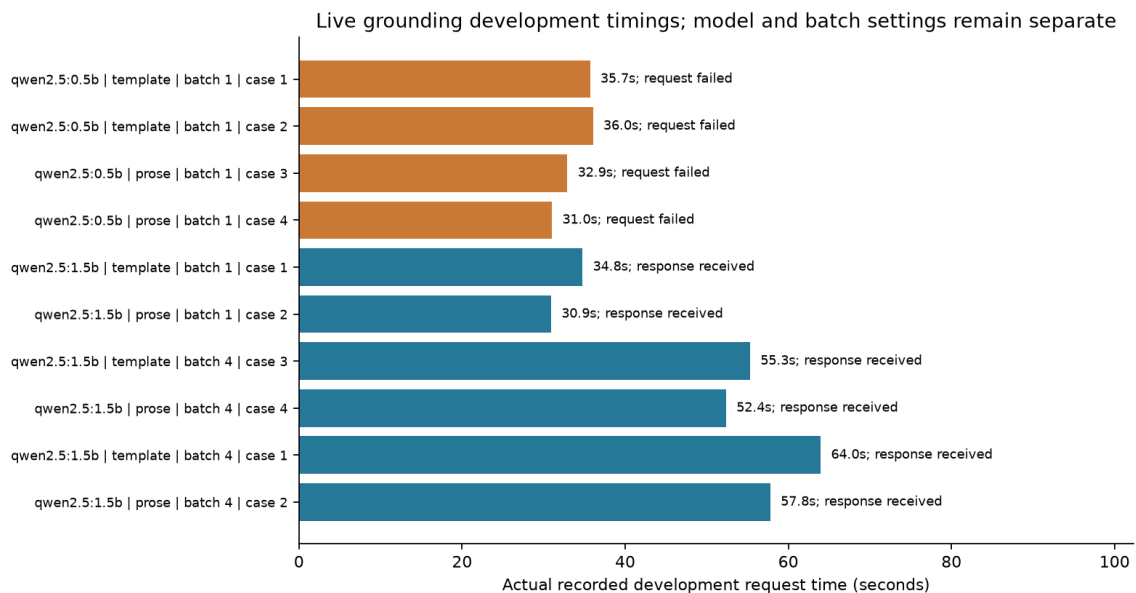


Figure E.11. Development extraction timings for the candidate model profiles. Receiving a response does not establish extraction accuracy or policy benefit.

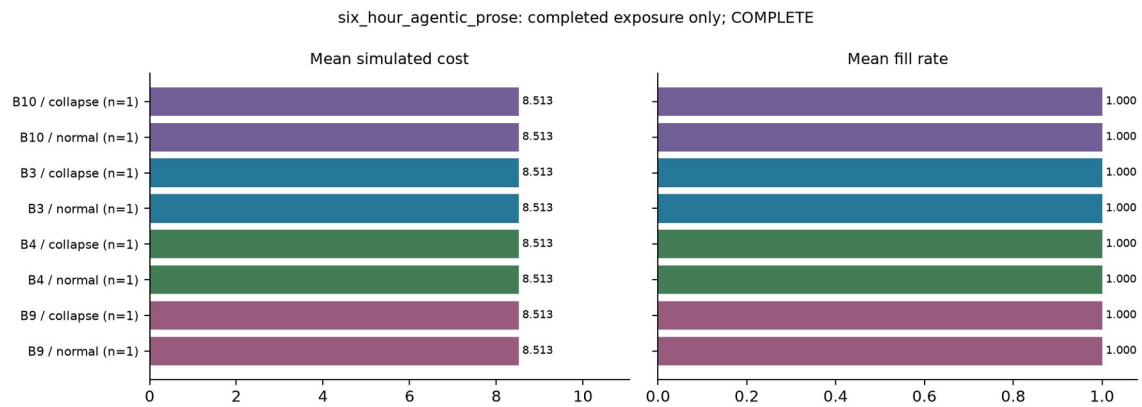


Figure E.12. Two-day controlled-prose feasibility: all decisions were held. Initial stock served demand; low purchase cost is not an LLM saving.

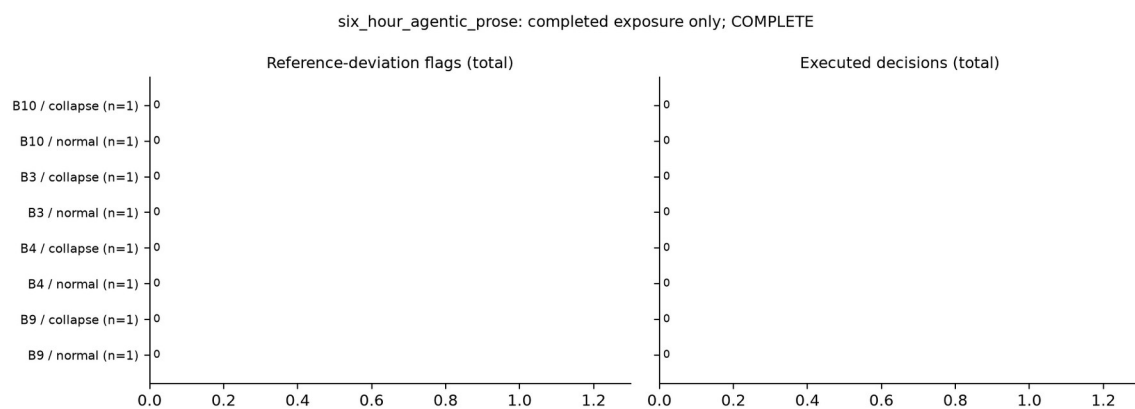


Figure E.13. Two-day controlled-prose feasibility: reference deviations and execution. No live-model proposal reached the optimizer.

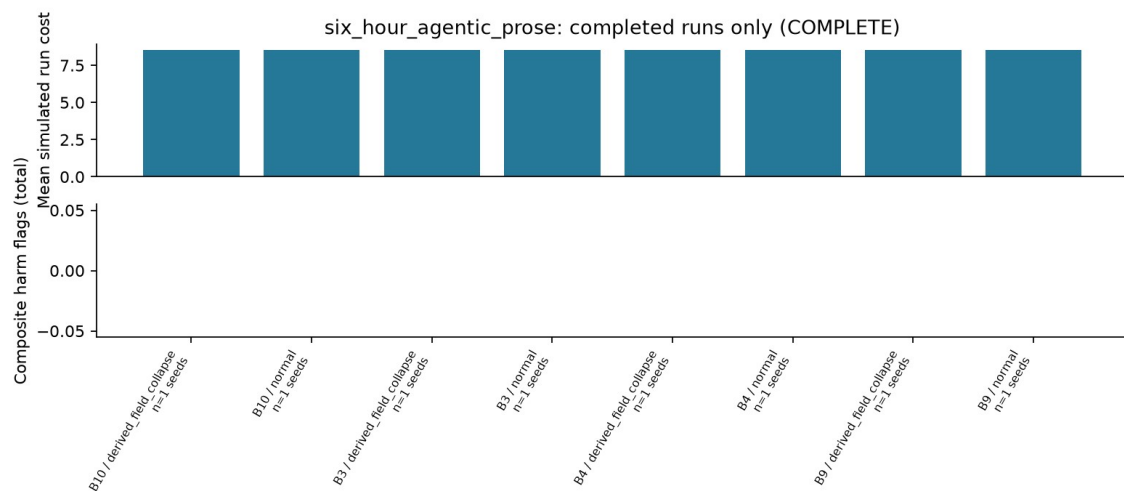


Figure E.14. Two-day controlled-prose feasibility summary. The horizon and document carrier differ from the numerical and V2 comparisons.

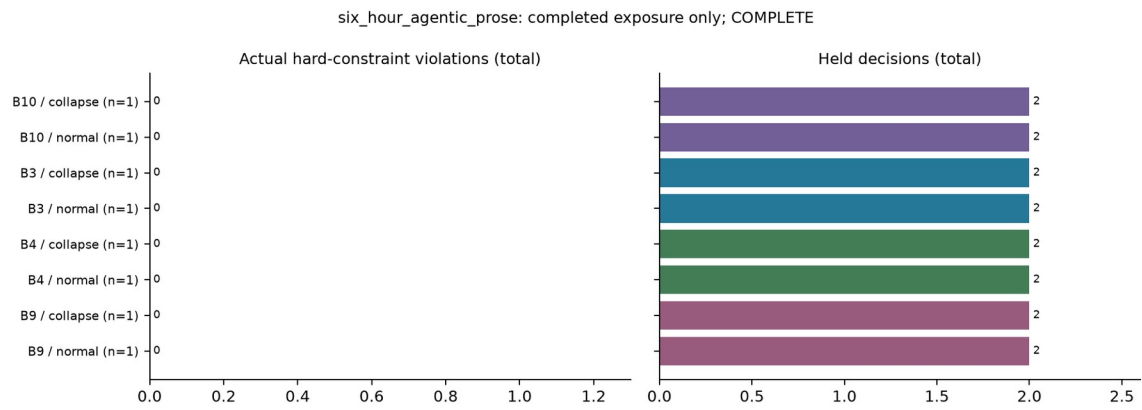


Figure E.15. Two-day controlled-prose feasibility: violations and holds. A zero violation count in all-held cases does not demonstrate executable replenishment.

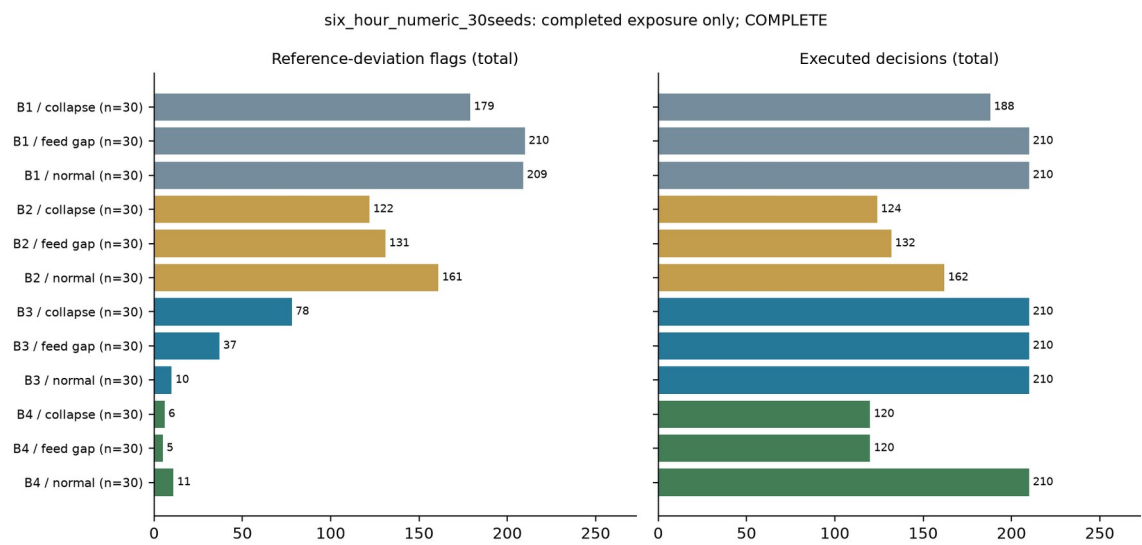


Figure E.16. Thirty-seed numerical benchmark: clean-reference deviations and execution. Reference deviation is not the same as an actual hard violation.

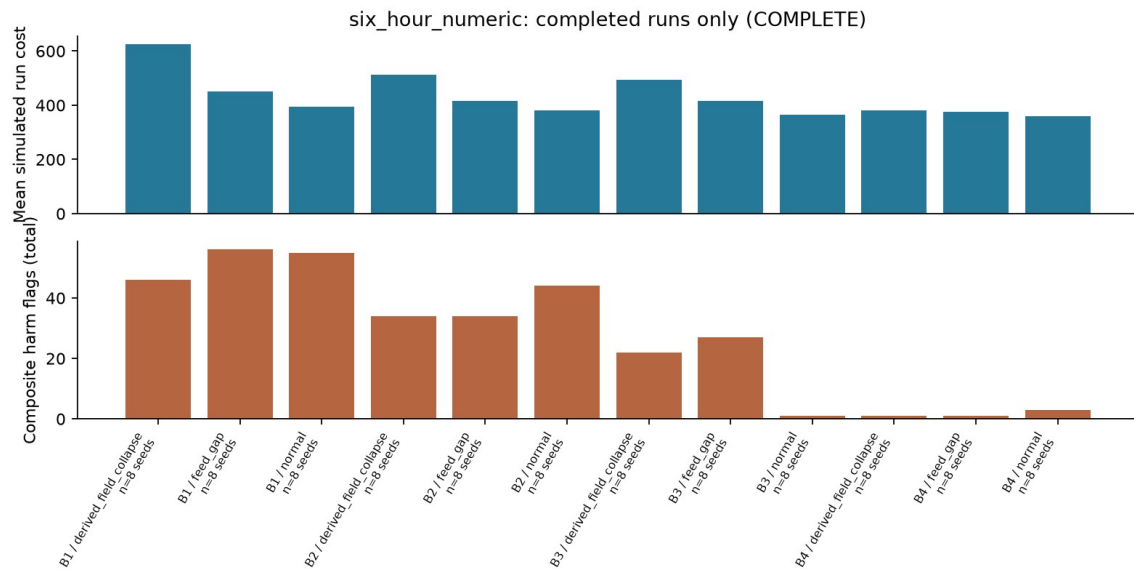


Figure E.17. Initial eight-seed numerical phase only. These 96 runs are a subset of the later verified 360-run union, not additional observations to pool.

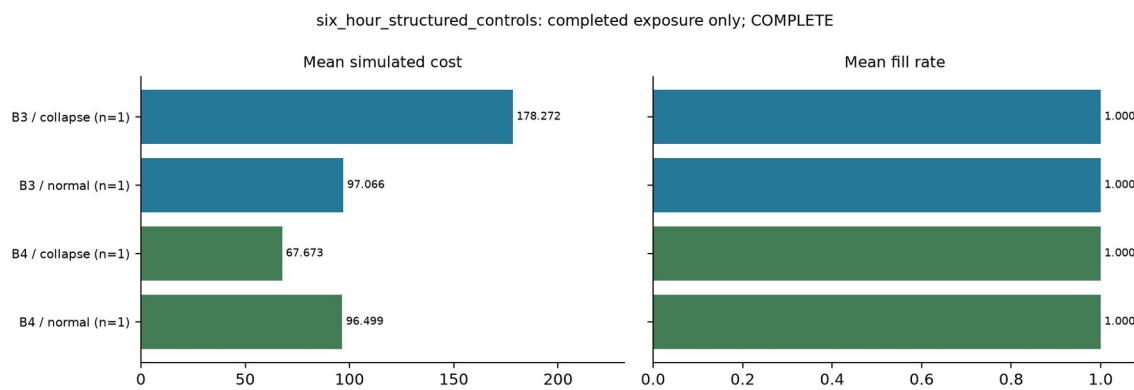


Figure E.18. Two-day structured-rule controls: cost and service. Orders create expenditure for future stock, without a terminal salvage credit.

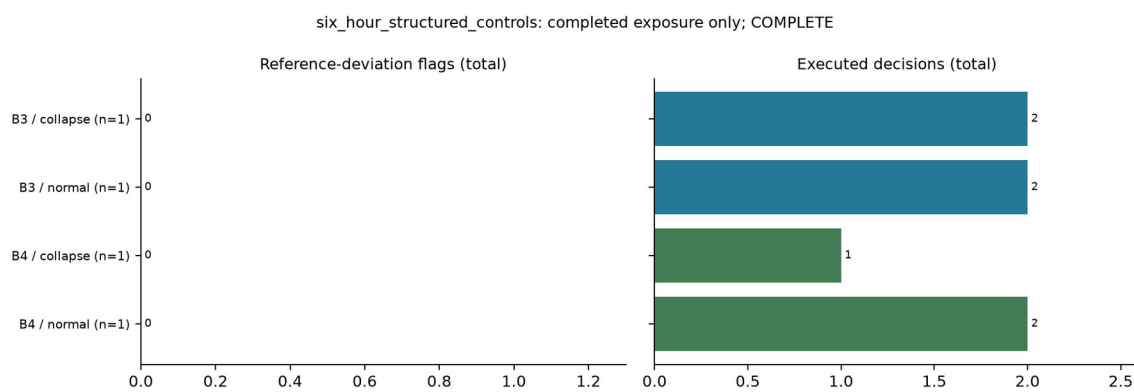


Figure E.19. Two-day structured-rule controls: reference deviations and execution.

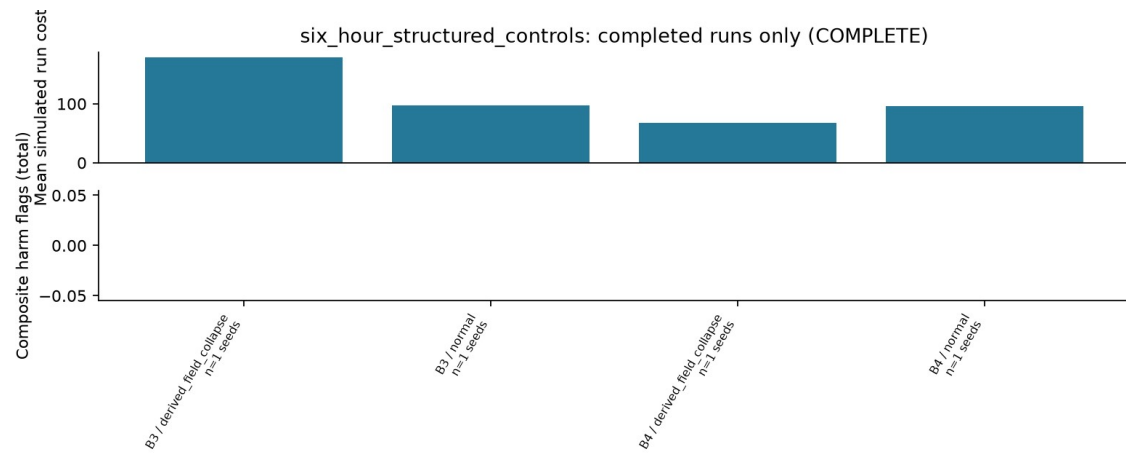


Figure E.20. Two-day structured-rule control summary; one seed and two scenarios.

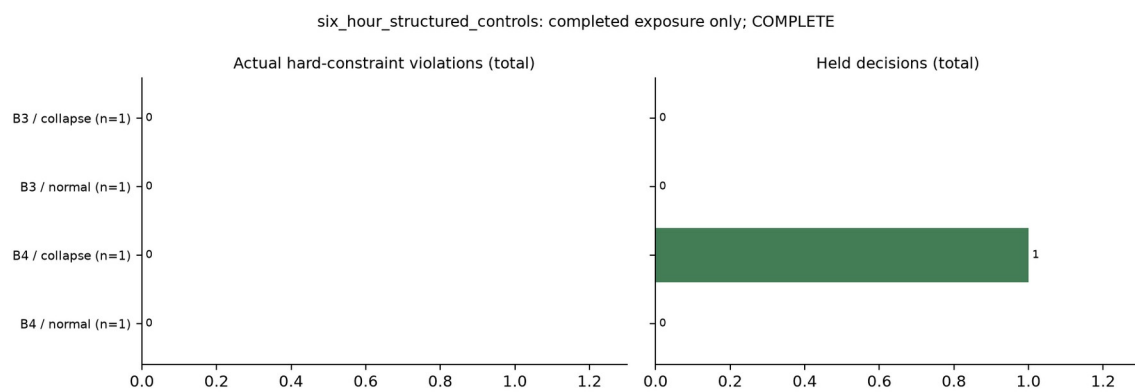


Figure E.21. Two-day structured-rule controls: hard violations and holds. These are deterministic controls, not model-generated decisions.

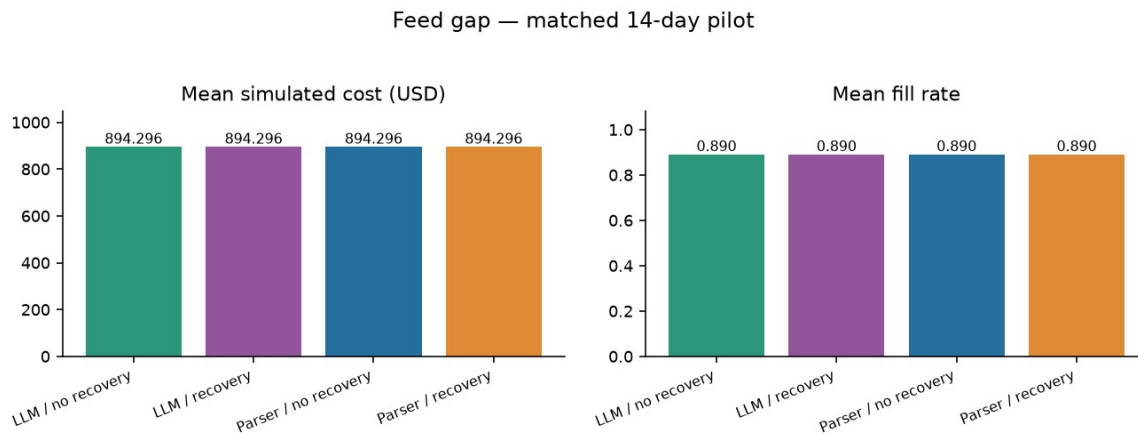


Figure E.22. V2 stale-feed cost and service. Recovery cannot repair absent current evidence; all parser/LLM matched outcomes remain identical.

Capacity cut — inventory and latency

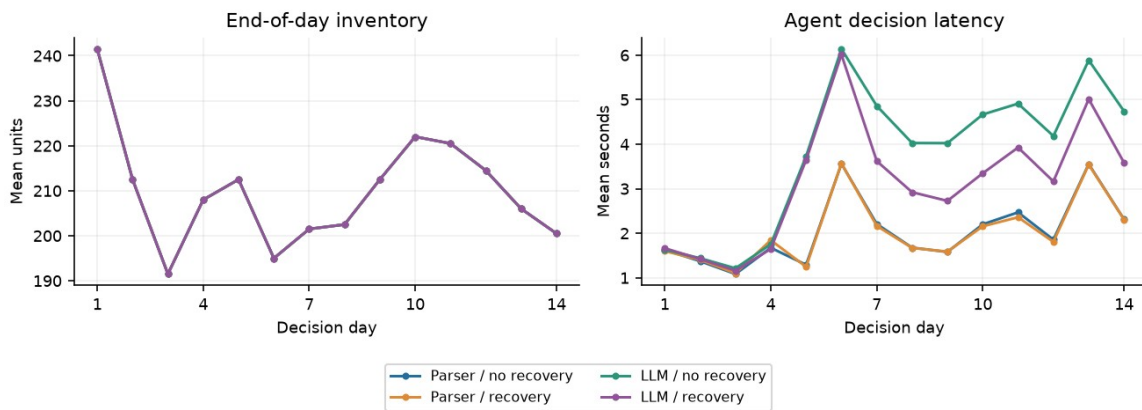


Figure E.23. V2 capacity-cut inventory and latency. Changed constraints are checked before numerical planning.

Feed gap — inventory and latency

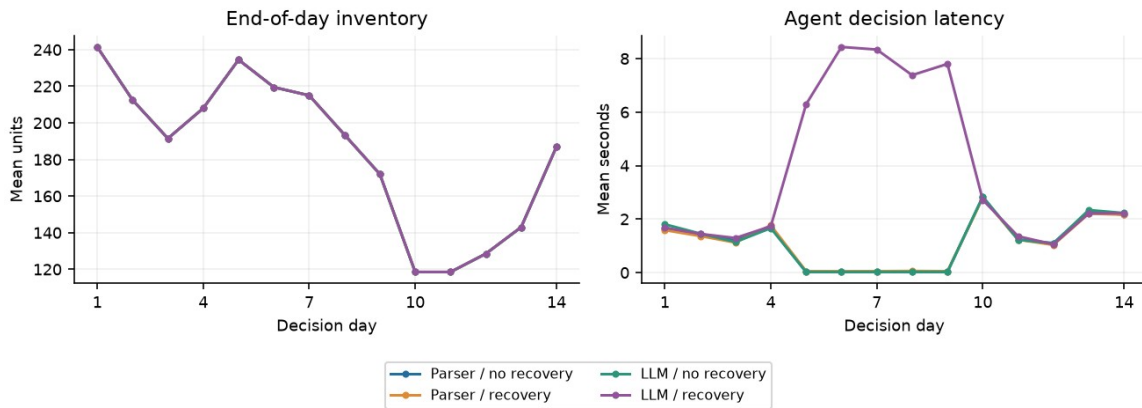


Figure E.24. V2 stale-feed inventory and latency. Held days must be assessed alongside service loss and review needs.

Normal — inventory and latency

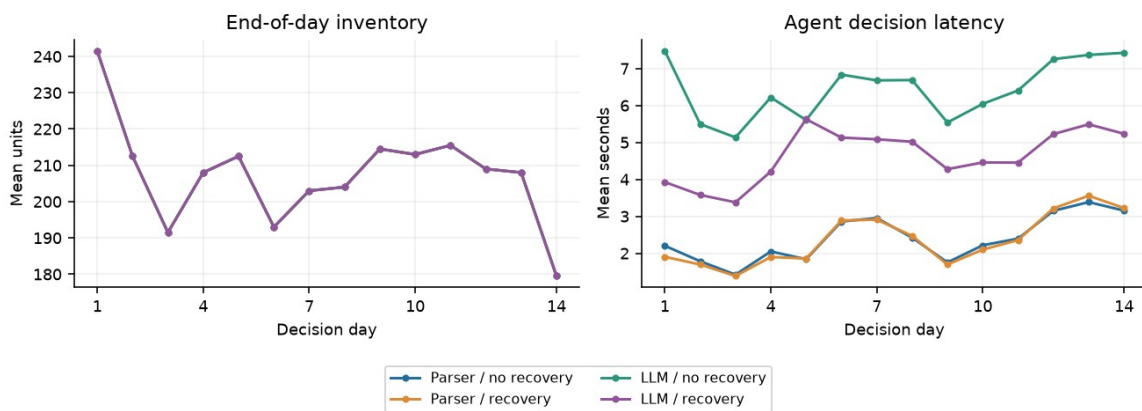


Figure E.25. V2 normal-operation inventory and latency. A language interface adds inference work without a measured operating advantage on this corpus.

Appendix F Complete Historical M5 Results

These tables retain every completed M5 run as historical evidence: 360 seven-day numerical controls, eight two-day prose feasibility runs, four two-day structured controls and 32 fourteen-day V2 runs. The stages are reported separately because their horizons and interventions differ. Simulated USD is the original M5 convention; it is not a currency conversion for FreshRetailNet.

Table F.1. Initial numerical study: B1, Derived-field collapse; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	580.844	94.091	1.429	261.857	73.648	0.791	0	0	7
1	600.789	91.818	3.810	222.857	43.670	0.906	0	0	6
2	605.788	87.727	4.286	236.429	35.049	0.816	0	0	6
3	680.328	95.909	0.952	266.143	73.244	0.793	0	0	6
4	678.667	95.000	1.905	298.429	54.686	0.700	0	0	6
5	635.819	93.636	1.905	277.143	77.946	0.743	0	0	4
6	541.950	97.727	0.952	241.429	51.508	0.891	0	0	6
7	674.733	91.818	2.381	285.143	72.593	0.708	0	0	5
8	676.918	92.273	1.429	280.571	65.194	0.724	0	0	6
9	667.022	85.455	4.286	228.000	55.514	0.825	0	0	5
10	580.279	94.545	2.857	246.857	71.252	0.843	0	0	4
11	656.122	95.000	1.429	296.429	51.062	0.705	0	0	4
12	657.478	94.545	1.905	273.571	58.458	0.760	0	0	5
13	587.079	90.909	3.333	255.714	43.648	0.782	0	0	7
14	591.957	92.727	1.429	250.000	45.033	0.816	0	0	5
15	539.382	92.273	3.810	217.286	31.523	0.934	0	0	7
16	534.790	94.091	0.952	252.571	53.041	0.820	0	0	6
17	628.433	90.455	3.333	250.857	29.274	0.793	0	0	6

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
18	580.608	91.364	2.381	223.714	41.356	0.898	0	0	7
19	613.120	92.727	1.905	261.571	81.347	0.780	0	0	6
20	736.048	93.636	2.857	293.714	69.236	0.701	0	0	7
21	604.112	93.636	3.333	259.429	46.882	0.794	0	0	5
22	605.147	92.273	2.381	257.143	34.685	0.789	0	0	6
23	576.364	98.182	1.429	239.429	44.377	0.902	0	0	7
24	629.771	90.909	3.810	252.000	50.949	0.794	0	0	6
25	622.803	90.909	1.905	247.714	43.441	0.807	0	0	7
26	567.485	91.818	3.333	209.286	28.716	0.965	0	0	7
27	637.631	93.182	1.429	271.571	48.974	0.755	0	0	7
28	653.200	91.364	1.429	259.143	54.502	0.776	0	0	7
29	687.348	93.182	1.429	249.143	58.575	0.823	0	0	6

Table F.2. Initial numerical study: B1, Feed gap; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	453.547	92.273	1.905	208.714	4.390	0.973	0	0	7
1	516.076	88.636	4.286	194.857	0.962	1.001	0	0	7
2	460.242	87.273	4.762	211.000	1.205	0.910	0	0	7
3	482.642	92.727	2.857	195.000	2.719	1.046	0	0	7
4	472.057	87.273	4.286	213.143	2.546	0.901	0	0	7
5	384.216	93.636	1.905	245.429	3.943	0.839	0	0	7
6	395.782	94.091	1.905	201.000	6.578	1.030	0	0	7
7	436.732	90.455	3.333	210.714	0.966	0.944	0	0	7
8	467.024	91.364	1.905	233.857	4.068	0.859	0	0	7
9	480.188	85.000	5.238	184.000	1.063	1.016	0	0	7
10	431.353	91.364	4.286	197.143	2.519	1.020	0	0	7

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
11	448.667	92.727	2.381	248.429	1.279	0.821	0	0	7
12	456.807	89.091	5.238	211.714	2.035	0.926	0	0	7
13	422.681	90.909	3.333	219.143	1.194	0.913	0	0	7
14	433.970	88.636	2.381	203.000	1.042	0.961	0	0	7
15	447.725	90.455	5.238	177.714	2.699	1.120	0	0	7
16	410.086	91.818	2.381	205.000	3.609	0.985	0	0	7
17	427.502	91.364	2.857	236.429	1.726	0.850	0	0	7
18	388.154	91.364	2.381	209.286	9.036	0.960	0	0	7
19	436.058	92.727	1.905	223.286	2.137	0.914	0	0	7
20	499.305	93.182	3.333	207.429	2.468	0.988	0	0	7
21	389.929	93.182	3.810	201.286	2.176	1.018	0	0	7
22	443.170	92.273	2.381	229.571	1.071	0.884	0	0	7
23	399.386	96.364	3.333	179.571	2.503	1.181	0	0	7
24	466.739	90.000	4.286	218.857	3.149	0.905	0	0	7
25	457.127	90.455	2.381	217.571	0.676	0.915	0	0	7
26	469.554	90.455	3.333	177.143	0.687	1.123	0	0	7
27	419.355	90.000	2.857	207.714	1.036	0.953	0	0	7
28	445.311	85.909	4.286	174.429	0.583	1.084	0	0	7
29	454.122	90.455	3.333	173.857	2.151	1.145	0	0	7

Table F.3. Initial numerical study: B1, Normal; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	427.941	94.091	1.429	218.000	3.412	0.950	0	0	7
1	426.077	91.818	3.810	193.143	1.867	1.046	0	0	7
2	433.769	87.727	4.286	213.429	1.733	0.904	0	0	6
3	376.813	94.545	2.381	183.571	0.788	1.133	0	0	7

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
4	375.439	94.091	2.381	215.143	2.194	0.962	0	0	7
5	375.320	93.636	1.905	244.000	1.689	0.844	0	0	7
6	347.486	97.273	1.429	200.857	1.856	1.065	0	0	7
7	395.907	91.818	2.381	218.286	3.565	0.925	0	0	7
8	391.873	92.273	1.429	232.714	1.719	0.872	0	0	7
9	459.809	84.545	4.286	179.857	0.950	1.034	0	0	7
10	391.769	93.182	3.333	201.143	3.580	1.019	0	0	7
11	360.191	95.000	1.429	247.143	1.719	0.846	0	0	7
12	420.493	90.000	3.810	212.286	1.689	0.933	0	0	7
13	382.606	90.909	3.333	219.286	1.147	0.912	0	0	7
14	380.727	91.364	2.381	203.857	1.435	0.986	0	0	7
15	413.080	90.000	5.714	170.143	1.752	1.164	0	0	7
16	345.932	93.636	1.429	203.714	3.835	1.011	0	0	7
17	423.607	90.455	3.333	234.571	1.124	0.848	0	0	7
18	395.513	91.364	2.381	209.571	3.593	0.959	0	0	7
19	417.142	92.727	1.905	219.714	0.928	0.928	0	0	7
20	437.309	93.636	2.857	203.286	2.497	1.013	0	0	7
21	402.809	91.818	2.857	199.857	2.368	1.011	0	0	7
22	400.968	92.273	2.381	226.571	1.877	0.896	0	0	7
23	378.786	98.182	1.429	185.429	1.855	1.165	0	0	7
24	432.461	90.000	4.286	218.714	1.529	0.905	0	0	7
25	402.264	90.909	1.905	215.571	1.018	0.928	0	0	7
26	429.032	90.909	4.286	177.429	1.653	1.127	0	0	7
27	364.765	92.727	1.905	206.000	0.758	0.990	0	0	7
28	394.891	89.091	3.333	175.143	1.905	1.119	0	0	7

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
29	410.619	91.364	2.857	171.857	2.674	1.170	0	0	7

Table F.4. Initial numerical study: B2, Derived-field collapse; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	442.620	90.455	2.857	234.571	55.452	0.848	0	0	5
1	550.061	82.273	6.190	203.857	46.002	0.888	0	0	6
2	531.775	81.818	6.190	216.286	39.000	0.832	0	0	4
3	509.095	92.273	2.381	204.429	47.238	0.993	0	0	5
4	551.962	88.636	2.857	256.571	62.758	0.760	0	0	4
5	558.581	83.636	4.286	245.857	76.143	0.748	0	0	3
6	455.358	89.091	5.238	216.571	45.838	0.905	0	0	4
7	509.624	92.273	1.905	240.857	39.663	0.843	0	0	3
8	584.858	87.273	2.857	261.714	69.405	0.734	0	0	3
9	599.004	81.364	5.238	204.429	33.558	0.876	0	0	4
10	522.896	85.455	5.238	220.143	48.942	0.854	0	0	3
11	565.885	81.364	4.762	256.857	33.643	0.697	0	0	5
12	593.948	84.545	5.714	247.429	51.897	0.752	0	0	4
13	533.309	85.000	5.238	237.000	54.527	0.789	0	0	3
14	611.986	78.182	4.762	221.571	45.233	0.776	0	0	4
15	531.119	86.364	6.190	194.714	28.095	0.976	0	0	5
16	552.544	85.000	4.286	236.000	48.989	0.792	0	0	6
17	593.748	80.455	5.714	234.000	41.604	0.756	0	0	3
18	563.209	80.000	5.714	208.286	55.463	0.845	0	0	3
19	524.788	86.364	4.286	236.000	70.471	0.805	0	0	3
20	568.176	86.818	4.286	239.429	29.763	0.798	0	0	5
21	575.580	83.636	5.714	233.000	39.441	0.790	0	0	4

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
22	559.922	83.636	4.762	222.571	40.494	0.827	0	0	3
23	504.483	90.455	2.857	209.143	35.658	0.952	0	0	5
24	591.026	82.727	5.714	242.571	74.770	0.750	0	0	4
25	573.241	85.000	3.810	227.429	44.564	0.822	0	0	4
26	566.234	80.909	6.667	188.857	46.698	0.943	0	0	5
27	589.727	80.909	4.286	230.857	36.604	0.771	0	0	5
28	539.287	87.727	4.286	223.000	32.282	0.865	0	0	4
29	515.095	91.818	2.857	192.143	49.129	1.051	0	0	3

Table F.5. Initial numerical study: B2, Feed gap; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	369.753	80.455	5.714	189.714	6.850	0.933	0	0	3
1	481.586	75.455	8.095	173.286	3.804	0.958	0	0	6
2	405.321	79.091	6.667	189.286	2.144	0.919	0	0	6
3	437.455	78.636	9.048	139.429	3.602	1.241	0	0	4
4	466.180	75.000	7.143	183.000	6.347	0.902	0	0	5
5	356.344	82.727	4.762	215.571	8.936	0.844	0	0	2
6	389.796	81.364	7.143	184.571	5.355	0.970	0	0	6
7	429.411	77.727	5.238	177.714	6.806	0.962	0	0	2
8	308.103	85.455	3.810	215.000	5.497	0.874	0	0	2
9	557.206	66.818	10.000	153.000	11.101	0.961	0	0	3
10	452.514	74.545	9.048	177.714	5.391	0.923	0	0	4
11	405.040	79.091	6.190	217.429	6.634	0.800	0	0	5
12	456.908	77.727	8.095	178.429	3.181	0.958	0	0	5
13	394.538	80.455	5.714	200.857	3.357	0.881	0	0	5
14	498.013	70.909	8.095	180.429	7.996	0.865	0	0	2

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
15	509.456	71.818	12.857	148.571	1.166	1.063	0	0	7
16	412.066	79.545	7.619	180.000	6.258	0.972	0	0	3
17	396.831	80.000	6.190	214.429	6.517	0.821	0	0	5
18	421.931	80.000	5.714	194.286	4.813	0.906	0	0	5
19	406.748	80.909	5.238	197.143	18.343	0.903	0	0	2
20	419.921	80.909	5.714	173.429	8.279	1.026	0	0	5
21	524.975	68.182	10.476	170.143	4.487	0.882	0	0	4
22	352.414	82.273	5.714	196.857	5.839	0.919	0	0	4
23	507.455	75.000	7.619	154.571	6.505	1.067	0	0	7
24	466.837	75.909	7.619	201.143	6.301	0.830	0	0	4
25	366.120	81.364	6.190	191.000	3.678	0.937	0	0	4
26	564.985	68.636	10.952	150.571	8.831	1.003	0	0	5
27	512.146	76.818	6.190	191.857	5.748	0.881	0	0	6
28	450.603	76.364	9.524	153.000	2.548	1.098	0	0	6
29	481.324	74.091	10.000	124.000	7.990	1.315	0	0	4

Table F.6. Initial numerical study: B2, Normal; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	301.064	88.636	4.286	194.857	1.965	1.001	0	0	6
1	419.940	81.818	6.667	176.143	2.077	1.022	0	0	7
2	363.764	81.818	6.190	192.000	1.947	0.938	0	0	5
3	473.391	77.273	8.571	145.714	2.735	1.167	0	0	5
4	368.462	83.636	5.714	186.143	1.201	0.988	0	0	6
5	356.494	82.727	4.762	215.571	1.785	0.844	0	0	4
6	362.434	82.273	6.667	184.143	2.940	0.983	0	0	6
7	401.815	83.182	4.286	181.429	4.236	1.009	0	0	5

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
8	321.382	85.455	3.810	217.571	1.283	0.864	0	0	5
9	435.857	77.727	8.095	157.857	3.011	1.083	0	0	5
10	355.952	83.636	6.667	188.143	3.392	0.978	0	0	4
11	391.441	79.545	5.714	219.000	1.164	0.799	0	0	6
12	385.545	80.455	8.095	182.143	0.954	0.972	0	0	5
13	328.913	85.000	5.238	202.857	0.923	0.922	0	0	5
14	451.455	75.455	7.619	184.429	1.518	0.900	0	0	5
15	467.125	77.273	10.476	155.286	2.172	1.095	0	0	6
16	390.689	81.818	6.190	184.143	3.216	0.978	0	0	5
17	412.291	80.000	6.190	214.429	2.569	0.821	0	0	5
18	417.287	80.000	5.714	194.286	2.162	0.906	0	0	5
19	413.252	80.909	5.238	197.143	3.210	0.903	0	0	4
20	356.139	85.909	4.762	174.429	2.476	1.084	0	0	7
21	440.575	77.727	6.667	180.714	1.579	0.946	0	0	6
22	332.206	82.273	5.714	199.571	1.953	0.907	0	0	4
23	445.634	81.818	5.714	161.857	0.935	1.112	0	0	7
24	415.406	80.909	6.667	204.000	1.779	0.873	0	0	6
25	357.617	84.091	4.762	192.429	0.590	0.961	0	0	6
26	475.234	75.455	9.524	153.571	1.407	1.081	0	0	6
27	448.054	77.727	5.714	192.143	1.158	0.890	0	0	6
28	433.716	76.364	9.048	154.857	1.903	1.085	0	0	4
29	465.747	78.182	8.095	131.714	2.544	1.306	0	0	5

Table F.7. Initial numerical study: B3, Derived-field collapse; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	439.866	95.909	1.905	217.571	13.219	0.970	0	0	4

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
1	489.219	88.182	3.810	172.714	9.275	1.123	0	0	2
2	481.347	91.364	2.857	184.714	8.423	1.088	0	0	3
3	593.887	95.909	1.429	253.429	32.880	0.833	0	0	2
4	535.276	92.273	1.905	229.857	25.378	0.883	0	0	2
5	435.261	90.909	2.857	219.143	14.608	0.913	0	0	4
6	482.820	91.364	3.810	202.857	10.897	0.991	0	0	3
7	488.040	98.182	1.429	261.714	41.732	0.825	0	0	2
8	475.448	94.545	2.381	239.286	25.149	0.869	0	0	3
9	549.855	94.091	1.429	223.714	24.415	0.925	0	0	2
10	491.177	96.364	0.952	219.714	22.533	0.965	0	0	1
11	485.073	91.818	2.381	235.714	19.124	0.857	0	0	1
12	475.349	96.818	0.476	242.000	38.702	0.880	0	0	2
13	391.486	93.636	1.429	215.000	19.859	0.958	0	0	2
14	512.646	93.636	2.381	214.143	14.187	0.962	0	0	3
15	512.774	94.545	0.952	217.143	20.231	0.958	0	0	3
16	504.385	93.636	0.952	226.143	25.361	0.911	0	0	3
17	455.242	90.909	3.810	208.143	7.777	0.961	0	0	3
18	448.193	90.909	4.286	185.714	11.151	1.077	0	0	3
19	458.832	91.818	3.810	209.571	17.728	0.964	0	0	4
20	579.008	92.727	3.333	229.571	35.758	0.889	0	0	3
21	554.454	98.636	0.476	284.000	30.370	0.764	0	0	3
22	447.659	92.727	2.381	210.143	22.027	0.971	0	0	4
23	559.739	95.000	2.381	228.000	32.030	0.917	0	0	3
24	434.741	96.818	1.905	227.286	7.129	0.937	0	0	2
25	452.407	94.091	1.905	204.857	15.000	1.010	0	0	3

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
26	534.844	92.273	2.857	181.571	12.105	1.118	0	0	1
27	464.362	93.636	1.905	208.143	10.386	0.990	0	0	3
28	596.584	93.636	0.952	247.714	33.902	0.832	0	0	3
29	581.406	95.000	0.952	222.143	27.574	0.941	0	0	1

Table F.8. Initial numerical study: B3, Feed gap; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	326.618	94.545	2.857	181.857	3.413	1.144	0	2	2
1	523.324	83.636	6.190	154.571	3.069	1.190	0	2	2
2	419.805	88.636	3.333	170.000	1.065	1.147	0	3	0
3	529.969	89.091	4.286	167.857	2.130	1.168	0	3	1
4	437.249	88.636	6.190	169.429	0.284	1.151	0	3	0
5	299.497	92.727	2.857	192.571	2.486	1.059	0	2	1
6	414.194	89.091	4.762	169.571	2.604	1.156	0	2	1
7	371.522	97.727	1.905	199.857	3.001	1.076	0	2	1
8	254.657	97.273	1.905	190.857	1.993	1.121	0	0	4
9	497.763	89.091	3.810	169.429	1.826	1.157	0	3	0
10	413.192	94.545	2.381	188.429	2.149	1.104	0	1	2
11	349.793	91.818	3.333	202.714	1.446	0.996	0	3	0
12	425.813	86.818	6.667	166.857	2.907	1.145	0	3	1
13	323.446	92.727	1.905	189.000	1.906	1.079	0	3	0
14	485.673	84.545	6.667	176.286	1.013	1.055	0	2	1
15	549.822	90.455	3.333	189.286	7.208	1.051	0	1	2
16	390.510	89.545	3.333	167.571	6.772	1.176	0	3	0
17	418.278	90.909	3.333	190.714	1.972	1.049	0	3	0
18	395.155	87.273	4.762	173.143	2.569	1.109	0	2	1

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
19	436.883	88.182	4.762	181.286	1.877	1.070	0	3	2
20	418.359	90.455	5.238	165.571	1.711	1.202	0	2	2
21	509.666	90.455	3.333	211.714	7.502	0.940	0	1	3
22	345.444	90.000	4.286	166.714	3.868	1.188	0	3	1
23	461.991	90.909	3.333	157.857	4.337	1.267	0	2	1
24	410.656	90.455	4.762	202.571	8.565	0.982	0	0	3
25	412.484	86.818	5.714	175.571	1.594	1.088	0	2	1
26	503.949	89.545	4.762	145.286	3.136	1.356	0	2	1
27	468.920	88.182	3.810	177.429	2.154	1.093	0	2	1
28	477.102	91.364	2.381	178.000	2.144	1.129	0	2	1
29	495.359	93.182	2.381	161.000	2.906	1.273	0	1	2

Table F.9. Initial numerical study: B3, Normal; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	320.042	98.182	0.952	179.286	3.138	1.205	0	0	0
1	386.882	93.182	1.905	146.429	3.429	1.400	0	0	1
2	378.872	91.818	3.810	159.714	3.407	1.265	0	0	0
3	396.710	95.455	1.905	161.714	1.953	1.299	0	0	0
4	384.803	95.000	1.905	162.857	2.490	1.283	0	0	0
5	298.615	93.182	2.381	190.571	4.864	1.076	0	0	0
6	381.362	92.727	1.905	164.857	5.980	1.237	0	0	0
7	362.470	97.273	1.905	206.857	7.629	1.035	0	0	0
8	271.920	97.273	1.429	184.143	1.285	1.162	0	0	0
9	481.100	91.364	2.381	182.286	2.696	1.103	0	0	1
10	409.881	94.545	1.905	184.000	6.548	1.130	0	0	0
11	364.008	92.273	2.381	200.143	1.547	1.014	0	0	0

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
12	354.187	92.273	2.381	172.000	2.388	1.180	0	0	0
13	339.547	92.273	2.857	185.857	6.262	1.092	0	0	0
14	442.746	89.545	4.762	181.429	1.376	1.086	0	0	0
15	431.136	94.545	0.952	167.143	0.978	1.244	0	0	1
16	349.777	93.636	0.952	167.000	3.585	1.234	0	0	1
17	315.878	94.545	3.333	185.286	2.050	1.123	0	0	0
18	333.663	93.182	3.810	168.143	4.469	1.219	0	0	1
19	378.203	94.091	2.381	176.571	3.519	1.172	0	0	1
20	383.219	94.091	2.381	172.286	3.940	1.201	0	0	0
21	428.192	96.364	1.429	202.714	6.337	1.046	0	0	1
22	320.705	95.455	3.333	165.000	5.749	1.273	0	0	1
23	443.282	92.727	2.857	163.429	4.801	1.248	0	0	0
24	324.717	97.727	0.952	200.857	3.774	1.070	0	0	0
25	310.064	96.364	1.905	171.000	1.536	1.240	0	0	1
26	395.072	92.273	3.333	142.857	3.897	1.421	0	0	0
27	393.204	92.273	2.381	175.714	1.103	1.155	0	0	0
28	441.395	90.909	1.905	163.000	2.735	1.227	0	0	1
29	439.198	94.091	1.905	157.857	2.949	1.311	0	0	0

Table F.10. Initial numerical study: B4, Derived-field collapse; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	303.053	95.909	1.429	184.571	7.241	1.143	3	0	0
1	396.408	87.273	4.286	153.143	11.240	1.254	3	0	1
2	366.133	88.182	4.762	169.857	5.108	1.142	3	0	0
3	424.469	89.545	4.286	160.571	8.745	1.227	3	0	0
4	386.805	89.545	2.381	165.000	10.447	1.194	3	0	0

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
5	318.722	89.545	3.810	198.714	8.105	0.991	3	0	0
6	384.435	87.273	6.190	171.286	7.347	1.121	3	0	0
7	461.449	84.545	5.714	182.000	15.848	1.022	3	0	0
8	266.395	93.636	2.857	198.571	6.640	1.037	3	0	0
9	417.995	91.364	3.333	159.714	16.684	1.258	3	0	0
10	429.800	88.636	4.762	176.857	23.623	1.103	3	0	0
11	325.677	90.909	2.857	204.000	8.837	0.980	3	0	0
12	338.401	90.909	3.333	165.714	15.270	1.207	3	0	0
13	301.053	92.273	2.381	185.286	9.396	1.096	3	0	0
14	356.771	90.455	3.810	170.571	10.808	1.167	3	0	0
15	403.855	92.727	2.381	153.857	14.442	1.326	3	0	1
16	343.798	90.455	2.857	171.143	9.744	1.163	3	0	0
17	352.806	89.091	4.286	193.143	4.652	1.015	3	0	0
18	333.957	89.091	4.762	175.571	8.520	1.116	3	0	0
19	359.715	90.455	4.762	180.286	9.432	1.104	3	0	1
20	444.973	85.455	5.714	159.571	16.258	1.178	3	0	0
21	402.489	95.909	1.905	193.857	15.724	1.088	3	0	0
22	293.166	92.273	2.857	173.286	7.032	1.171	3	0	1
23	382.948	93.636	3.333	154.286	10.912	1.335	3	0	0
24	294.758	97.273	1.429	202.000	8.510	1.059	3	0	1
25	306.391	94.091	1.905	175.000	5.627	1.183	3	0	1
26	399.746	90.909	3.810	140.857	20.239	1.420	3	0	0
27	423.914	85.000	5.238	168.286	12.272	1.111	3	0	0
28	400.945	91.364	2.857	150.143	16.582	1.339	3	0	0
29	461.650	87.273	5.238	142.857	18.639	1.344	3	0	0

Table F.11. Initial numerical study: B4, Feed gap; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	306.083	95.909	1.429	184.429	7.337	1.144	3	0	0
1	391.109	87.273	4.286	153.429	10.667	1.251	3	0	1
2	366.133	88.182	4.762	169.857	5.108	1.142	3	0	0
3	424.469	89.545	4.286	160.571	8.745	1.227	3	0	0
4	386.805	89.545	2.381	165.000	10.447	1.194	3	0	0
5	299.313	90.909	2.857	199.571	8.105	1.002	3	0	0
6	384.435	87.273	6.190	171.286	7.347	1.121	3	0	0
7	455.897	85.000	5.714	181.571	15.693	1.030	3	0	0
8	266.395	93.636	2.857	198.571	6.640	1.037	3	0	0
9	417.715	91.364	3.333	159.857	16.684	1.257	3	0	0
10	433.089	88.182	4.762	177.286	23.260	1.094	3	0	0
11	330.077	90.909	2.857	204.000	9.047	0.980	3	0	0
12	338.401	90.909	3.333	165.714	15.270	1.207	3	0	0
13	301.053	92.273	2.381	185.286	9.396	1.096	3	0	0
14	356.771	90.455	3.810	170.571	10.808	1.167	3	0	0
15	403.855	92.727	2.381	153.857	14.442	1.326	3	0	0
16	343.800	90.455	2.857	171.143	9.744	1.163	3	0	0
17	352.806	89.091	4.286	193.143	4.652	1.015	3	0	0
18	333.957	89.091	4.762	175.571	8.520	1.116	3	0	0
19	359.715	90.455	4.762	180.286	9.432	1.104	3	0	1
20	445.536	85.455	5.714	159.286	16.258	1.180	3	0	0
21	398.958	96.364	1.429	194.857	15.838	1.088	3	0	0
22	293.166	92.273	2.857	173.286	7.032	1.171	3	0	1
23	382.948	93.636	3.333	154.286	10.912	1.335	3	0	1

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
24	282.546	97.273	1.429	202.429	7.776	1.057	3	0	1
25	306.391	94.091	1.905	175.000	5.627	1.183	3	0	0
26	380.429	90.909	3.810	140.714	21.374	1.421	3	0	0
27	421.714	85.000	5.238	168.286	12.267	1.111	3	0	0
28	400.660	91.364	2.857	150.286	16.582	1.337	3	0	0
29	469.650	86.818	5.714	142.857	18.606	1.337	3	0	0

Table F.12. Initial numerical study: B4, Normal; thirty seeds, seven days.

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
0	306.748	98.636	0.952	173.571	3.639	1.250	0	0	0
1	396.605	92.273	2.381	147.000	3.579	1.381	0	0	0
2	368.676	92.727	2.857	159.429	2.051	1.280	0	0	1
3	396.710	95.455	1.905	161.714	1.953	1.299	0	0	0
4	362.752	95.455	1.905	160.714	2.266	1.307	0	0	2
5	302.643	92.727	2.857	192.000	4.987	1.062	0	0	0
6	379.987	92.727	1.905	164.857	5.653	1.237	0	0	0
7	367.838	97.273	1.905	206.857	9.867	1.035	0	0	0
8	268.700	97.273	1.429	184.714	1.359	1.159	0	0	0
9	483.927	91.364	2.381	183.286	1.528	1.097	0	0	1
10	411.304	94.545	1.905	182.857	6.634	1.137	0	0	0
11	373.358	91.364	2.381	201.286	1.547	0.999	0	0	1
12	354.187	92.273	2.381	172.000	2.388	1.180	0	0	0
13	339.547	92.273	2.857	185.857	6.262	1.092	0	0	0
14	445.841	89.545	4.762	180.857	1.456	1.089	0	0	0
15	418.413	94.545	0.952	167.000	1.238	1.246	0	0	0
16	345.342	93.636	0.952	167.143	2.342	1.232	0	0	0

Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
17	311.474	94.545	3.333	185.429	1.802	1.122	0	0	0
18	349.115	91.818	3.333	168.429	4.516	1.199	0	0	0
19	384.601	93.636	2.381	176.571	3.346	1.167	0	0	1
20	382.430	94.091	2.381	171.571	3.900	1.206	0	0	0
21	430.336	96.364	1.429	199.571	7.362	1.062	0	0	0
22	320.534	95.455	3.333	165.714	5.999	1.267	0	0	1
23	431.231	93.636	2.381	163.429	4.777	1.260	0	0	0
24	319.875	97.727	0.952	201.143	3.789	1.069	0	0	1
25	310.064	96.364	1.905	171.000	1.536	1.240	0	0	1
26	398.448	92.273	3.333	143.857	4.218	1.411	0	0	0
27	386.998	92.727	2.381	171.429	1.025	1.190	0	0	1
28	439.788	90.909	2.381	162.429	2.542	1.231	0	0	1
29	432.529	94.545	1.429	158.571	2.949	1.312	0	0	0

Table F.13. Prose feasibility; two-day window, seed zero. All decisions were held.

Policy	Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
B10	Derived-field collapse	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B10	Normal	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B3	Derived-field collapse	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B3	Normal	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B4	Derived-field collapse	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B4	Normal	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B9	Derived-field collapse	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0
B9	Normal	0	8.513	100.000	0.000	237.500	0.000	0.232	2	0	0

Table F.14. Structured-rule controls; two-day window, seed zero.

Policy	Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
B3	Derived-field collapse	0	178.272	100.000	0.000	231.000	10.467	0.238	0	0	0
B3	Normal	0	97.066	100.000	0.000	224.000	1.526	0.246	0	0	0
B4	Derived-field collapse	0	67.673	100.000	0.000	231.000	5.817	0.238	1	0	0
B4	Normal	0	96.499	100.000	0.000	225.000	1.526	0.244	0	0	0

Table F.15. V2 L0 (llm_v2_no_recovery); two seeds, fourteen days.

Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
Capacity cut	13	750.366	95.556	2.619	205.286	4.546	2.095	0	0	1
Capacity cut	29	859.773	96.000	2.143	214.857	11.992	2.011	1	0	0
Derived-field collapse	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0
Derived-field collapse	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Feed gap	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0
Feed gap	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Normal	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Normal	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0

Table F.16. V2 L1 (llm_v2_recovery); two seeds, fourteen days.

Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
Capacity cut	13	750.366	95.556	2.619	205.286	4.546	2.095	0	0	1
Capacity cut	29	859.773	96.000	2.143	214.857	11.992	2.011	1	0	0
Derived-field collapse	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Derived-field collapse	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0
Feed gap	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0

Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
Feed gap	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Normal	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Normal	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0

Table F.17. V2 P0 (parser_no_recovery); two seeds, fourteen days.

Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
Capacity cut	13	750.366	95.556	2.619	205.286	4.546	2.095	0	0	1
Capacity cut	29	859.773	96.000	2.143	214.857	11.992	2.011	1	0	0
Derived-field collapse	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0
Derived-field collapse	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Feed gap	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0
Feed gap	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Normal	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Normal	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0

Table F.18. V2 P1 (parser_recovery); two seeds, fourteen days.

Scenario	Seed	Cost USD	Fill %	Stockout %	Mean stock	Bullwhip	Turns	Holds	Hard viol.	Ref. dev.
Capacity cut	13	750.366	95.556	2.619	205.286	4.546	2.095	0	0	1
Capacity cut	29	859.773	96.000	2.143	214.857	11.992	2.011	1	0	0
Derived-field collapse	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Derived-field collapse	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0
Feed gap	13	906.030	86.000	7.381	172.429	19.709	2.244	5	0	0
Feed gap	29	882.563	92.000	4.286	196.643	22.010	2.105	6	0	0
Normal	13	760.716	95.556	2.619	204.214	5.147	2.106	0	0	0
Normal	29	832.295	96.444	1.667	210.857	13.135	2.058	1	0	0

Table F.19. All sixteen recorded forecast backtest cases; thirty bottom series and 112 aggregate nodes. Lower error scores are better; coverage is reported against a 95% interval target.

Model	Origin index	Role	WRMSSE	Weighted pinball	CRPS	Coverage %
croston_sba	1800	Overlapping descriptive	1.20383	0.36674	0.62790	94.405
croston_sba	1828	Overlapping descriptive	1.42780	0.42589	0.64551	94.643
Native GRU/NB	1800	Overlapping descriptive	1.79618	0.40671	0.58237	93.929
Native GRU/NB	1828	Overlapping descriptive	1.71381	0.43887	0.62141	96.071
croston_sba	1760	Pre-study holdout	1.26410	0.58463	0.58649	97.381
croston_sba	1788	Pre-study holdout	1.33625	0.38161	0.63009	95.357
Native GRU/NB	1760	Pre-study holdout	1.44460	0.42540	0.51009	97.738
Native GRU/NB	1788	Pre-study holdout	2.01835	0.47638	0.64160	93.929
lightgbm	1760	Pre-study holdout	1.28532	0.47244	0.53955	97.143
lightgbm	1788	Pre-study holdout	1.80343	0.49123	0.65094	93.095
seasonal_naive	1760	Pre-study holdout	1.63697	0.84973	0.78446	91.071
seasonal_naive	1788	Pre-study holdout	1.57705	0.46101	0.77301	92.262
lightgbm	1800	Overlapping descriptive	1.49309	0.41168	0.61588	93.929
lightgbm	1828	Overlapping descriptive	1.85932	0.53388	0.65519	95.119
seasonal_naive	1800	Overlapping descriptive	1.53790	0.55032	0.87648	88.571
seasonal_naive	1828	Overlapping descriptive	1.59260	0.55467	0.80829	91.429